

and $b = \hat{\mathbf{R}} \cdot \hat{\mathbf{z}} = \cos \theta$. Upon inserting these expressions for a and b in Eq. (3.62) and replacing $\hat{\mathbf{r}}$ with Eq. (3.56a), we have

$$\hat{\mathbf{R}} = \hat{\mathbf{x}} \sin \theta \cos \phi + \hat{\mathbf{y}} \sin \theta \sin \phi + \hat{\mathbf{z}} \cos \theta.$$
(3.64a)

A similar procedure can be followed to obtain the expression for $\hat{\boldsymbol{\theta}}$:

$$\hat{\boldsymbol{\theta}} = \hat{\mathbf{x}} \cos \theta \cos \phi + \hat{\mathbf{y}} \cos \theta \sin \phi - \hat{\mathbf{z}} \sin \theta.$$
(3.64b)

Finally $\hat{\boldsymbol{\phi}}$ is given by Eq. (3.56b) as

$$\hat{\boldsymbol{\phi}} = -\hat{\mathbf{x}} \sin \phi + \hat{\mathbf{y}} \cos \phi.$$
(3.64c)

Equations (3.64a) through (3.64c) can be solved simultaneously to give the expressions for $(\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}})$ in terms of $(\hat{\mathbf{R}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}})$:

$$\hat{\mathbf{x}} = \hat{\mathbf{R}} \sin \theta \cos \phi + \hat{\boldsymbol{\theta}} \cos \theta \cos \phi - \hat{\boldsymbol{\phi}} \sin \phi,$$
(3.65a)

$$\hat{\mathbf{y}} = \hat{\mathbf{R}} \sin \theta \sin \phi + \hat{\boldsymbol{\theta}} \cos \theta \sin \phi + \hat{\boldsymbol{\phi}} \cos \phi,$$
(3.65b)

$$\hat{\mathbf{z}} = \hat{\mathbf{R}} \cos \theta - \hat{\boldsymbol{\theta}} \sin \theta.$$
(3.65c)

Equations (3.64a) to (3.65c) also can be used to transform Cartesian components (A_x, A_y, A_z) of vector $\mathbf{A}$ into their spherical counterparts (A_R, A_θ, A_ϕ) , and vice versa, by replacing $(\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}, \hat{\mathbf{R}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}})$ with $(A_x, A_y, A_z, A_R, A_\theta, A_\phi)$.

Example 3-9:
Cartesian to Spherical Transformation

Express vector $\mathbf{A} = \hat{\mathbf{x}}(x + y) + \hat{\mathbf{y}}(y - x) + \hat{\mathbf{z}}z$ in spherical coordinates.

Solution: Using the transformation relation for A_R given in Table 3-2, we have

$$\begin{aligned} A_R &= A_x \sin \theta \cos \phi + A_y \sin \theta \sin \phi + A_z \cos \theta \\ &= (x + y) \sin \theta \cos \phi + (y - x) \sin \theta \sin \phi + z \cos \theta. \end{aligned}$$

Using the expressions for x , y , and z given by Eq. (3.61c), we have

$$\begin{aligned} A_R &= (R \sin \theta \cos \phi + R \sin \theta \sin \phi) \sin \theta \cos \phi \\ &\quad + (R \sin \theta \sin \phi - R \sin \theta \cos \phi) \sin \theta \sin \phi + R \cos^2 \theta \\ &= R \sin^2 \theta (\cos^2 \phi + \sin^2 \phi) + R \cos^2 \theta \\ &= R \sin^2 \theta + R \cos^2 \theta = R. \end{aligned}$$

Similarly,

$$\begin{aligned} A_\theta &= (x + y) \cos \theta \cos \phi + (y - x) \cos \theta \sin \phi - z \sin \theta, \\ A_\phi &= -(x + y) \sin \phi + (y - x) \cos \phi, \end{aligned}$$

and following the procedure used with A_R , we obtain

$$A_\theta = 0, \quad A_\phi = -R \sin \theta.$$

Hence,

$$\mathbf{A} = \hat{\mathbf{R}} A_R + \hat{\boldsymbol{\theta}} A_\theta + \hat{\boldsymbol{\phi}} A_\phi = \hat{\mathbf{R}} R - \hat{\boldsymbol{\phi}} R \sin \theta.$$

3-3.3 Cylindrical to Spherical Transformations

Transformations between cylindrical and spherical coordinates can be realized by combining the transformation relations of the preceding two subsections. The results are given in Table 3-2.

3-3.4 Distance between Two Points

In Cartesian coordinates, the distance d between two points $P_1 = (x_1, y_1, z_1)$ and $P_2 = (x_2, y_2, z_2)$ is given by Eq. (3.12) as

$$d = |\mathbf{R}_{12}| = [(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]^{1/2}.$$
(3.66)

Upon using Eq. (3.52) to convert the Cartesian coordinates of P_1 and P_2 into their cylindrical equivalents, we have

$$\begin{aligned} d &= [(r_2 \cos \phi_2 - r_1 \cos \phi_1)^2 \\ &\quad + (r_2 \sin \phi_2 - r_1 \sin \phi_1)^2 + (z_2 - z_1)^2]^{1/2} \\ &= [r_2^2 + r_1^2 - 2r_1 r_2 \cos(\phi_2 - \phi_1) + (z_2 - z_1)^2]^{1/2}. \end{aligned}$$
(3.67)

(cylindrical)

A similar transformation using Eqs. (3.61a) through (3.61c) leads to an expression for d in terms of the spherical coordinates of P_1 and P_2 :

$$\begin{aligned} d &= \{R_2^2 + R_1^2 - 2R_1 R_2 [\cos \theta_2 \cos \theta_1 \\ &\quad + \sin \theta_1 \sin \theta_2 \cos(\phi_2 - \phi_1)]\}^{1/2}. \end{aligned}$$
(3.68)

(spherical)

和 $b = \hat{\mathbf{R}} \cdot \hat{\mathbf{z}} = \cos \theta$ 。在等式中插入 a 和 b 的这些表达式后： (3.62) 并用等式替换 $\hat{\mathbf{r}}$ 。 (3.56a) ， 我们有

$$\hat{\mathbf{R}} = \hat{\mathbf{x}} \sin \theta \cos \phi + \hat{\mathbf{y}} \sin \theta \sin \phi + \hat{\mathbf{z}} \cos \theta.$$
(3.64a)

可以遵循类似的过程来获得 $\hat{\boldsymbol{\theta}}$ 的表达式：

$$\hat{\boldsymbol{\theta}} = \hat{\mathbf{x}} \cos \theta \cos \phi + \hat{\mathbf{y}} \cos \theta \sin \phi - \hat{\mathbf{z}} \sin \theta.$$
(3.64b)

最后, $\hat{\boldsymbol{\phi}}$ 由等式给出。 (3.56b) 为

$$\hat{\boldsymbol{\phi}} = -\hat{\mathbf{x}} \sin \phi + \hat{\mathbf{y}} \cos \phi.$$
(3.64c)

方程 (3.64a) 到 (3.64c) 可以同时求解, 以 $(\hat{\mathbf{R}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}})$ 的形式给出 $(\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}})$ 的表达式：

$$\hat{\mathbf{x}} = \hat{\mathbf{R}} \sin \theta \cos \phi + \hat{\boldsymbol{\theta}} \cos \theta \cos \phi - \hat{\boldsymbol{\phi}} \sin \phi,$$
(3.65a)

$$\hat{\mathbf{y}} = \hat{\mathbf{R}} \sin \theta \sin \phi + \hat{\boldsymbol{\theta}} \cos \theta \sin \phi + \hat{\boldsymbol{\phi}} \cos \phi,$$
(3.65b)

$$\hat{\mathbf{z}} = \hat{\mathbf{R}} \cos \theta - \hat{\boldsymbol{\theta}} \sin \theta.$$
(3.65c)

方程 (3.64a) 至 (3.65c) 也可用于将向量 $\mathbf{A}$ 的笛卡尔分量 (A_x, A_y, A_z) 转换为它们的球面分量 (A_R, A_θ, A_ϕ) , 反之亦然, 通过替换 $(\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}})$, $\hat{\mathbf{R}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}}$ 与 $(A_x, A_y, A_z, A_R, A_\theta, A_\phi)$ 。

示例 3-9: 笛卡尔到球面的变换

在球坐标中表达向量 $\mathbf{A} = \hat{\mathbf{x}}(x + y) + \hat{\mathbf{y}}(y - x) + \hat{\mathbf{z}}z$ 。

解：利用表3-2中给出的 A_R 的变换关系， 我们有

$$\begin{aligned} A_R &= A_x \sin \theta \cos \phi + A_y \sin \theta \sin \phi + A_z \cos \theta \\ &= (x + y) \sin \theta \cos \phi + (y - x) \sin \theta \sin \phi + z \cos \theta. \end{aligned}$$

使用方程式给出的 x、y 和 z 表达式。 (3.61c) ， 我们有

$$\begin{aligned} A_R &= (R \sin \theta \cos \phi + R \sin \theta \sin \phi) \sin \theta \cos \phi \\ &\quad + (R \sin \theta \sin \phi - R \sin \theta \cos \phi) \sin \theta \sin \phi + R \cos^2 \theta \\ &= R \sin^2 \theta (\cos^2 \phi + \sin^2 \phi) + R \cos^2 \theta \\ &= R \sin^2 \theta + R \cos^2 \theta = R. \end{aligned}$$

相似地,

$$\begin{aligned} A_\theta &= (x + y) \cos \theta \cos \phi + (y - x) \cos \theta \sin \phi - z \sin \theta, \\ A_\phi &= -(x + y) \sin \phi + (y - x) \cos \phi, \end{aligned}$$

并按照 A_R 使用的过程， 我们得到

$$A_\theta = 0, \quad A_\phi = -R \sin \theta.$$

因此,

$$\mathbf{A} = \hat{\mathbf{R}} A_R + \hat{\boldsymbol{\theta}} A_\theta + \hat{\boldsymbol{\phi}} A_\phi = \hat{\mathbf{R}} R - \hat{\boldsymbol{\phi}} R \sin \theta.$$

3-3.3 圆柱到球面的变换

结合前两小节的变换关系， 可以实现柱坐标和球坐标之间的变换。 结果如表3-2所示。

3-3.4 两点之间的距离

在笛卡尔坐标系中， 两点 $P_1 = (x_1, y_1, z_1)$ 和 $P_2 = (x_2, y_2, z_2)$ 之间的距离 d 由等式给出： (3.12) 作为

$$d = |\mathbf{R}_{12}| = [(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]^{1/2}.$$
(3.66)

使用方程式后。 (3.52) 将 P_1 和 P_2 的笛卡尔坐标转换为它们的圆柱坐标， 我们有

$$\begin{aligned} d &= [(r_2 \cos \phi_2 - r_1 \cos \phi_1)^2 \\ &\quad + (r_2 \sin \phi_2 - r_1 \sin \phi_1)^2 + (z_2 - z_1)^2]^{1/2} \\ &= [r_2^2 + r_1^2 - 2r_1 r_2 \cos(\phi_2 - \phi_1) + (z_2 - z_1)^2]^{1/2}. \end{aligned}$$
(3.67)

(cylindrical)

使用等式进行类似的转换。 (3.61a) 到 (3.61c) 得出 d 用 P_1 和 P_2 球坐标表示的表达式：

$$\begin{aligned} d &= \{R_2^2 + R_1^2 - 2R_1 R_2 [\cos \theta_2 \cos \theta_1 \\ &\quad + \sin \theta_1 \sin \theta_2 \cos(\phi_2 - \phi_1)]\}^{1/2}. \end{aligned}$$
(3.68)

(spherical)

Example 3-10: Vector Component

At a given point in space, vectors **A** and **B** are given in cylindrical coordinates by

$$\mathbf{A} = \hat{\mathbf{r}}2 + \hat{\boldsymbol{\phi}}3 - \hat{\mathbf{z}},$$

$$\mathbf{B} = \hat{\mathbf{r}} + \hat{\mathbf{z}}.$$

Determine (a) the scalar component of **B**, or projection, in the direction of **A**, (b) the vector component of **B** in the direction of **A**, and (c) the vector component of **B** perpendicular to **A**.

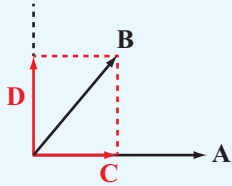


Figure 3-19 Vectors **A**, **B**, **C**, and **D** of Example 3-10.

Solution: (a) Let us denote the scalar component of **B** in the direction of **A** as *C*, as shown in Fig. 3-19. Thus,

$$C = \mathbf{B} \cdot \hat{\mathbf{a}} = \mathbf{B} \cdot \frac{\mathbf{A}}{|\mathbf{A}|} = (\hat{\mathbf{r}} + \hat{\mathbf{z}}) \cdot \frac{(\hat{\mathbf{r}}2 + \hat{\boldsymbol{\phi}}3 - \hat{\mathbf{z}})}{\sqrt{4+9+1}} = \frac{2-1}{\sqrt{14}} = 0.267.$$

(b) The vector component of **B** in the direction of **A** is given by the product of the scalar component *C* and the unit vector **â**:

$$\mathbf{C} = \hat{\mathbf{a}}C = \frac{\mathbf{A}}{|\mathbf{A}|} C = \frac{(\hat{\mathbf{r}}2 + \hat{\boldsymbol{\phi}}3 - \hat{\mathbf{z}})}{\sqrt{14}} \times 0.267$$

$$= \hat{\mathbf{r}}0.143 + \hat{\boldsymbol{\phi}}0.214 - \hat{\mathbf{z}}0.071.$$

(c) The vector component of **B** perpendicular to **A** is equal to **B** minus **C**:

$$\mathbf{D} = \mathbf{B} - \mathbf{C} = (\hat{\mathbf{r}} + \hat{\mathbf{z}}) - (\hat{\mathbf{r}}0.143 + \hat{\boldsymbol{\phi}}0.214 - \hat{\mathbf{z}}0.071)$$

$$= \hat{\mathbf{r}}0.857 - \hat{\boldsymbol{\phi}}0.214 + \hat{\mathbf{z}}0.929.$$

Concept Question 3-7: Why do we use more than one coordinate system?

Concept Question 3-8: Why is it that the base vectors (**x̂**, **ŷ**, **ẑ**) are independent of the location of a point, but **r̂** and **φ̂** are not?

Concept Question 3-9: What are the cyclic relations for the base vectors in (a) Cartesian coordinates, (b) cylindrical coordinates, and (c) spherical coordinates?

Concept Question 3-10: How is the position vector of a point in cylindrical coordinates related to its position vector in spherical coordinates?

Exercise 3-7: Point *P* = (2√3, π/3, −2) is given in cylindrical coordinates. Express *P* in spherical coordinates.

Answer: *P* = (4, 2π/3, π/3). (See EM.)

Exercise 3-8: Transform vector

$$\mathbf{A} = \hat{\mathbf{x}}(x+y) + \hat{\mathbf{y}}(y-x) + \hat{\mathbf{z}}z$$

from Cartesian to cylindrical coordinates.

Answer: $\mathbf{A} = \hat{\mathbf{r}}r - \hat{\boldsymbol{\phi}}r + \hat{\mathbf{z}}z$. (See EM.)

3-4 Gradient of a Scalar Field

When dealing with a scalar physical quantity whose magnitude depends on a single variable, such as the temperature *T* as a function of height *z*, the rate of change of *T* with height can be described by the derivative *dT/dz*. However, if *T* is also a function of *x* and *y*, its spatial rate of change becomes more difficult to describe because we now have to deal with three separate variables. The differential change in *T* along *x*, *y*, and *z* can be described in terms of the partial derivatives of *T* with respect to the three coordinate variables, but it is not immediately obvious as to how we should combine the three partial derivatives so as to describe the spatial rate of change of *T* along a specified direction. Furthermore, many of the quantities we deal with in electromagnetics are vectors; therefore, both their magnitudes and directions may vary with spatial position. To this end, we introduce three fundamental operators to describe the differential spatial variations of scalars and vectors: the *gradient*, *divergence*, and *curl* operators. The gradient operator applies to scalar fields and is the subject of the present section. The other two operators, which apply to vector fields, are discussed in succeeding sections.

例 3-10：向量分量

在空间中的给定点，向量 **A** 和 **B** 在柱坐标中给出：

$$\mathbf{A} = \hat{\mathbf{r}}2 + \hat{\boldsymbol{\phi}}3 - \hat{\mathbf{z}},$$

$$\mathbf{B} = \hat{\mathbf{r}} + \hat{\mathbf{z}}.$$

确定 (a) **B** 在 **A** 方向上的标量分量或投影，(b) **B** 在 **A** 方向上的矢量分量，以及 (c) **B** 垂直于 **A** 的矢量分量。

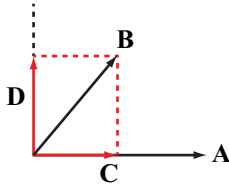


图 3-19 例 3-10 中的向量 **A**、**B**、**C** 和 **D**。

解：(a) 将 **B** 在 **A** 方向上的标量分量表示为 *C*，如图 3-19 所示。因此，

$$C = \mathbf{B} \cdot \hat{\mathbf{a}} = \mathbf{B} \cdot \frac{\mathbf{A}}{|\mathbf{A}|} = (\hat{\mathbf{r}} + \hat{\mathbf{z}}) \cdot \frac{(\hat{\mathbf{r}}2 + \hat{\boldsymbol{\phi}}3 - \hat{\mathbf{z}})}{\sqrt{4+9+1}} = \frac{2-1}{\sqrt{14}} = 0.267.$$

(b) **B** 在 **A** 方向上的向量分量由标量分量 *C* 和单位向量 **â** 的乘积给出：

$$\mathbf{C} = \hat{\mathbf{a}}C = \frac{\mathbf{A}}{|\mathbf{A}|} C = \frac{(\hat{\mathbf{r}}2 + \hat{\boldsymbol{\phi}}3 - \hat{\mathbf{z}})}{\sqrt{14}} \times 0.267$$

$$= \hat{\mathbf{r}}0.143 + \hat{\boldsymbol{\phi}}0.214 - \hat{\mathbf{z}}0.071.$$

(c) **B** 垂直于 **A** 的矢量分量等于 **B** 减去 **C**：

$$\mathbf{D} = \mathbf{B} - \mathbf{C} = (\hat{\mathbf{r}} + \hat{\mathbf{z}}) - (\hat{\mathbf{r}}0.143 + \hat{\boldsymbol{\phi}}0.214 - \hat{\mathbf{z}}0.071)$$

$$= \hat{\mathbf{r}}0.857 - \hat{\boldsymbol{\phi}}0.214 + \hat{\mathbf{z}}0.929.$$

概念问题 3-7：为什么我们使用多个坐标系？

Concept Question 3-8: Why is it that the base vectors (**x̂**, **ŷ**, **ẑ**) are independent of the location of a point, but **r̂** and **φ̂** are not?

Concept Question 3-9: What are the cyclic relations for the base vectors in (a) Cartesian coordinates, (b) cylindrical coordinates, and (c) spherical coordinates?

Concept Question 3-10: How is the position vector of a point in cylindrical coordinates related to its position vector in spherical coordinates?

Exercise 3-7: Point *P* = (2√3, π/3, −2) is given in cylindrical coordinates. Express *P* in spherical coordinates.

Answer: *P* = (4, 2π/3, π/3). (See EM.)

Exercise 3-8: Transform vector

$$\mathbf{A} = \hat{\mathbf{x}}(x+y) + \hat{\mathbf{y}}(y-x) + \hat{\mathbf{z}}z$$

from Cartesian to cylindrical coordinates.

Answer: $\mathbf{A} = \hat{\mathbf{r}}r - \hat{\boldsymbol{\phi}}r + \hat{\mathbf{z}}z$. (See EM.)

3-4 标量场的梯度

当处理大小取决于单个变量的标量物理量时，例如温度 *T* 作为高度 *z* 的函数，*T* 随高度的变化率可以用导数 *dT/dz* 来描述。然而，如果 *T* 也是 *x* 和 *y* 的函数，则其空间变化率变得更难以描述，因为我们现在必须处理三个单独的变量。*T* 沿 *x*、*y* 和 *z* 的微分变化可以用 *T* 对三个坐标变量的偏导数来描述，但如何结合这三个偏导数来描述 *T* 沿特定方向的空间变化率并不是立即显而易见的。此外，我们在电磁学中处理的许多量都是矢量。因此，它们的大小和方向都可能随空间位置而变化。为此，我们引入了三个基本算子来描述标量和向量的微分空间变化：梯度、散度和旋度算子。梯度算子适用于标量场，是本节的主题。其他两个适用于向量域的运算符将在后续部分中讨论。

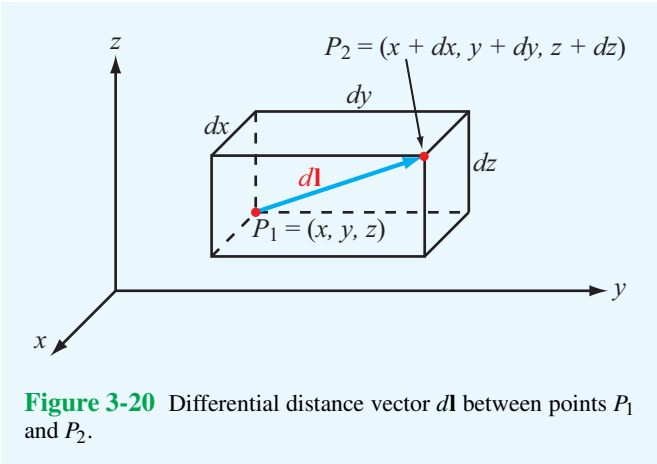


Figure 3-20 Differential distance vector $d\mathbf{l}$ between points P_1 and P_2 .

Suppose that $T_1 = T(x, y, z)$ is the temperature at point $P_1 = (x, y, z)$ in some region of space, and

$$T_2 = T(x + dx, y + dy, z + dz)$$

is the temperature at a nearby point $P_2 = (x + dx, y + dy, z + dz)$ (**Fig. 3-20**). The differential distances dx , dy , and dz are the components of the differential distance vector $d\mathbf{l}$. That is,

$$d\mathbf{l} = \hat{\mathbf{x}} \, dx + \hat{\mathbf{y}} \, dy + \hat{\mathbf{z}} \, dz. \quad (3.69)$$

From differential calculus, the temperature difference between points P_1 and P_2 , $dT = T_2 - T_1$, is

$$dT = \frac{\partial T}{\partial x} dx + \frac{\partial T}{\partial y} dy + \frac{\partial T}{\partial z} dz. \quad (3.70)$$

Because $dx = \hat{\mathbf{x}} \cdot d\mathbf{l}$, $dy = \hat{\mathbf{y}} \cdot d\mathbf{l}$, and $dz = \hat{\mathbf{z}} \cdot d\mathbf{l}$, Eq. (3.70) can be rewritten as

$$\begin{aligned} dT &= \hat{\mathbf{x}} \frac{\partial T}{\partial x} \cdot d\mathbf{l} + \hat{\mathbf{y}} \frac{\partial T}{\partial y} \cdot d\mathbf{l} + \hat{\mathbf{z}} \frac{\partial T}{\partial z} \cdot d\mathbf{l} \\ &= \left[\hat{\mathbf{x}} \frac{\partial T}{\partial x} + \hat{\mathbf{y}} \frac{\partial T}{\partial y} + \hat{\mathbf{z}} \frac{\partial T}{\partial z} \right] \cdot d\mathbf{l}. \end{aligned} \quad (3.71)$$

The vector inside the square brackets in Eq. (3.71) relates the change in temperature dT to a vector change in direction $d\mathbf{l}$. This vector is called the **gradient** of T (or **grad** T for short) and denoted ∇T :

$$\nabla T = \text{grad } T = \hat{\mathbf{x}} \frac{\partial T}{\partial x} + \hat{\mathbf{y}} \frac{\partial T}{\partial y} + \hat{\mathbf{z}} \frac{\partial T}{\partial z}. \quad (3.72)$$

Equation (3.71) then can be expressed as

$$dT = \nabla T \cdot d\mathbf{l}. \quad (3.73)$$

The symbol ∇ is called the **del** or **gradient operator** and is defined as

$$\nabla = \hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z} \quad \text{(Cartesian)}. \quad (3.74)$$

► Whereas the gradient operator itself has no physical meaning, it attains a physical meaning once it operates on a scalar quantity, and the result of the operation is a vector with magnitude equal to the maximum rate of change of the physical quantity per unit distance and pointing in the direction of maximum increase. ◀

With $d\mathbf{l} = \hat{\mathbf{a}}_l dl$, where $\hat{\mathbf{a}}_l$ is the unit vector of $d\mathbf{l}$, the **directional derivative** of T along $\hat{\mathbf{a}}_l$ is

$$\frac{dT}{dl} = \nabla T \cdot \hat{\mathbf{a}}_l. \quad (3.75)$$

We can find the difference $(T_2 - T_1)$, where $T_1 = T(x_1, y_1, z_1)$ and $T_2 = T(x_2, y_2, z_2)$ are the values of T at points

$$P_1 = (x_1, y_1, z_1) \quad \text{and} \quad P_2 = (x_2, y_2, z_2)$$

not necessarily infinitesimally close to one another, by integrating both sides of Eq. (3.73). Thus,

$$T_2 - T_1 = \int_{P_1}^{P_2} \nabla T \cdot d\mathbf{l}. \quad (3.76)$$

Example 3-11: Directional Derivative

Find the directional derivative of $T = x^2 + y^2z$ along direction $\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 - \hat{\mathbf{z}}2$ and evaluate it at $(1, -1, 2)$.

Solution: First, we find the gradient of T :

$$\nabla T = \left(\hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z} \right) (x^2 + y^2z) = \hat{\mathbf{x}}2x + \hat{\mathbf{y}}2yz + \hat{\mathbf{z}}y^2.$$

We denote $\mathbf{l}$ as the given direction,

$$\mathbf{l} = \hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 - \hat{\mathbf{z}}2.$$

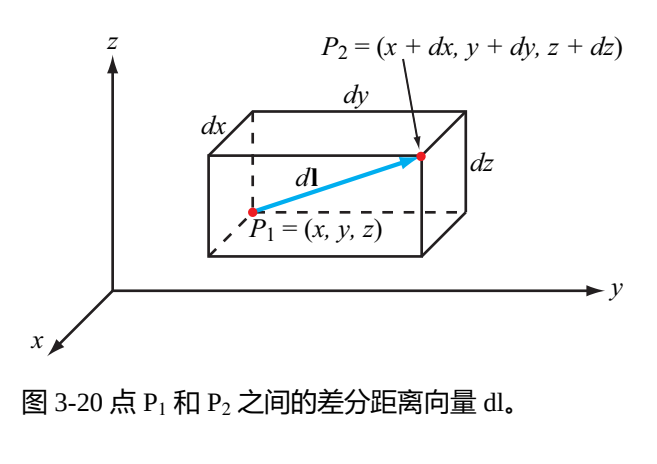


图 3-20 点 P_1 和 P_2 之间的差分距离向量 $d\mathbf{l}$ 。

假设 $T_1 = T(x, y, z)$ 是空间某个区域中 $P_1 = (x, y, z)$ 点的温度，并且

$$T_2 = T(x + dx, y + dy, z + dz)$$

是附近点 $P_2 = (x + dx, y + dy, z + dz)$ (的温度3-20)。差分距离 dx 、 dy 、 dz 是差分距离矢量 $d\mathbf{l}$ 的分量。那是，

$$d\mathbf{l} = \hat{\mathbf{x}} \, dx + \hat{\mathbf{y}} \, dy + \hat{\mathbf{z}} \, dz. \quad (3.69)$$

根据微分学，点 P_1 和 P_2 之间的温度差 $dT = T_2 - T_1$ 为

$$dT = \frac{\partial T}{\partial x} dx + \frac{\partial T}{\partial y} dy + \frac{\partial T}{\partial z} dz. \quad (3.70)$$

因为 $dx = \hat{\mathbf{x}} \cdot d\mathbf{l}$ 、 $dy = \hat{\mathbf{y}} \cdot d\mathbf{l}$ 和 $dz = \hat{\mathbf{z}} \cdot d\mathbf{l}$ ，等式：(3.70) 可以重写为

$$\begin{aligned} dT &= \hat{\mathbf{x}} \frac{\partial T}{\partial x} \cdot d\mathbf{l} + \hat{\mathbf{y}} \frac{\partial T}{\partial y} \cdot d\mathbf{l} + \hat{\mathbf{z}} \frac{\partial T}{\partial z} \cdot d\mathbf{l} \\ &= \left[\hat{\mathbf{x}} \frac{\partial T}{\partial x} + \hat{\mathbf{y}} \frac{\partial T}{\partial y} + \hat{\mathbf{z}} \frac{\partial T}{\partial z} \right] \cdot d\mathbf{l}. \end{aligned} \quad (3.71)$$

方程中方括号内的向量。(3.71) 将温度 dT 的变化与方向 $d\mathbf{l}$ 上的矢量变化联系起来。该向量称为 T 的梯度（或简称 $\text{grad } T$ ），记为 ∇T ：

$$\nabla T = \text{grad } T = \hat{\mathbf{x}} \frac{\partial T}{\partial x} + \hat{\mathbf{y}} \frac{\partial T}{\partial y} + \hat{\mathbf{z}} \frac{\partial T}{\partial z}. \quad (3.72)$$

则方程(3.71)可表示为

$$dT = \nabla T \cdot d\mathbf{l}. \quad (3.73)$$

符号 ∇ 称为 **del** 或**梯度运算符**，定义为

$$\nabla = \hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z} \quad \text{(Cartesian)}. \quad (3.74)$$

► 梯度算子本身没有物理意义，但它对标量进行运算后就具有物理意义，运算结果是一个向量，其大小等于该物理量单位距离的最大变化率，并指向最大增加的方向。 ◀

对于 $d\mathbf{l} = \hat{\mathbf{a}}_l dl$ ，其中 $\hat{\mathbf{a}}_l$ 是 $d\mathbf{l}$ 的单位向量， T 沿 $\hat{\mathbf{a}}_l$ 的方向导数为

$$\frac{dT}{dl} = \nabla T \cdot \hat{\mathbf{a}}_l. \quad (3.75)$$

我们可以求出差值 $(T_2 - T_1)$ ，其中 $T_1 = T(x_1, y_1, z_1)$ 和 $T_2 = T(x_2, y_2, z_2)$ 是 T 在点处的值

$$P_1 = (x_1, y_1, z_1) \quad \text{and} \quad P_2 = (x_2, y_2, z_2)$$

通过对方程两边求积分，不一定彼此无限小地接近。(3.73)。因此，

$$T_2 - T_1 = \int_{P_1}^{P_2} \nabla T \cdot d\mathbf{l}. \quad (3.76)$$

例 3-11：方向导数

求 $T = x^2 + y^2z$ 沿方向 $\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 - \hat{\mathbf{z}}2$ 的方向导数，并在 $(1, -1, 2)$ 处对其求值。

解决方案：首先，我们求 T 的梯度：

$$\nabla T = \left(\hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z} \right) (x^2 + y^2z) = \hat{\mathbf{x}}2x + \hat{\mathbf{y}}2yz + \hat{\mathbf{z}}y^2.$$

我们将 $\mathbf{l}$ 表示为给定方向，

$$\mathbf{l} = \hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 - \hat{\mathbf{z}}2.$$

Its unit vector is

$$\hat{\mathbf{a}}_l = \frac{\mathbf{l}}{|\mathbf{l}|} = \frac{\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 - \hat{\mathbf{z}}2}{\sqrt{2^2 + 3^2 + 2^2}} = \frac{\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 - \hat{\mathbf{z}}2}{\sqrt{17}}.$$

Application of Eq. (3.75) gives

$$\begin{aligned} \frac{dT}{dl} &= \nabla T \cdot \hat{\mathbf{a}}_l = (\hat{\mathbf{x}}2x + \hat{\mathbf{y}}2yz + \hat{\mathbf{z}}y^2) \cdot \left(\frac{\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 - \hat{\mathbf{z}}2}{\sqrt{17}} \right) \\ &= \frac{4x + 6yz - 2y^2}{\sqrt{17}}. \end{aligned}$$

At $(1, -1, 2)$,

$$\left. \frac{dT}{dl} \right|_{(1,-1,2)} = \frac{4 - 12 - 2}{\sqrt{17}} = \frac{-10}{\sqrt{17}}.$$

3-4.1 Gradient Operator in Cylindrical and Spherical Coordinates

Even though Eq. (3.73) was derived using Cartesian coordinates, it should have counterparts in other coordinate systems. To convert Eq. (3.72) into cylindrical coordinates (r, ϕ, z) , we start by restating the coordinate relations

$$r = \sqrt{x^2 + y^2}, \quad \tan \phi = \frac{y}{x}. \quad (3.77)$$

From differential calculus,

$$\frac{\partial T}{\partial x} = \frac{\partial T}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial T}{\partial \phi} \frac{\partial \phi}{\partial x} + \frac{\partial T}{\partial z} \frac{\partial z}{\partial x}. \quad (3.78)$$

Since z is orthogonal to x and $\partial z / \partial x = 0$, the last term in Eq. (3.78) vanishes. From the coordinate relations given by Eq. (3.77), it follows that

$$\frac{\partial r}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}} = \cos \phi, \quad (3.79a)$$

$$\frac{\partial \phi}{\partial x} = -\frac{1}{r} \sin \phi. \quad (3.79b)$$

Hence,

$$\frac{\partial T}{\partial x} = \cos \phi \frac{\partial T}{\partial r} - \frac{\sin \phi}{r} \frac{\partial T}{\partial \phi}. \quad (3.80)$$

This expression can be used to replace the coefficient of $\hat{\mathbf{x}}$ in Eq. (3.72), and a similar procedure can be followed to obtain an expression for $\partial T / \partial y$ in terms of r and ϕ . If, in addition, we use the relations $\hat{\mathbf{x}} = \hat{\mathbf{r}} \cos \phi - \hat{\phi} \sin \phi$ and $\hat{\mathbf{y}} = \hat{\mathbf{r}} \sin \phi + \hat{\phi} \cos \phi$ [from Eqs. (3.57a) and (3.57b)], then Eq. (3.72) becomes

$$\nabla T = \hat{\mathbf{r}} \frac{\partial T}{\partial r} + \hat{\phi} \frac{1}{r} \frac{\partial T}{\partial \phi} + \hat{\mathbf{z}} \frac{\partial T}{\partial z}. \quad (3.81)$$

Hence, the gradient operator in cylindrical coordinates can be expressed as

$$\nabla = \hat{\mathbf{r}} \frac{\partial}{\partial r} + \hat{\phi} \frac{1}{r} \frac{\partial}{\partial \phi} + \hat{\mathbf{z}} \frac{\partial}{\partial z}, \quad (\text{cylindrical})$$

A similar procedure leads to the expression for the gradient in spherical coordinates:

$$\nabla = \hat{\mathbf{R}} \frac{\partial}{\partial R} + \hat{\theta} \frac{1}{R} \frac{\partial}{\partial \theta} + \hat{\phi} \frac{1}{R \sin \theta} \frac{\partial}{\partial \phi}. \quad (3.83)$$

3-4.2 Properties of the Gradient Operator

For any two scalar functions U and V , the following relations apply:

$$(1) \quad \nabla(U + V) = \nabla U + \nabla V, \quad (3.84a)$$

$$(2) \quad \nabla(UV) = U \nabla V + V \nabla U, \quad (3.84b)$$

$$(3) \quad \nabla V^n = nV^{n-1} \nabla V, \quad \text{for any } n. \quad (3.84c)$$

Example 3-12: Calculating the Gradient

Find the gradient of each of the following scalar functions and then evaluate it at the given point.

(a) $V_1 = 24V_0 \cos(\pi y/3) \sin(2\pi z/3)$ at $(3, 2, 1)$ in Cartesian coordinates,

(b) $V_2 = V_0 e^{-2r} \sin 3\phi$ at $(1, \pi/2, 3)$ in cylindrical coordinates,

(c) $V_3 = V_0 (a/R) \cos 2\theta$ at $(2a, 0, \pi)$ in spherical coordinates.

Solution: (a) Using Eq. (3.72) for ∇ ,

$$\begin{aligned} \nabla V_1 &= \hat{\mathbf{x}} \frac{\partial V_1}{\partial x} + \hat{\mathbf{y}} \frac{\partial V_1}{\partial y} + \hat{\mathbf{z}} \frac{\partial V_1}{\partial z} \\ &= -\hat{\mathbf{y}} 8\pi V_0 \sin \frac{\pi y}{3} \sin \frac{2\pi z}{3} + \hat{\mathbf{z}} 16\pi V_0 \cos \frac{\pi y}{3} \cos \frac{2\pi z}{3} \\ &= 8\pi V_0 \left[-\hat{\mathbf{y}} \sin \frac{\pi y}{3} \sin \frac{2\pi z}{3} + \hat{\mathbf{z}} 2 \cos \frac{\pi y}{3} \cos \frac{2\pi z}{3} \right]. \end{aligned}$$

At $(3, 2, 1)$,

$$\nabla V_1 = 8\pi V_0 \left[-\hat{\mathbf{y}} \sin^2 \frac{2\pi}{3} + \hat{\mathbf{z}} 2 \cos^2 \frac{2\pi}{3} \right] = \pi V_0 [-\hat{\mathbf{y}} 6 + \hat{\mathbf{z}} 4].$$

其单位向量为

$$\hat{\mathbf{a}}_l = \frac{\mathbf{l}}{|\mathbf{l}|} = \frac{\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 - \hat{\mathbf{z}}2}{\sqrt{2^2 + 3^2 + 2^2}} = \frac{\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 - \hat{\mathbf{z}}2}{\sqrt{17}}.$$

方程的应用(3.75) 给出

$$\begin{aligned} \frac{dT}{dl} &= \nabla T \cdot \hat{\mathbf{a}}_l = (\hat{\mathbf{x}}2x + \hat{\mathbf{y}}2yz + \hat{\mathbf{z}}y^2) \cdot \left(\frac{\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 - \hat{\mathbf{z}}2}{\sqrt{17}} \right) \\ &= \frac{4x + 6yz - 2y^2}{\sqrt{17}}. \end{aligned}$$

在 $(1, -1, 2)$ 处,

$$\left. \frac{dT}{dl} \right|_{(1,-1,2)} = \frac{4 - 12 - 2}{\sqrt{17}} = \frac{-10}{\sqrt{17}}.$$

3-4.1 柱坐标系和球坐标系中的梯度算子

即使方程。 (3.73) 是使用笛卡尔坐标导出的, 它在其他坐标系中应该有对应的。转换方程。 (3.72) 转化为柱坐标 (r, ϕ, z) , 我们首先重申坐标关系

$$r = \sqrt{x^2 + y^2}, \quad \tan \phi = \frac{y}{x}. \quad (3.77)$$

由微分学可知,

$$\frac{\partial T}{\partial x} = \frac{\partial T}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial T}{\partial \phi} \frac{\partial \phi}{\partial x} + \frac{\partial T}{\partial z} \frac{\partial z}{\partial x}. \quad (3.78)$$

由于 z 与 x 和 $\partial z / \partial x = 0$ 正交, 因此方程中的最后一项(3.78) 消失。根据方程给出的坐标关系: (3.77), 由此可知

$$\frac{\partial r}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}} = \cos \phi, \quad (3.79a)$$

$$\frac{\partial \phi}{\partial x} = -\frac{1}{r} \sin \phi. \quad (3.79b)$$

因此,

$$\frac{\partial T}{\partial x} = \cos \phi \frac{\partial T}{\partial r} - \frac{\sin \phi}{r} \frac{\partial T}{\partial \phi}. \quad (3.80)$$

该表达式可用于替换式 (1) 中的 $\hat{\mathbf{x}}$ 系数。 (3.72), 并且可以遵循类似的过程来获得以 r 和 ϕ 表示的 $\partial T / \partial y$ 的表达式。此外, 如果我们使用关系式 $\hat{\mathbf{x}} = \hat{\mathbf{r}} \cos \phi - \hat{\phi} \sin \phi$ 和 $\hat{\mathbf{y}} = \hat{\mathbf{r}} \sin \phi + \hat{\phi} \cos \phi$ [来自等式: (3.57a) 和 (3.57b)], 则方程(3.72) 变为

$$\nabla T = \hat{\mathbf{r}} \frac{\partial T}{\partial r} + \hat{\phi} \frac{1}{r} \frac{\partial T}{\partial \phi} + \hat{\mathbf{z}} \frac{\partial T}{\partial z}. \quad (3.81)$$

因此, 柱坐标系中的梯度算子可以表示为

$$\nabla = \hat{\mathbf{r}} \frac{\partial}{\partial r} + \hat{\phi} \frac{1}{r} \frac{\partial}{\partial \phi} + \hat{\mathbf{z}} \frac{\partial}{\partial z}, \quad (\text{cylindrical})$$

$$(3.82)$$

类似的过程可得出球坐标中的梯度表达式:

$$\nabla = \hat{\mathbf{R}} \frac{\partial}{\partial R} + \hat{\theta} \frac{1}{R} \frac{\partial}{\partial \theta} + \hat{\phi} \frac{1}{R \sin \theta} \frac{\partial}{\partial \phi}. \quad (3.83)$$

(spherical)

3-4.2 梯度算子的性质

对于任意两个标量函数 U 和 V , 以下关系适用:

$$(1) \quad \nabla(U + V) = \nabla U + \nabla V, \quad (3.84a)$$

$$(2) \quad \nabla(UV) = U \nabla V + V \nabla U, \quad (3.84b)$$

$$(3) \quad \nabla V^n = nV^{n-1} \nabla V, \quad \text{for any } n. \quad (3.84c)$$

例 3-12: 计算梯度

求以下每个标量函数的梯度, 然后在给定点对其求值。

(a) 笛卡尔坐标中 $(3, 2, 1)$ 处的 $V_1 = 24V_0 \cos(\pi y/3) \sin(2\pi z/3)$, (b) 圆柱坐标中 $(1, \pi/2, 3)$ 处的 $V_2 = V_0 e^{-2r} \sin 3\phi$ nates, (c) $V_3 = V_0 (a/R) \cos 2\theta$ 在球坐标 $(2a, 0, \pi)$ 处。

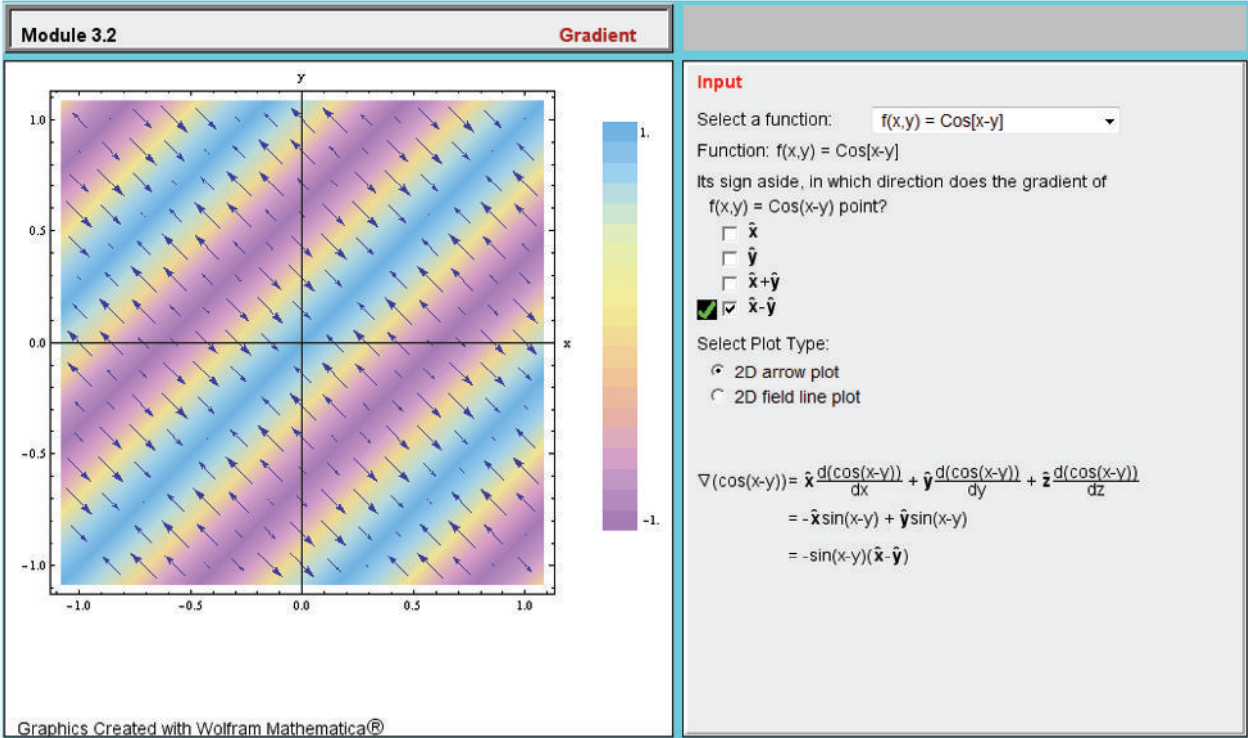
解决方案: (a) 使用等式。 (3.72) 对于 ∇ ,

$$\begin{aligned} \nabla V_1 &= \hat{\mathbf{x}} \frac{\partial V_1}{\partial x} + \hat{\mathbf{y}} \frac{\partial V_1}{\partial y} + \hat{\mathbf{z}} \frac{\partial V_1}{\partial z} \\ &= -\hat{\mathbf{y}} 8\pi V_0 \sin \frac{\pi y}{3} \sin \frac{2\pi z}{3} + \hat{\mathbf{z}} 16\pi V_0 \cos \frac{\pi y}{3} \cos \frac{2\pi z}{3} \\ &= 8\pi V_0 \left[-\hat{\mathbf{y}} \sin \frac{\pi y}{3} \sin \frac{2\pi z}{3} + \hat{\mathbf{z}} 2 \cos \frac{\pi y}{3} \cos \frac{2\pi z}{3} \right]. \end{aligned}$$

在 $(3, 2, 1)$ 处,

$$\nabla V_1 = 8\pi V_0 \left[-\hat{\mathbf{y}} \sin^2 \frac{2\pi}{3} + \hat{\mathbf{z}} 2 \cos^2 \frac{2\pi}{3} \right] = \pi V_0 [-\hat{\mathbf{y}} 6 + \hat{\mathbf{z}} 4].$$

Module 3.2 Gradient Select a scalar function $f(x,y,z)$, evaluate its gradient, and display both in an appropriate 2-D plane.



(b) The function V_2 is expressed in terms of cylindrical variables. Hence, we need to use Eq. (3.82) for ∇ :

$$\begin{aligned}\nabla V_2 &= \left(\hat{\mathbf{r}} \frac{\partial}{\partial r} + \hat{\boldsymbol{\phi}} \frac{1}{r} \frac{\partial}{\partial \phi} + \hat{\mathbf{z}} \frac{\partial}{\partial z} \right) V_0 e^{-2r} \sin 3\phi \\ &= -\hat{\mathbf{r}} 2V_0 e^{-2r} \sin 3\phi + \hat{\boldsymbol{\phi}} (3V_0 e^{-2r} \cos 3\phi) / r \\ &= \left[-\hat{\mathbf{r}} 2 \sin 3\phi + \hat{\boldsymbol{\phi}} \frac{3 \cos 3\phi}{r} \right] V_0 e^{-2r}.\end{aligned}$$

At $(1, \pi/2, 3)$, $r = 1$ and $\phi = \pi/2$. Hence,

$$\nabla V_2 = \left[-\hat{\mathbf{r}} 2 \sin \frac{3\pi}{2} + \hat{\boldsymbol{\phi}} 3 \cos \frac{3\pi}{2} \right] V_0 e^{-2} = \hat{\mathbf{r}} 2V_0 e^{-2} = \hat{\mathbf{r}} 0.27V_0.$$

(c) As V_3 is expressed in spherical coordinates, we apply Eq. (3.83) to V_3 :

$$\begin{aligned}\nabla V_3 &= \left(\hat{\mathbf{r}} \frac{\partial}{\partial R} + \hat{\boldsymbol{\theta}} \frac{1}{R} \frac{\partial}{\partial \theta} + \hat{\boldsymbol{\phi}} \frac{1}{R \sin \theta} \frac{\partial}{\partial \phi} \right) V_0 \left(\frac{a}{R} \right) \cos 2\theta \\ &= -\hat{\mathbf{r}} \frac{V_0 a}{R^2} \cos 2\theta - \hat{\boldsymbol{\theta}} \frac{2V_0 a}{R^2} \sin 2\theta \\ &= -[\hat{\mathbf{r}} \cos 2\theta + \hat{\boldsymbol{\theta}} 2 \sin 2\theta] \frac{V_0 a}{R^2}.\end{aligned}$$

At $(2a, 0, \pi)$, $R = 2a$ and $\theta = 0$, which yields

$$\nabla V_3 = -\hat{\mathbf{r}} \frac{V_0}{4a}.$$

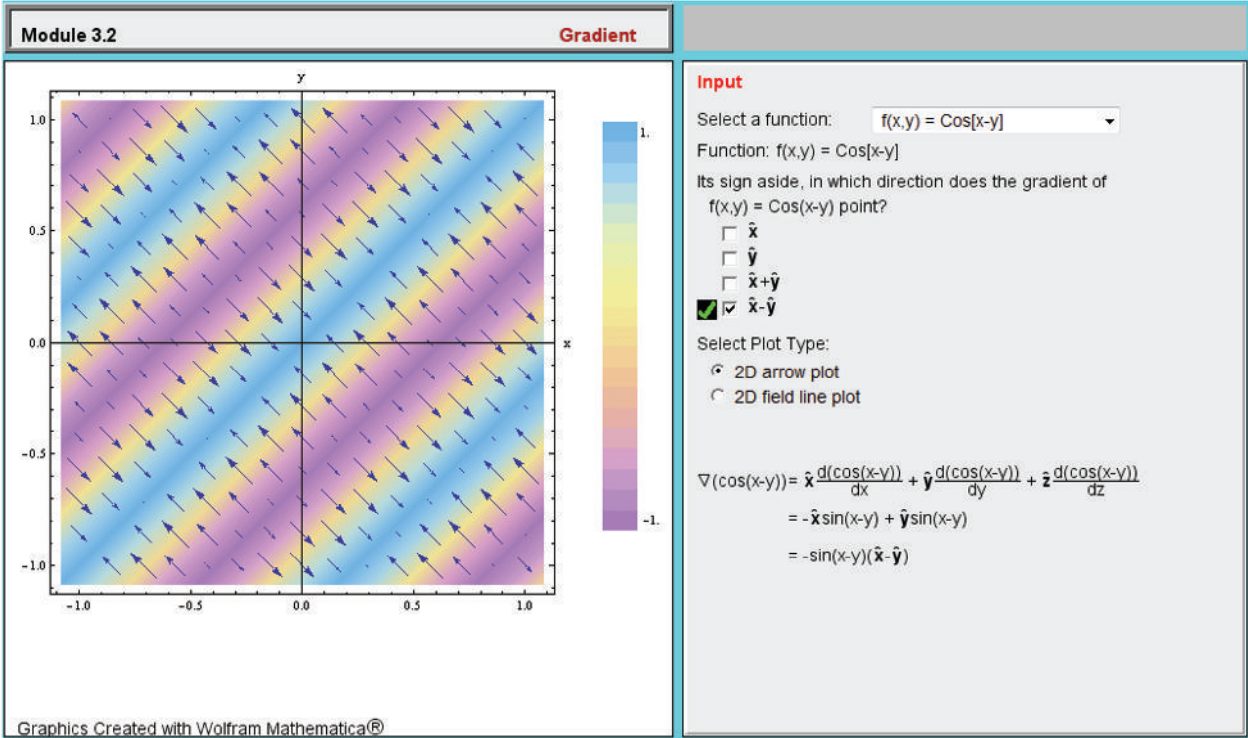
Exercise 3-9: Given $V = x^2y + xy^2 + xz^2$, (a) find the gradient of V , and (b) evaluate it at $(1, -1, 2)$.

Answer: (a) $\nabla V = \hat{\mathbf{x}}(2xy + y^2 + z^2) + \hat{\mathbf{y}}(x^2 + 2xy) + \hat{\mathbf{z}}2xz$, (b) $\nabla V|_{(1,-1,2)} = \hat{\mathbf{x}}3 - \hat{\mathbf{y}} + \hat{\mathbf{z}}4$. (See [EM](#).)

Exercise 3-10: Find the directional derivative of $V = rz^2 \cos 2\phi$ along the direction of $\mathbf{A} = \hat{\mathbf{r}}2 - \hat{\mathbf{z}}$ and evaluate it at $(1, \pi/2, 2)$.

Answer: $(dV/dl)|_{(1,\pi/2,2)} = -4/\sqrt{5}$. (See [EM](#).)

模块 3.2 梯度 选择标量函数 $f(x, y, z)$ ，计算其梯度，并将两者显示在适当的二维平面中。



(b) 函数 V_2 用圆柱变量表示。因此，我们需要使用方程 (3.82) 对于 ∇ ：

$$\begin{aligned}\nabla V_2 &= \left(\hat{\mathbf{r}} \frac{\partial}{\partial r} + \hat{\boldsymbol{\phi}} \frac{1}{r} \frac{\partial}{\partial \phi} + \hat{\mathbf{z}} \frac{\partial}{\partial z} \right) V_0 e^{-2r} \sin 3\phi \\ &= -\hat{\mathbf{r}} 2V_0 e^{-2r} \sin 3\phi + \hat{\boldsymbol{\phi}} (3V_0 e^{-2r} \cos 3\phi) / r \\ &= \left[-\hat{\mathbf{r}} 2 \sin 3\phi + \hat{\boldsymbol{\phi}} \frac{3 \cos 3\phi}{r} \right] V_0 e^{-2r}.\end{aligned}$$

在 $(1, \pi/2, 3)$ 处， $r = 1$ 和 $\phi = \pi/2$ 。因此，

$$\nabla V_2 = \left[-\hat{\mathbf{r}} 2 \sin \frac{3\pi}{2} + \hat{\boldsymbol{\phi}} 3 \cos \frac{3\pi}{2} \right] V_0 e^{-2} = \hat{\mathbf{r}} 2V_0 e^{-2} = \hat{\mathbf{r}} 0.27V_0.$$

(c) 由于 V_3 以球坐标表示，因此我们应用等式：(3.83) 至 V_3 ：

$$\begin{aligned}\nabla V_3 &= \left(\hat{\mathbf{r}} \frac{\partial}{\partial R} + \hat{\boldsymbol{\theta}} \frac{1}{R} \frac{\partial}{\partial \theta} + \hat{\boldsymbol{\phi}} \frac{1}{R \sin \theta} \frac{\partial}{\partial \phi} \right) V_0 \left(\frac{a}{R} \right) \cos 2\theta \\ &= -\hat{\mathbf{r}} \frac{V_0 a}{R^2} \cos 2\theta - \hat{\boldsymbol{\theta}} \frac{2V_0 a}{R^2} \sin 2\theta \\ &= -[\hat{\mathbf{r}} \cos 2\theta + \hat{\boldsymbol{\theta}} 2 \sin 2\theta] \frac{V_0 a}{R^2}.\end{aligned}$$

在 $(2a, 0, \pi)$ 处， $R = 2a$ 和 $\theta = 0$ ，得出

$$\nabla V_3 = -\hat{\mathbf{r}} \frac{V_0}{4a}.$$

练习 3-9：给定 $V = x^2y + xy^2 + xz^2$ ，(a) 求 V 的梯度，(b) 在 $(1, -1, 2)$ 处评估它。

答案：(a) $\nabla V = \hat{\mathbf{x}}(2xy + y^2 + z^2) + \hat{\mathbf{y}}(x^2 + 2xy) + \hat{\mathbf{z}}2xz$ ，(b) $\nabla V|_{(1,-1,2)} = \hat{\mathbf{x}}3 - \hat{\mathbf{y}} + \hat{\mathbf{z}}4$ 。（参见 [EM](#)。）

练习 3-10：求 $V = rz^2 \cos 2\phi$ 沿 $\mathbf{A} = \hat{\mathbf{r}}2 - \hat{\mathbf{z}}$ 方向的方向导数，并在 $(1, \pi/2, 2)$ 处对其求值。

Answer: $(dV/dl)|_{(1,\pi/2,2)} = -4/\sqrt{5}$ 。（参见 [EM](#)。）

Technology Brief 5:
Global Positioning System

The Global Positioning System (GPS), initially developed in the 1980s by the U.S. Department of Defense as a navigation tool for military use, has evolved into a system with numerous civilian applications, including vehicle tracking, aircraft navigation, map displays in automobiles and hand-held cell phones (**Fig. TF5-1**), and topographic mapping. The overall GPS comprises three segments. The **space segment** consists of 24 satellites (**Fig. TF5-2**), each circling Earth every 12 hours at an orbital altitude of about 12,000 miles and transmitting continuous coded time signals. All satellite transmitters broadcast coded messages at two specific frequencies: 1.57542 GHz and 1.22760 GHz. The **user segment** consists of hand-held or vehicle-mounted receivers that determine their own locations by receiving and processing multiple satellite signals. The third segment is a network of five **ground stations** distributed around the world that monitor the satellites and provide them with updates on their precise orbital information.

► GPS provides a location inaccuracy of about 30 m both horizontally and vertically, but it can be improved to within 1 m by **differential GPS**. (See final section.) ◀

Principle of Operation

The **triangulation technique** allows the determination of the location (x_0, y_0, z_0) of any object in 3-D space from knowledge of the distances d_1, d_2 , and d_3 between that object and three other independent points in space of known locations (x_1, y_1, z_1) to (x_3, y_3, z_3) . In GPS, the distances are established by measuring the times it takes the signals to travel from the satellites to the GPS receivers, and then multiplying them by the speed of light $c = 3 \times 10^8$ m/s. Time synchronization is achieved by using atomic clocks. The satellites use very precise clocks, accurate to 3 nanoseconds (3×10^{-9} s), but receivers use less accurate, inexpensive, ordinary quartz clocks. Consequently, the receiver clock may have an unknown **time offset error** t_0 relative to the satellite clocks. *To correct for the time error of a GPS receiver, a signal from a fourth satellite is needed.*

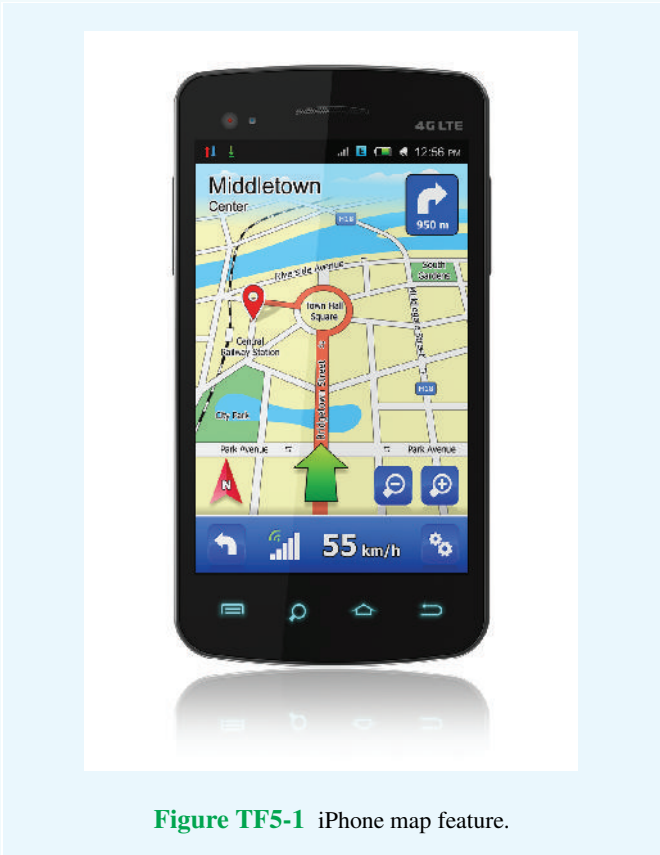


Figure TF5-1 iPhone map feature.

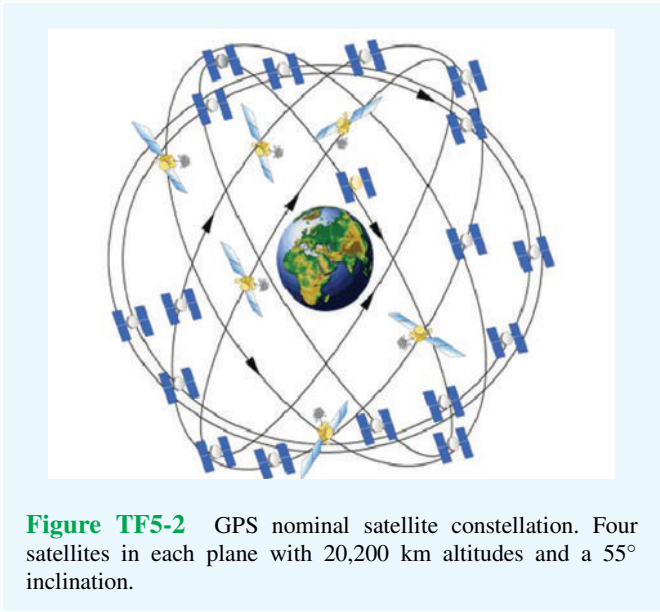


Figure TF5-2 GPS nominal satellite constellation. Four satellites in each plane with 20,200 km altitudes and a 55° inclination.

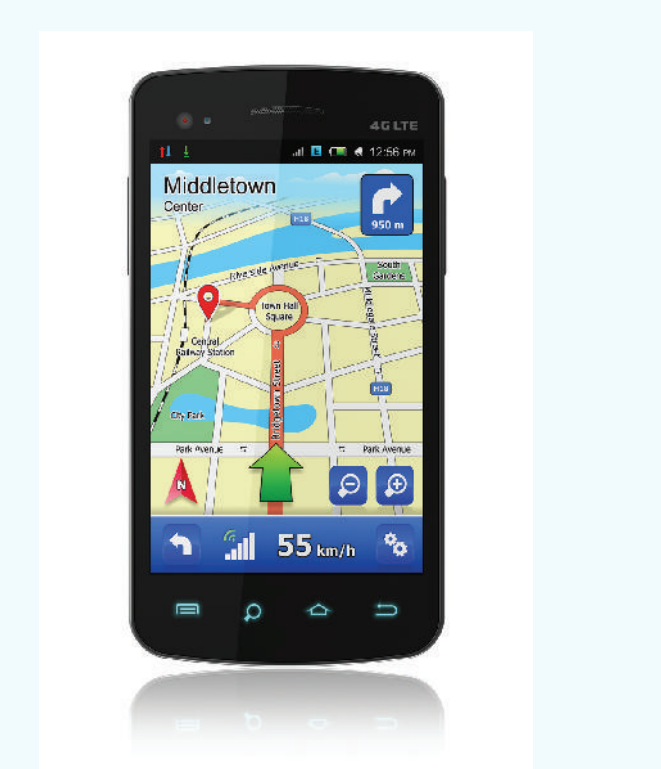
技术简介 5：全球定位系
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全球定位系统 (GPS) 最初由美国国防部于 20 世纪 80 年代开发，作为军事用途的导航工具，现已发展成为具有多种民用应用的系统，包括车辆跟踪、飞机导航、汽车和手持手机中的地图显示 (图 TF5-1) 以及地形测绘。整个 GPS 包括三个部分。太空部分由 24 颗卫星组成 (图 TF5-2)，每颗卫星每 12 小时绕地球一周，轨道高度约为 12,000 英里，并传输连续编码时间信号。所有卫星发射机都以两个特定频率广播编码消息：1.57542 GHz 和 1.22760 GHz。用户部分由手持式或车载接收器组成，它们通过接收和处理多个卫星信号来确定自己的位置。第三部分是由分布在世界各地的五个地面站组成的网络，用于监视卫星并为其提供最新的精确轨道信息。

► GPS 的水平和垂直定位误差约为 30 m，但差分 GPS 可以将其改善到 1 m 以内。(参见最后一节。)

工作原理

三角测量技术允许根据已知位置 (x_1, y_1, z_1) 到 (x_3) 的空间中的其他三个独立点与该对象之间的距离 d_1, d_2 和 d_3 的知识，确定 3D 空间中任何对象的位置 (x_0, y_0, z_0) 。在 GPS 中，距离是通过测量信号从卫星传输到 GPS 接收器所需的时间，然后乘以光速 $c = 3 \times 10^8$ m/s 来确定的。时间同步是通过使用原子钟来实现的。卫星使用非常精确的时钟，精确到 3 纳秒 (3×10^{-9} s)，但接收器使用不太精确、廉价的普通石英钟。因此，接收器时钟相对于卫星时钟可能具有未知的偏移误差 t_0 。为了纠正 GPS 接收器的时间误差，需要来自第四颗卫星的信号。



图TF5-1 iPhone地图功能。

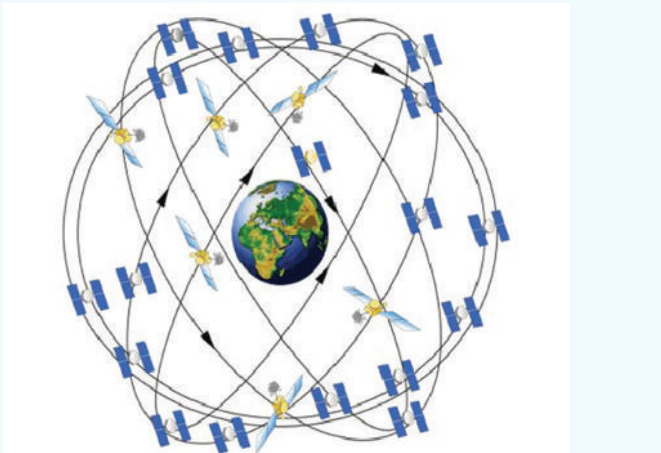


图 TF5-2 GPS 标称卫星星座。每个平面有四颗卫星，高度为 20,200 公里，倾角为 55°。

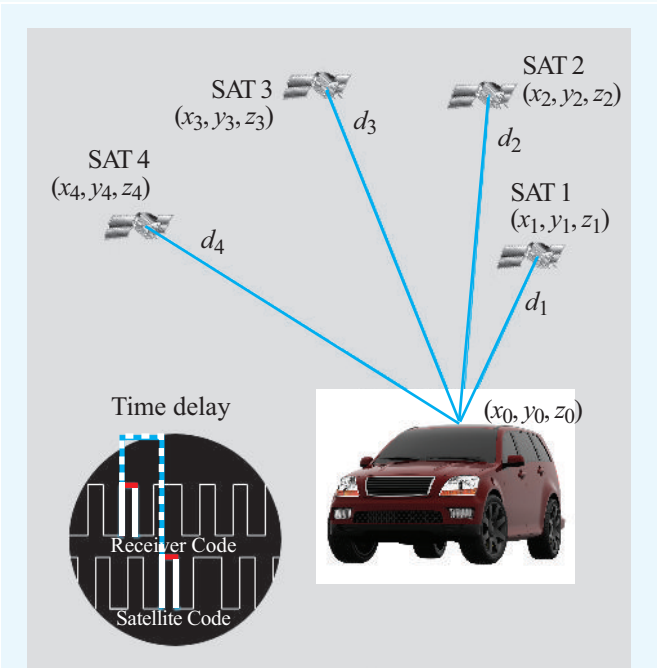


Figure TF5-3 Automobile GPS receiver at location (x_0, y_0, z_0) .

similar process applies to satellites 2 to 4, leading to four equations:

$$\begin{aligned}d_1^2 &= (x_1 - x_0)^2 + (y_1 - y_0)^2 + (z_1 - z_0)^2 = c^2 [(t_1 + t_0)]^2, \\d_2^2 &= (x_2 - x_0)^2 + (y_2 - y_0)^2 + (z_2 - z_0)^2 = c^2 [(t_2 + t_0)]^2, \\d_3^2 &= (x_3 - x_0)^2 + (y_3 - y_0)^2 + (z_3 - z_0)^2 = c^2 [(t_3 + t_0)]^2, \\d_4^2 &= (x_4 - x_0)^2 + (y_4 - y_0)^2 + (z_4 - z_0)^2 = c^2 [(t_4 + t_0)]^2.\end{aligned}$$

The four satellites report their coordinates (x_1, y_1, z_1) to (x_4, y_4, z_4) to the GPS receiver, and the time delays t_1 to t_4 are measured directly by the receiver. The unknowns are (x_0, y_0, z_0) , the coordinates of the GPS receiver, and the time offset of its clock t_0 . Simultaneous solution of the four equations provides the desired location information.

Differential GPS

The 30 m GPS position inaccuracy is attributed to several factors, including **time-delay errors** (due to the difference between the speed of light and the actual signal speed in the troposphere) that depend on the receiver's location on Earth, delays due to signal reflections by tall buildings, and satellites' locations misreporting errors.

► **Differential GPS**, or **DGPS**, uses a stationary reference receiver at a location with known coordinates. ◀

By calculating the difference between its location on the basis of the GPS estimate and its true location, the **reference receiver** establishes coordinate correction factors and transmits them to all DGPS receivers in the area. Application of the correction information usually reduces the location inaccuracy down to about 1 m.

The GPS receiver of the automobile in Fig. TF5-3 is at distances d_1 to d_4 from the GPS satellites. Each satellite sends a message identifying its orbital coordinates (x_1, y_1, z_1) for satellite 1, and so on for the other satellites, together with a binary-coded sequence common to all satellites. The GPS receiver generates the same binary sequence (Fig. TF5-3), and by comparing its code with the one received from satellite 1, it determines the time t_1 corresponding to travel time over the distance d_1 . A

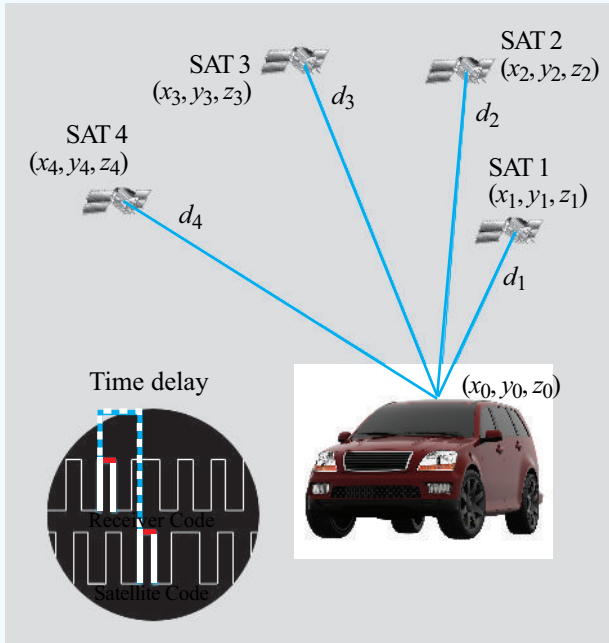


图 TF5-3 位置 (x_0, y_0, z_0) 处的汽车 GPS 接收器。

类似的过程适用于卫星 2 至 4，从而得出四个方程：

$$\begin{aligned}d_1^2 &= (x_1 - x_0)^2 + (y_1 - y_0)^2 + (z_1 - z_0)^2 = c^2 [(t_1 + t_0)]^2, \\d_2^2 &= (x_2 - x_0)^2 + (y_2 - y_0)^2 + (z_2 - z_0)^2 = c^2 [(t_2 + t_0)]^2, \\d_3^2 &= (x_3 - x_0)^2 + (y_3 - y_0)^2 + (z_3 - z_0)^2 = c^2 [(t_3 + t_0)]^2, \\d_4^2 &= (x_4 - x_0)^2 + (y_4 - y_0)^2 + (z_4 - z_0)^2 = c^2 [(t_4 + t_0)]^2.\end{aligned}$$

四颗卫星将它们的坐标 (x_1, y_1, z_1) 到 (x_4, y_4, z_4) 报告给 GPS 接收器，并且时间延迟 t_1 到 t_4 由接收器直接测量。未知数为 (x_0, y_0, z_0) 、GPS 接收器的坐标及其时钟的时间偏移 t_0 。四个方程的同时求解提供了所需的位置信息。

差分GPS

30 m GPS 位置误差归因于多种因素，包括取决于接收器在地球上的位置的时延误差（由于光速与对流层中实际信号速度之间的差异）、高层建筑反射信号造成的延迟以及卫星位置误报误差。

► 差分 GPS 或 DGPS 在已知坐标的位置使用固定参考接收器。 ◀

图TF5-3中汽车的GPS接收器距GPS卫星的距离为 d_1 至 d_4 。每颗卫星发送一条消息，标识卫星 1 的轨道坐标 (x_1, y_1, z_1) ，其他卫星依此类推，以及所有卫星共用的二进制编码序列。GPS接收器生成相同的二进制序列（图TF5-3），并通过将其代码与从卫星1接收到的代码进行比较，确定与经过距离 d_1 的行进时间相对应的时间 t_1 。一个

通过根据 GPS 估计计算其位置与其真实位置之间的差异，参考接收器建立坐标校正因子并将其传输到该区域中的所有 DGPS 接收器。校正信息的应用通常可将定位误差降低至约 1 m。

Exercise 3-11: The power density radiated by a star (**Fig. E3.11(a)**) decreases radially as $S(R) = S_0/R^2$, where R is the radial distance from the star and S_0 is a constant. Recalling that the gradient of a scalar function denotes the maximum rate of change of that function per unit distance and the direction of the gradient is along the direction of maximum increase, generate an arrow representation of ∇S .

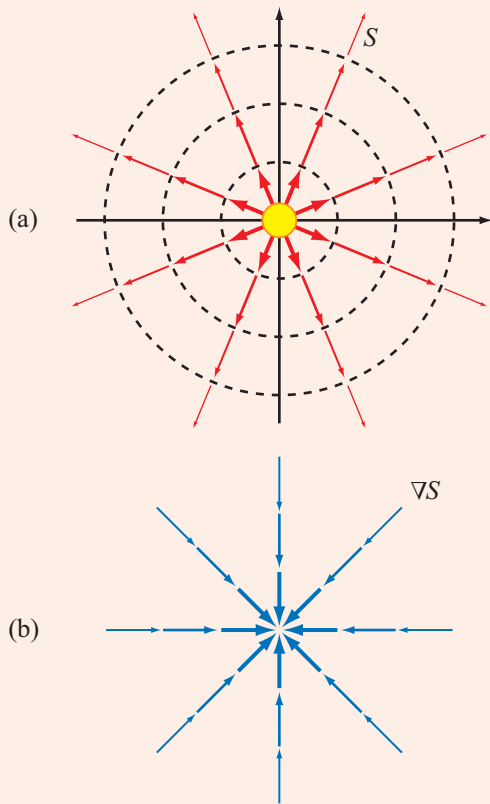


Figure E3.11

Answer: $\nabla S = -\hat{\mathbf{R}} 2S_0/R^3$ (**Fig. E3.11(b)**). (See [EM](#).)

Exercise 3-12: The graph in **Fig. E3.12(a)** depicts a gentle change in atmospheric temperature from T_1 over the sea to T_2 over land. The temperature profile is described by the function

$$T(x) = T_1 + \frac{T_2 - T_1}{e^{-x} + 1},$$

where x is measured in kilometers and $x = 0$ is the sea-land boundary. (a) In which direction does ∇T point and (b) at what value of x is it a maximum?

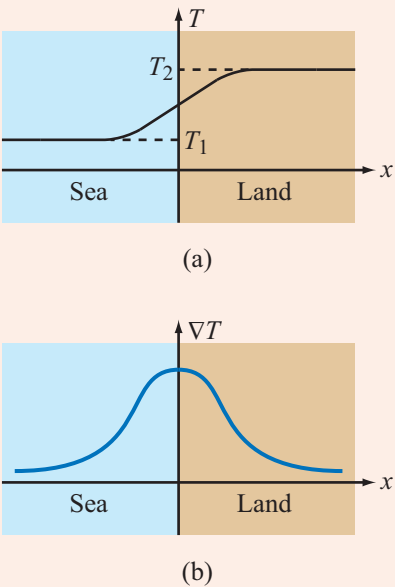


Figure E3.12

Answer: (a) $+\hat{\mathbf{x}}$; (b) at $x = 0$.

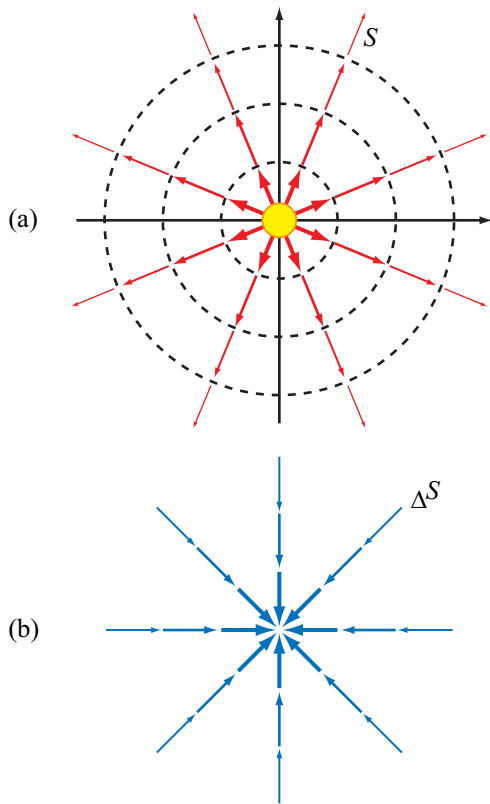
$$T(x) = T_1 + \frac{T_2 - T_1}{e^{-x} + 1},$$
$$\nabla T = \hat{\mathbf{x}} \frac{\partial T}{\partial x} = \hat{\mathbf{x}} \frac{e^{-x}(T_2 - T_1)}{(e^{-x} + 1)^2}.$$

(See [EM](#).)

3-5 Divergence of a Vector Field

From our brief introduction of Coulomb's law in Chapter 1, we know that an isolated, positive point charge q induces an electric field $\mathbf{E}$ in the space around it with the direction of $\mathbf{E}$ being outward away from the charge. Also, the strength (magnitude) of $\mathbf{E}$ is proportional to q and decreases with distance R from the charge as $1/R^2$. In a graphical presentation, a vector field is usually represented by **field lines**, as shown in **Fig. 3-21**. The arrowhead denotes the direction of the field at the point where the field line is drawn, and the length of the line provides a qualitative depiction of the field's magnitude.

练习 3-11: 恒星辐射的功率密度 (图 E3.11(a)) 随着 $S(R) = S_0/R^2$ 径向减小, 其中 R 是距恒星的径向距离, S_0 是常数。回想一下标量函数的梯度表示该函数每单位距离的最大变化率, 并且梯度的方向沿着最大增加的方向, 生成 ∇S 的箭头表示。



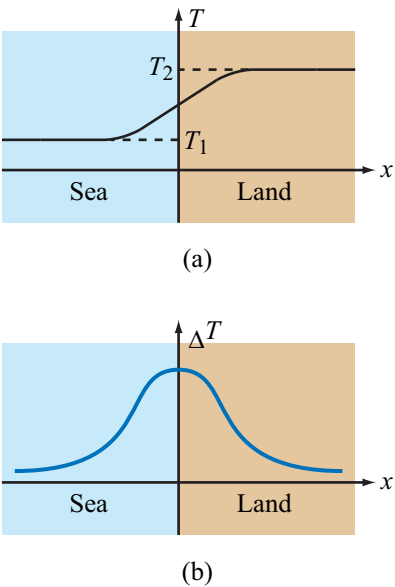
图E3.11

答案: $\nabla S = -\hat{\mathbf{R}} 2S_0/R^3$ (图E3.11(b)). (参见 [EM](#)。)

练习3-12: 图E3.12(a)中的图表描绘了大气温度从海上的 T_1 到陆地上的 T_2 的缓慢变化。温度曲线由函数描述

$$T(x) = T_1 + \frac{T_2 - T_1}{e^{-x} + 1},$$

其中 x 以公里为单位, $x = 0$ 是海陆边界。(a) ∇T 指向哪个方向, 以及 (b) x 的最大值在多少处?



图E3.12

答案: (a) $+\hat{\mathbf{x}}$; (b) 在 $x = 0$ 处。

$$T(x) = T_1 + \frac{T_2 - T_1}{e^{-x} + 1},$$
$$\nabla T = \hat{\mathbf{x}} \frac{\partial T}{\partial x} = \hat{\mathbf{x}} \frac{e^{-x}(T_2 - T_1)}{(e^{-x} + 1)^2}.$$

(参见 [EM](#)。)

3-5 矢量场的散度

从第一章对库仑定律的简要介绍中, 我们知道一个孤立的正点电荷 q 在其周围的空间中感应出一个电场 $\mathbf{E}$, $\mathbf{E}$ 的方向远离电荷。此外, $\mathbf{E}$ 的强度 (大小) 与 q 成正比, 并随着距电荷的距离 R 的增加而减小, 即 $1/R^2$ 。在图形表示中, 矢量场通常用场线表示, 如图 3-21 所示。箭头表示绘制场线处的场方向, 线的长度提供了场大小的定性描述。

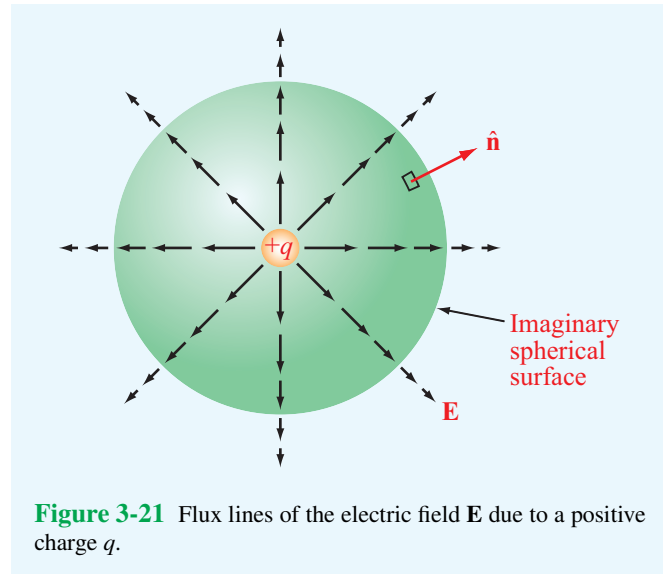


Figure 3-21 Flux lines of the electric field $\mathbf{E}$ due to a positive charge q .

At a surface boundary, **flux density** is defined as the amount of outward flux crossing a unit surface ds :

$$\text{Flux density of } \mathbf{E} = \frac{\mathbf{E} \cdot d\mathbf{s}}{|ds|} = \frac{\mathbf{E} \cdot \hat{\mathbf{n}} ds}{ds} = \mathbf{E} \cdot \hat{\mathbf{n}}, \quad (3.85)$$

where $\hat{\mathbf{n}}$ is the normal to ds . The **total flux** outwardly crossing a closed surface S , such as the enclosed surface of the imaginary sphere outlined in **Fig. 3-21**, is

$$\text{Total flux} = \oint_S \mathbf{E} \cdot d\mathbf{s}. \quad (3.86)$$

Let us now consider the case of a differential rectangular parallelepiped, such as a cube, whose edges align with the Cartesian axes shown in **Fig. 3-22**. The edges are of lengths Δx along x , Δy along y , and Δz along z . A vector field $\mathbf{E}(x, y, z)$ exists in the region of space containing the parallelepiped, and we wish to determine the flux of $\mathbf{E}$ through its total surface S . Since S includes six faces, we need to sum up the fluxes through all of them, and by definition, the flux through any face is the *outward* flux from the volume ΔV through that face.

Let $\mathbf{E}$ be defined as

$$\mathbf{E} = \hat{\mathbf{x}}E_x + \hat{\mathbf{y}}E_y + \hat{\mathbf{z}}E_z. \quad (3.87)$$

The area of the face marked 1 in **Fig. 3-22** is $\Delta y \Delta z$, and its unit vector $\hat{\mathbf{n}}_1 = -\hat{\mathbf{x}}$. Hence, the outward flux F_1 through face 1 is

$$\begin{aligned} F_1 &= \int_{\text{Face 1}} \mathbf{E} \cdot \hat{\mathbf{n}}_1 ds \\ &= \int_{\text{Face 1}} (\hat{\mathbf{x}}E_x + \hat{\mathbf{y}}E_y + \hat{\mathbf{z}}E_z) \cdot (-\hat{\mathbf{x}}) dy dz \\ &\approx -E_x(1) \Delta y \Delta z, \end{aligned} \quad (3.88)$$

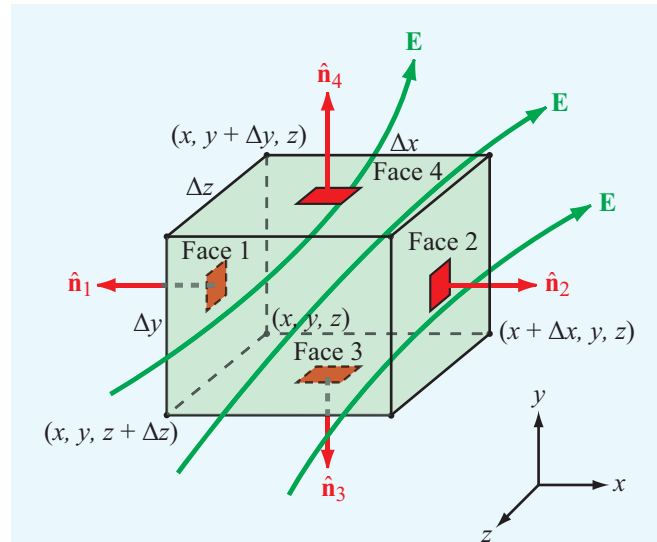


Figure 3-22 Flux lines of a vector field $\mathbf{E}$ passing through a differential rectangular parallelepiped of volume $\Delta V = \Delta x \Delta y \Delta z$.

where $E_x(1)$ is the value of E_x at the center of face 1. Approximating E_x over face 1 by its value at the center is justified by the assumption that the differential volume under consideration is very small.

Similarly, the flux out of face 2 (with $\hat{\mathbf{n}}_2 = \hat{\mathbf{x}}$) is

$$F_2 = E_x(2) \Delta y \Delta z, \quad (3.89)$$

where $E_x(2)$ is the value of E_x at the center of face 2. Over a differential separation Δx between the centers of faces 1 and 2, $E_x(2)$ is related to $E_x(1)$ by

$$E_x(2) = E_x(1) + \frac{\partial E_x}{\partial x} \Delta x, \quad (3.90)$$

where we have ignored higher-order terms involving $(\Delta x)^2$ and higher powers because their contributions are negligibly small when Δx is very small. Substituting Eq. (3.90) into Eq. (3.89) gives

$$F_2 = \left[E_x(1) + \frac{\partial E_x}{\partial x} \Delta x \right] \Delta y \Delta z. \quad (3.91)$$

The sum of the fluxes out of faces 1 and 2 is obtained by adding Eqs. (3.88) and (3.91),

$$F_1 + F_2 = \frac{\partial E_x}{\partial x} \Delta x \Delta y \Delta z. \quad (3.92a)$$

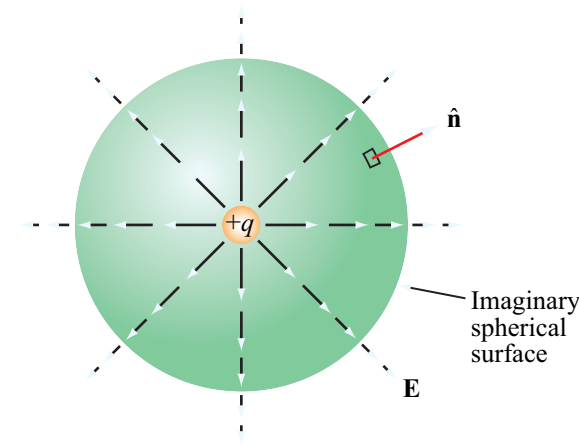


图 3-21 正电荷 q 产生的电场 $\mathbf{E}$ 的磁通线。

在表面边界处，通量密度定义为穿过单位表面 ds 的向外通量：

$$\text{Flux density of } \mathbf{E} = \frac{\mathbf{E} \cdot d\mathbf{s}}{|ds|} = \frac{\mathbf{E} \cdot \hat{\mathbf{n}} ds}{ds} = \mathbf{E} \cdot \hat{\mathbf{n}}, \quad (3.85)$$

其中 $\hat{\mathbf{n}}$ 是 ds 的法线。向外穿过封闭表面 S (如图 3-21 所示的假想球体的封闭表面) 的总通量为

$$\text{Total flux} = \oint_S \mathbf{E} \cdot d\mathbf{s}. \quad (3.86)$$

现在让我们考虑微分长方体的情况，例如立方体，其边缘与笛卡尔轴对齐，如图 3-22 所示。边沿 x 的长度为 Δx ，沿 y 的长度为 Δy ，沿 z 的长度为 Δz 。向量场 $\mathbf{E}(x, y, z)$ 存在于包含平行六面体的空间区域中，我们希望确定 $\mathbf{E}$ 通过其总表面 S 的通量。由于 S 包括六个面，我们需要对通过所有面的通量求和，根据定义，通过任何面的通量是从体积 ΔV 通过该面的向外通量。

令 $\mathbf{E}$ 定义为

$$\mathbf{E} = \hat{\mathbf{x}}E_x + \hat{\mathbf{y}}E_y + \hat{\mathbf{z}}E_z. \quad (3.87)$$

图3-22中标记为1的面的面积为 $\Delta y \Delta z$ ，其单位向量为 $\hat{\mathbf{n}}_1 = -\hat{\mathbf{x}}$ 。因此，通过面1的向外通量 F_1 为

$$\begin{aligned} F_1 &= \int_{\text{Face 1}} \mathbf{E} \cdot \hat{\mathbf{n}}_1 ds \\ &= \int_{\text{Face 1}} (\hat{\mathbf{x}}E_x + \hat{\mathbf{y}}E_y + \hat{\mathbf{z}}E_z) \cdot (-\hat{\mathbf{x}}) dy dz \\ &\approx -E_x(1) \Delta y \Delta z, \end{aligned} \quad (3.88)$$

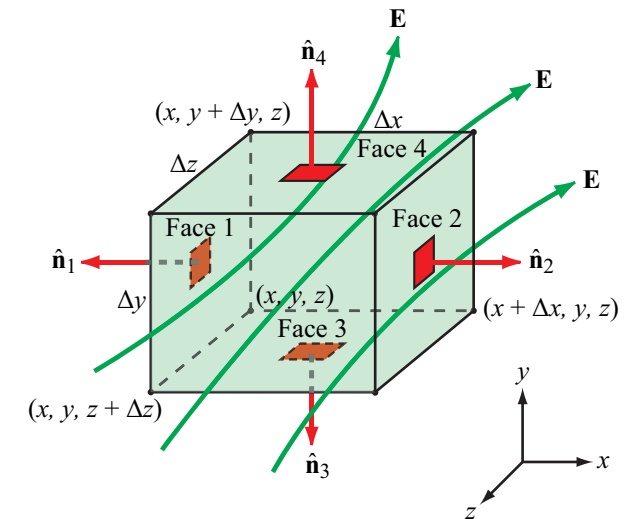


图 3-22 穿过体积为 $\Delta V = \Delta x \Delta y \Delta z$ 的微分长方体的向量场 $\mathbf{E}$ 的磁通线。

其中 $E_x(1)$ 是面 1 中心处的 E_x 值。通过假设所考虑的微分体积非常小，可以通过面 1 中心处的值来近似 E_x 。

类似地，面 2 (具有 $\hat{\mathbf{n}}_2 = \hat{\mathbf{x}}$) 的通量为

$$F_2 = E_x(2) \Delta y \Delta z, \quad (3.89)$$

其中 $E_x(2)$ 是面 2 中心处的 E_x 值。在面 1 和 2 中心之间的差分间隔 Δx 上， $E_x(2)$ 与 $E_x(1)$ 的关系为

$$E_x(2) = E_x(1) + \frac{\partial E_x}{\partial x} \Delta x, \quad (3.90)$$

我们忽略了涉及 $(\Delta x)^2$ 和更高幂的高阶项，因为当 Δx 非常小时，它们的贡献可以忽略不计。代入方程式 (3.90) 代入方程。 (3.89) 给出

$$F_2 = \left[E_x(1) + \frac{\partial E_x}{\partial x} \Delta x \right] \Delta y \Delta z. \quad (3.91)$$

面 1 和面 2 的通量之和可通过添加等式获得。 (3.88) 和 (3.91) ,

$$F_1 + F_2 = \frac{\partial E_x}{\partial x} \Delta x \Delta y \Delta z. \quad (3.92a)$$

Repeating the same procedure to each of the other face pairs leads to

$$F_3 + F_4 = \frac{\partial E_y}{\partial y} \Delta x \Delta y \Delta z, \quad (3.92b)$$

$$F_5 + F_6 = \frac{\partial E_z}{\partial z} \Delta x \Delta y \Delta z. \quad (3.92c)$$

The sum of fluxes F_1 through F_6 gives the total flux through surface S of the parallelepiped:

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = \left(\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} \right) \Delta x \Delta y \Delta z = (\text{div } \mathbf{E}) \Delta \mathcal{V}, \quad (3.93)$$

where $\Delta \mathcal{V} = \Delta x \Delta y \Delta z$ and $\text{div } \mathbf{E}$ is a scalar function called the **divergence** of $\mathbf{E}$, specified in Cartesian coordinates as

$$\text{div } \mathbf{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}. \quad (3.94)$$

► By shrinking the volume $\Delta \mathcal{V}$ to zero, we define the **divergence** of $\mathbf{E}$ at a point as the net outward flux per unit volume over a closed incremental surface. ◀

Thus, from Eq. (3.93), we have

$$\text{div } \mathbf{E} \triangleq \lim_{\Delta \mathcal{V} \rightarrow 0} \frac{\oint_S \mathbf{E} \cdot d\mathbf{s}}{\Delta \mathcal{V}}, \quad (3.95)$$

where S encloses the elemental volume $\Delta \mathcal{V}$. Instead of denoting the divergence of $\mathbf{E}$ by $\text{div } \mathbf{E}$, it is common practice to denote it as $\nabla \cdot \mathbf{E}$. That is,

$$\nabla \cdot \mathbf{E} = \text{div } \mathbf{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} \quad (3.96)$$

for a vector $\mathbf{E}$ in Cartesian coordinates.

► From the definition of the divergence of $\mathbf{E}$ given by Eq. (3.95), field $\mathbf{E}$ has positive divergence if the net flux out of surface S is positive, which may be “viewed” as if volume $\Delta \mathcal{V}$ contains a **source** of field lines. If the divergence is negative, $\Delta \mathcal{V}$ may be viewed as containing a **sink** of field lines because the net flux is into $\Delta \mathcal{V}$. For a uniform field $\mathbf{E}$, the same amount of flux enters $\Delta \mathcal{V}$ as leaves it; hence, its divergence is zero and the field is said to be **divergenceless**. ◀

The divergence is a differential operator, it always operates on vectors, and the result of its operation is a scalar. This is in contrast with the gradient operator, which always operates on scalars and results in a vector. Expressions for the divergence of a vector in cylindrical and spherical coordinates are provided in Appendix C.

The divergence operator is distributive. That is, for any pair of vectors $\mathbf{E}_1$ and $\mathbf{E}_2$,

$$\nabla \cdot (\mathbf{E}_1 + \mathbf{E}_2) = \nabla \cdot \mathbf{E}_1 + \nabla \cdot \mathbf{E}_2. \quad (3.97)$$

If $\nabla \cdot \mathbf{E} = 0$, the vector field $\mathbf{E}$ is called **divergenceless**.

The result given by Eq. (3.93) for a differential volume $\Delta \mathcal{V}$ can be extended to relate the volume integral of $\nabla \cdot \mathbf{E}$ over any volume $\mathcal{V}$ to the flux of $\mathbf{E}$ through the closed surface S that bounds $\mathcal{V}$. That is,

$$\int_{\mathcal{V}} \nabla \cdot \mathbf{E} d\mathcal{V} = \oint_S \mathbf{E} \cdot d\mathbf{s}. \quad (3.98)$$

(divergence theorem)

This relationship, known as the **divergence theorem**, is used extensively in electromagnetics.

Example 3-13: Calculating the Divergence

Determine the divergence of each of the following vector fields and then evaluate them at the indicated points:

- (a) $\mathbf{E} = \hat{\mathbf{x}}3x^2 + \hat{\mathbf{y}}2z + \hat{\mathbf{z}}x^2z$ at $(2, -2, 0)$;
(b) $\mathbf{E} = \hat{\mathbf{R}}(a^3 \cos \theta / R^2) - \hat{\boldsymbol{\theta}}(a^3 \sin \theta / R^2)$ at $(a/2, 0, \pi)$.

Solution: (a)

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} \\ &= \frac{\partial}{\partial x}(3x^2) + \frac{\partial}{\partial y}(2z) + \frac{\partial}{\partial z}(x^2z) \\ &= 6x + 0 + x^2 = x^2 + 6x. \end{aligned}$$

$$\text{At } (2, -2, 0), \nabla \cdot \mathbf{E} \Big|_{(2, -2, 0)} = 16.$$

对其他每对面孔重复相同的过程会导致

$$F_3 + F_4 = \frac{\partial E_y}{\partial y} \Delta x \Delta y \Delta z, \quad (3.92b)$$

$$F_5 + F_6 = \frac{\partial E_z}{\partial z} \Delta x \Delta y \Delta z. \quad (3.92c)$$

通量 F_1 到 F_6 的总和给出了通过平行六面体表面 S 的总通量:

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = \left(\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} \right) \Delta x \Delta y \Delta z = (\text{div } \mathbf{E}) \Delta \mathcal{V}, \quad (3.93)$$

其中 $\Delta \mathcal{V} = \Delta x \Delta y \Delta z$ 和 $\text{div } \mathbf{E}$ 是称为 $\mathbf{E}$ 散度的标量函数, 在笛卡尔坐标中指定为

$$\text{div } \mathbf{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}. \quad (3.94)$$

► 通过将体积 $\Delta \mathcal{V}$ 缩小到零, 我们将 $\mathbf{E}$ 在某一点的散度定义为闭合增量表面上每单位体积的净向外通量。 ◀

因此, 从方程。 (3.93) , 我们有

$$\text{div } \mathbf{E} \triangleq \lim_{\Delta \mathcal{V} \rightarrow 0} \frac{\oint_S \mathbf{E} \cdot d\mathbf{s}}{\Delta \mathcal{V}}, \quad (3.95)$$

其中 S 包含元素体积 $\Delta \mathcal{V}$ 。通常的做法不是用 $\text{div } \mathbf{E}$ 来表示 $\mathbf{E}$ 的散度, 而是将其表示为 $\nabla \cdot \mathbf{E}$ 。也就是说,

$$\nabla \cdot \mathbf{E} = \text{div } \mathbf{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} \quad (3.96)$$

对于笛卡尔坐标系中的向量 $\mathbf{E}$ 。

► 根据方程给出的 $\mathbf{E}$ 散度的定义: (3.95) , 如果离开表面 S 的净通量为正, 则场 $\mathbf{E}$ 具有正散度, 这可以被“视为” 体积 $\Delta \mathcal{V}$ 包含场线源。如果散度为负, 则 $\Delta \mathcal{V}$ 可被视为包含磁力线汇, 因为净通量位于 $\Delta \mathcal{V}$ 中。对于均匀场 $\mathbf{E}$, 进入 $\Delta \mathcal{V}$ 的通量与离开它的通量相同; 因此, 它的散度为零, 并且该场被称为无散度。 ◀

散度是一个微分算子, 它总是对向量进行运算, 其运算结果是一个标量。这与梯度运算符相反, 梯度运算符始终对标量进行运算并产生向量。附录 C 提供了柱坐标和球坐标中向量散度的表达式。

散度算子是分布的。也就是说, 对于任意一对向量 $\mathbf{E}_1$ 和 $\mathbf{E}_2$,

$$\nabla \cdot (\mathbf{E}_1 + \mathbf{E}_2) = \nabla \cdot \mathbf{E}_1 + \nabla \cdot \mathbf{E}_2. \quad (3.97)$$

如果 $\nabla \cdot \mathbf{E} = 0$, 则向量场 $\mathbf{E}$ 称为无散。

方程给出的结果。对于微分体积 $\Delta \mathcal{V}$ 的方程 (3.93) 可以扩展为将任意体积 $\mathcal{V}$ 上的 $\nabla \cdot \mathbf{E}$ 的体积积分与 $\mathbf{E}$ 通过包围 $\mathcal{V}$ 的闭合表面 S 的通量联系起来。那是,

$$\int_{\mathcal{V}} \nabla \cdot \mathbf{E} d\mathcal{V} = \oint_S \mathbf{E} \cdot d\mathbf{s}. \quad (3.98)$$

(divergence theorem)

这种关系被称为散度定理, 广泛应用于电磁学中。

例 3-13: 计算散度

确定以下每个向量场的散度, 然后在指定点对其进行评估:

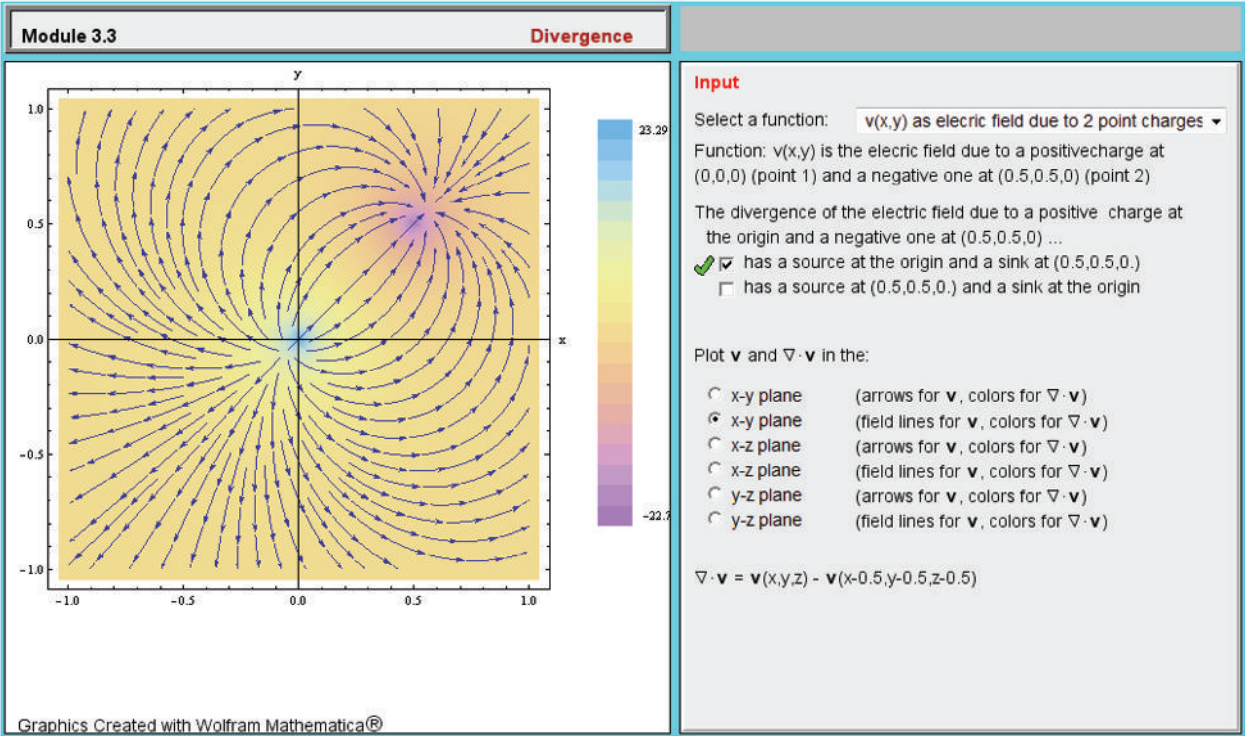
- (a) $\mathbf{E} = \hat{\mathbf{x}}3x^2 + \hat{\mathbf{y}}2z + \hat{\mathbf{z}}x^2z$ at $(2, -2, 0)$;
(b) $\mathbf{E} = \hat{\mathbf{R}}(a^3 \cos \theta / R^2) - \hat{\boldsymbol{\theta}}(a^3 \sin \theta / R^2)$ at $(a/2, 0, \pi)$.

解: (一)

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} \\ &= \frac{\partial}{\partial x}(3x^2) + \frac{\partial}{\partial y}(2z) + \frac{\partial}{\partial z}(x^2z) \\ &= 6x + 0 + x^2 = x^2 + 6x. \end{aligned}$$

$$\text{At } (2, -2, 0), \nabla \cdot \mathbf{E} \Big|_{(2, -2, 0)} = 16.$$

Module 3.3 Divergence Select a vector function $\mathbf{f}(x,y,z)$, evaluate its divergence, and display both in an appropriate 2-D plane.



(b) From the expression given in Appendix C for the divergence of a vector in spherical coordinates, it follows that

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{1}{R^2} \frac{\partial}{\partial R} (R^2 E_R) + \frac{1}{R \sin \theta} \frac{\partial}{\partial \theta} (E_\theta \sin \theta) \\ &\quad + \frac{1}{R \sin \theta} \frac{\partial E_\phi}{\partial \phi} \\ &= \frac{1}{R^2} \frac{\partial}{\partial R} (a^3 \cos \theta) + \frac{1}{R \sin \theta} \frac{\partial}{\partial \theta} \left(-\frac{a^3 \sin^2 \theta}{R^2} \right) \\ &= 0 - \frac{2a^3 \cos \theta}{R^3} = -\frac{2a^3 \cos \theta}{R^3}. \end{aligned}$$

At $R = a/2$ and $\theta = 0$, $\nabla \cdot \mathbf{E} \Big|_{(a/2,0,\pi)} = -16$.

Exercise 3-13: Given $\mathbf{A} = e^{-2y}(\hat{\mathbf{x}} \sin 2x + \hat{\mathbf{y}} \cos 2x)$, find $\nabla \cdot \mathbf{A}$.

Answer: $\nabla \cdot \mathbf{A} = 0$. (See [EM](#).)

Exercise 3-14: Given $\mathbf{A} = \hat{\mathbf{r}} r \cos \phi + \hat{\phi} r \sin \phi + \hat{\mathbf{z}} 3z$, find $\nabla \cdot \mathbf{A}$ at $(2,0,3)$.

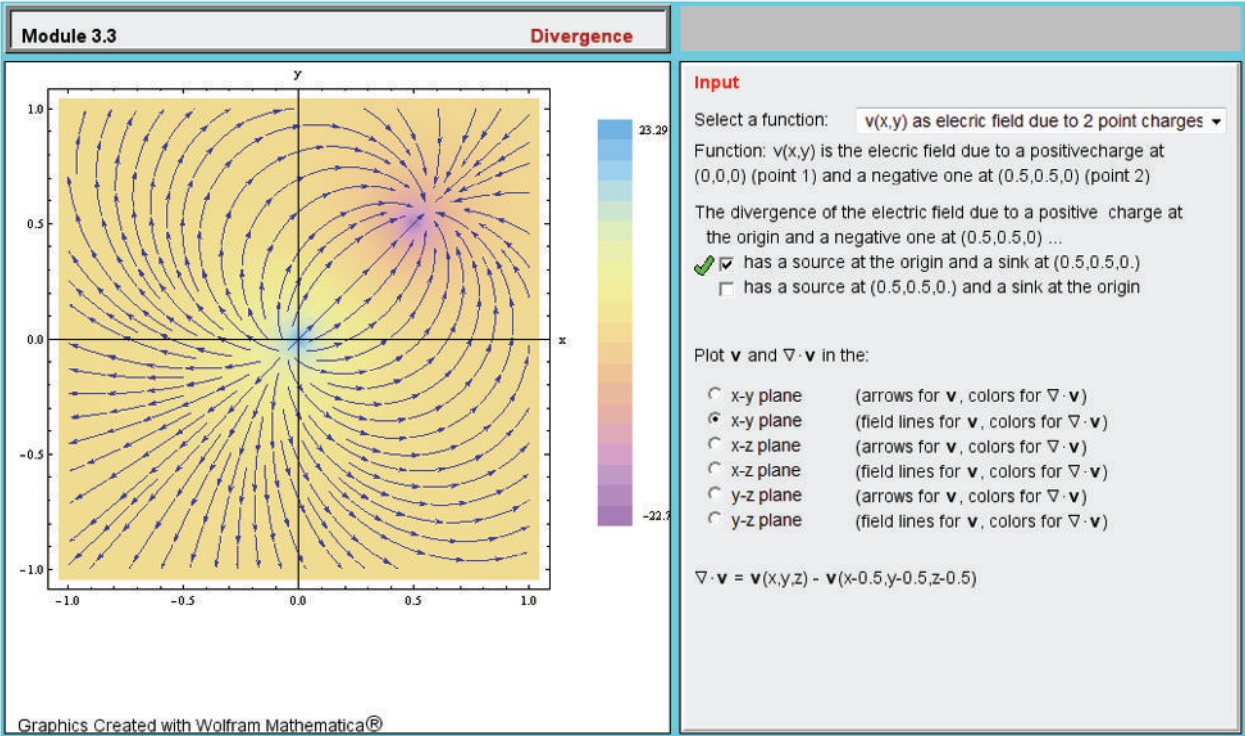
Answer: $\nabla \cdot \mathbf{A} = 6$. (See [EM](#).)

Exercise 3-15: If $\mathbf{E} = \hat{\mathbf{R}} R A R$ in spherical coordinates, calculate the flux of $\mathbf{E}$ through a spherical surface of radius a , centered at the origin.

Answer: $\oint_S \mathbf{E} \cdot d\mathbf{s} = 4\pi A a^3$. (See [EM](#).)

Exercise 3-16: Verify the divergence theorem by calculating the volume integral of the divergence of the field $\mathbf{E}$ of Exercise 3.15 over the volume bounded by the surface of radius a .

模块 3.3 散度 选择矢量函数 $\mathbf{f}(x, y, z)$ ，评估其散度，并将两者显示在适当的 2-D 平面中。



(b) 根据附录 C 给出的球坐标系向量散度表达式，可以得出：

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{1}{R^2} \frac{\partial}{\partial R} (R^2 E_R) + \frac{1}{R \sin \theta} \frac{\partial}{\partial \theta} (E_\theta \sin \theta) \\ &\quad + \frac{1}{R \sin \theta} \frac{\partial E_\phi}{\partial \phi} \\ &= \frac{1}{R^2} \frac{\partial}{\partial R} (a^3 \cos \theta) + \frac{1}{R \sin \theta} \frac{\partial}{\partial \theta} \left(-\frac{a^3 \sin^2 \theta}{R^2} \right) \\ &= 0 - \frac{2a^3 \cos \theta}{R^3} = -\frac{2a^3 \cos \theta}{R^3}. \end{aligned}$$

在 $R = a/2$ 和 $\theta = 0$ 、 $\nabla \cdot \mathbf{E} \Big|_{(a/2,0,\pi)} = -16$ 处。

练习 3-13：给定 $\mathbf{A} = e^{-2y}(\hat{\mathbf{x}} \sin 2x + \hat{\mathbf{y}} \cos 2x)$ ，求 $\nabla \cdot \mathbf{A}$ 。

答案： $\nabla \cdot \mathbf{A} = 0$ 。（参见 [EM](#)。）

练习 3-14：给定 $\mathbf{A} = \hat{\mathbf{r}} r \cos \phi + \hat{\phi} r \sin \phi + \hat{\mathbf{z}} 3z$ ，在 $(2, 0, 3)$ 处找到 $\nabla \cdot \mathbf{A}$ 。

Answer: $\nabla \cdot \mathbf{A} = 6$. (See [EM](#).)

练习3-15：如果球坐标中的 $\mathbf{E} = \hat{\mathbf{R}} R A R$ ，计算 $\mathbf{E}$ 穿过以原点为中心、半径为 a 的球面的通量。

Answer: $\oint_S \mathbf{E} \cdot d\mathbf{s} = 4\pi A a^3$. (See [EM](#).)

练习3-16：通过计算练习3.15中 $\mathbf{E}$ 场的散度在半径 a 的表面所包围的体积上的体积分来验证散度定理。

Exercise 3-17: The arrow representation in Fig. E3.17 represents the vector field $\mathbf{A} = \hat{x}x - \hat{y}y$. At a given point in space, $\mathbf{A}$ has a positive divergence $\nabla \cdot \mathbf{A}$ if the net flux flowing outward through the surface of an imaginary infinitesimal volume centered at that point is positive, $\nabla \cdot \mathbf{A}$ is negative if the net flux is into the volume, and $\nabla \cdot \mathbf{A} = 0$ if the same amount of flux enters into the volume as leaves it. Determine $\nabla \cdot \mathbf{A}$ everywhere in the x - y plane.

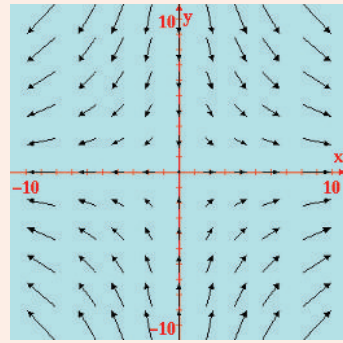


Figure E3.17

Answer: $\nabla \cdot \mathbf{A} = 0$ everywhere. (See E3.)

3-6 Curl of a Vector Field

So far we have defined and discussed two of the three fundamental operators used in vector analysis: the gradient of a scalar and the divergence of a vector. Now we introduce the **curl operator**. The curl of a vector field $\mathbf{B}$ describes its rotational property, or **circulation**. The circulation of $\mathbf{B}$ is defined as the line integral of $\mathbf{B}$ around a closed contour C :

$$\text{Circulation} = \oint_C \mathbf{B} \cdot d\mathbf{l}. \quad (3.99)$$

To gain a physical understanding of this definition, we consider two examples. The first is for a uniform field $\mathbf{B} = \hat{x}B_0$, whose field lines are as depicted in Fig. 3-23(a). For the rectangular contour $abcd$ shown in the figure, we have

$$\begin{aligned} \text{Circulation} &= \int_a^b \hat{x}B_0 \cdot \hat{x} dx + \int_b^c \hat{x}B_0 \cdot \hat{y} dy \\ &\quad + \int_c^d \hat{x}B_0 \cdot \hat{x} dx + \int_d^a \hat{x}B_0 \cdot \hat{y} dy \\ &= B_0 \Delta x - B_0 \Delta x = 0, \end{aligned} \quad (3.100)$$

where $\Delta x = b - a = c - d$ and, because $\hat{x} \cdot \hat{y} = 0$, the second and fourth integrals are zero. According to Eq. (3.100), *the circulation of a uniform field is zero*.

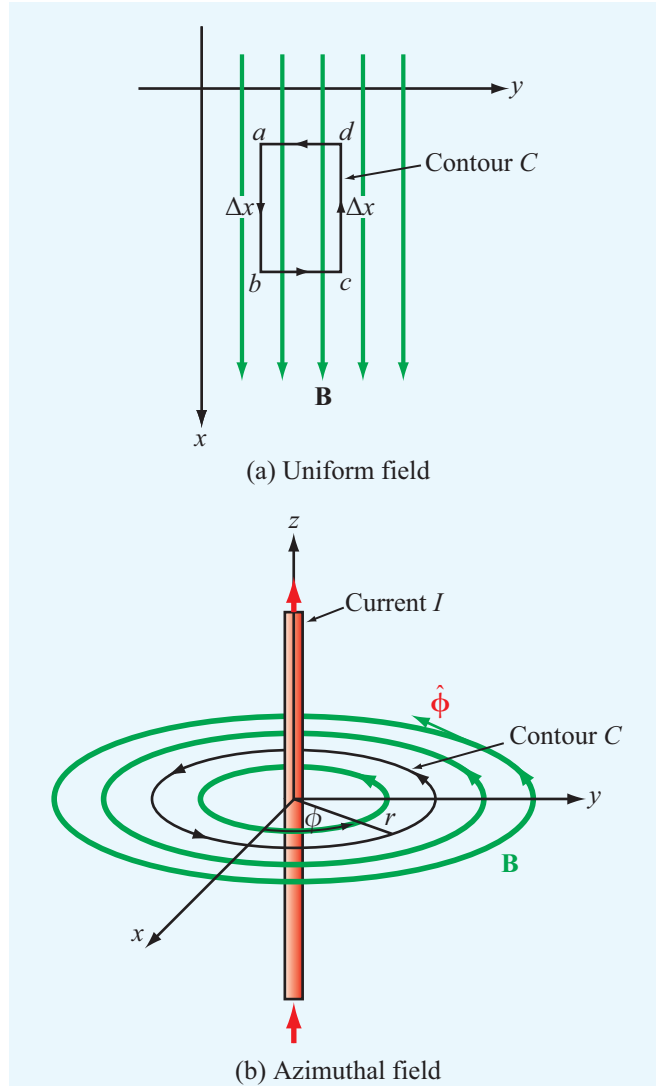


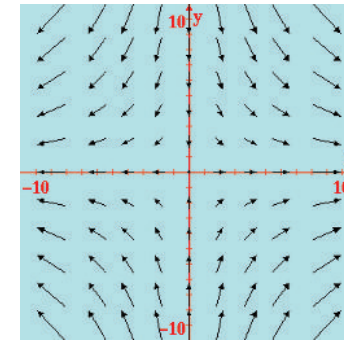
Figure 3-23 Circulation is zero for the uniform field in (a), but it is not zero for the azimuthal field in (b).

Next, we consider the magnetic flux density $\mathbf{B}$ induced by an infinite wire carrying a dc current I . If the current is in free space and it is oriented along the z direction, then from Eq. (1.13),

$$\mathbf{B} = \hat{\phi} \frac{\mu_0 I}{2\pi r}, \quad (3.101)$$

where μ_0 is the permeability of free space and r is the radial distance from the current in the x - y plane. The direction of $\mathbf{B}$ is along the azimuth unit vector $\hat{\phi}$. The field lines of $\mathbf{B}$ are concentric circles around the current, as shown in Fig. 3-23(b).

练习3-17: 图E3.17中的箭头表示向量场 $\mathbf{A} = \hat{x}x - \hat{y}y$ 。在空间中的给定点，如果通过以该点为中心的虚无限小体积表面向外流动的净通量为正，则 $\mathbf{A}$ 具有正散度 $\nabla \cdot \mathbf{A}$ ；如果净通量流入该体积，则 $\nabla \cdot \mathbf{A}$ 为负；如果进入该体积的通量与离开该体积的通量相同，则 $\nabla \cdot \mathbf{A} = 0$ 。确定 x - y 平面上任意位置的 $\nabla \cdot \mathbf{A}$ 。



图E3.17

答案: $\nabla \cdot \mathbf{A} = 0$ 到处都是。(参见E3.)

3-6 矢量场的旋度

到目前为止，我们已经定义并讨论了向量分析中使用的三个基本运算符中的两个：标量的梯度和向量的散度。现在我们介绍curl运算符。向量场 $\mathbf{B}$ 的旋度描述了它的旋转属性或循环。 $\mathbf{B}$ 的环流定义为 $\mathbf{B}$ 绕封闭轮廓 C 的线积分；

$$\text{Circulation} = \oint_C \mathbf{B} \cdot d\mathbf{l}. \quad (3.99)$$

为了获得对这个定义的物理理解，我们考虑两个例子。第一个是均匀场 $\mathbf{B} = \hat{x}B_0$ ，其场线如图 3-23(a) 所示。对于图中所示的矩形轮廓 $abcd$ ，我们有

$$\begin{aligned} \text{Circulation} &= \int_a^b \hat{x}B_0 \cdot \hat{x} dx + \int_b^c \hat{x}B_0 \cdot \hat{y} dy \\ &\quad + \int_c^d \hat{x}B_0 \cdot \hat{x} dx + \int_d^a \hat{x}B_0 \cdot \hat{y} dy \\ &= B_0 \Delta x - B_0 \Delta x = 0, \end{aligned} \quad (3.100)$$

其中 $\Delta x = b - a = c - d$ 并且，由于 $\hat{x} \cdot \hat{y} = 0$ ，第二个和第四个积分为零。根据方程 (3.100)，均匀场的环流为零。

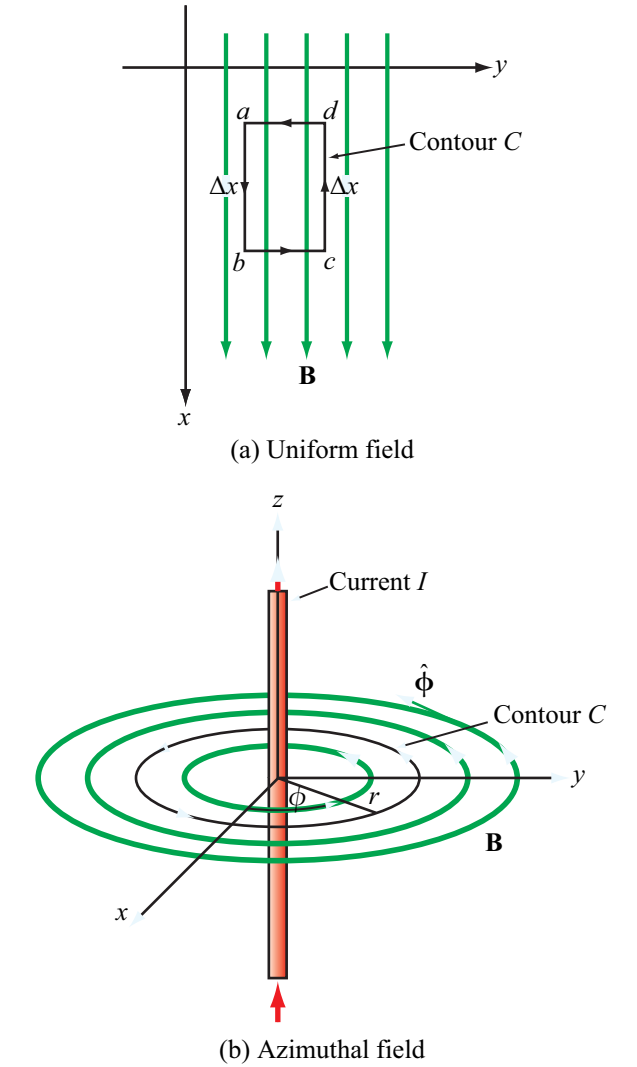


图 3-23 (a) 中均匀场的环流为零，但 (b) 中方位角场的环流不为零。

接下来，我们考虑由承载直流电流 I 的无限导线感应的磁通密度 $\mathbf{B}$ 。如果电流位于自由空间中并且沿 z 方向定向，则根据等式 (1.13)，

$$\mathbf{B} = \hat{\phi} \frac{\mu_0 I}{2\pi r}, \quad (3.101)$$

其中 μ_0 是自由空间的磁导率， r 是 x - y 平面中距电流的径向距离。 $\mathbf{B}$ 的方向沿着方位角单位向量 $\hat{\phi}$ 。 $\mathbf{B}$ 的磁力线是围绕电流的同心圆，如图 3-23(b) 所示。

For a circular contour C of radius r centered at the origin in the x - y plane, the differential length vector $d\mathbf{l} = \hat{\boldsymbol{\phi}} r d\phi$, and the circulation of $\mathbf{B}$ is

Circulation $= \oint_C \mathbf{B} \cdot d\mathbf{l} = \int_0^{2\pi} \hat{\boldsymbol{\phi}} \frac{\mu_0 I}{2\pi r} \cdot \hat{\boldsymbol{\phi}} r d\phi = \mu_0 I. \quad (3.102)$

In this case, the circulation is not zero. However, had the contour C been in the x - z or y - z planes, $d\mathbf{l}$ would not have had a $\hat{\boldsymbol{\phi}}$ component, and the integral would have yielded a zero circulation. Clearly, the circulation of $\mathbf{B}$ depends on the choice of contour and the direction in which it is traversed. To describe the circulation of a tornado, for example, we would like to choose our contour such that the circulation of the wind field is maximum, and we would like the circulation to have both a magnitude and a direction with the direction being toward the tornado's vortex. The curl operator embodies these properties. The curl of a vector field $\mathbf{B}$, denoted $\text{curl } \mathbf{B}$ or $\nabla \times \mathbf{B}$, is defined as

$$\nabla \times \mathbf{B} = \text{curl } \mathbf{B} = \lim_{\Delta s \rightarrow 0} \frac{1}{\Delta s} \left[\hat{\mathbf{n}} \oint_C \mathbf{B} \cdot d\mathbf{l} \right]_{\max}. \quad (3.103)$$

► **Curl $\mathbf{B}$** is the circulation of $\mathbf{B}$ per unit area, with the area Δs of the contour C being oriented such that the circulation is maximum. ◀

The direction of $\text{curl } \mathbf{B}$ is $\hat{\mathbf{n}}$, the unit normal of Δs , defined according to the right-hand rule with the four fingers of the right hand following the contour direction $d\mathbf{l}$ and the thumb pointing along $\hat{\mathbf{n}}$ (Fig. 3-24). When we use the notation $\nabla \times \mathbf{B}$ to denote $\text{curl } \mathbf{B}$, it should *not* be interpreted as the cross product of ∇ and $\mathbf{B}$.

For a vector $\mathbf{B}$ specified in Cartesian coordinates as

$$\mathbf{B} = \hat{\mathbf{x}}B_x + \hat{\mathbf{y}}B_y + \hat{\mathbf{z}}B_z, \quad (3.104)$$

it can be shown, through a rather lengthy derivation, that Eq. (3.103) leads to

$$\begin{aligned} \nabla \times \mathbf{B} &= \hat{\mathbf{x}} \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) + \hat{\mathbf{y}} \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) \\ &\quad + \hat{\mathbf{z}} \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right) \\ &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ B_x & B_y & B_z \end{vmatrix}. \end{aligned} \quad (3.105)$$

Expressions for $\nabla \times \mathbf{B}$ are given in Appendix C for the three orthogonal coordinate systems considered in this chapter.

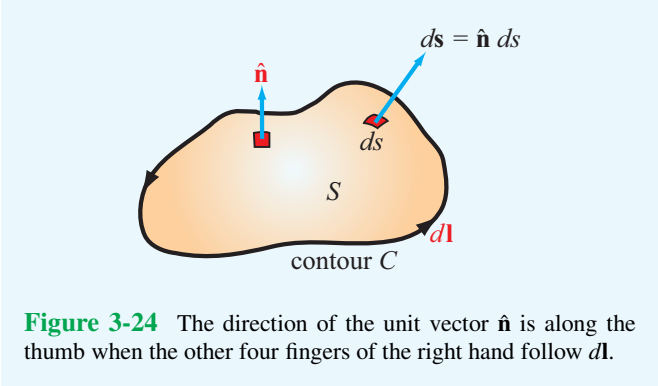


Figure 3-24 The direction of the unit vector $\hat{\mathbf{n}}$ is along the thumb when the other four fingers of the right hand follow $d\mathbf{l}$.

3-6.1 Vector Identities Involving the Curl

For any two vectors $\mathbf{A}$ and $\mathbf{B}$ and scalar V ,

(1) $\nabla \times (\mathbf{A} + \mathbf{B}) = \nabla \times \mathbf{A} + \nabla \times \mathbf{B}, \quad (3.106a)$

(2) $\nabla \cdot (\nabla \times \mathbf{A}) = 0, \quad (3.106b)$

(3) $\nabla \times (\nabla V) = 0. \quad (3.106c)$

3-6.2 Stokes's Theorem

► **Stokes's theorem** converts the surface integral of the curl of a vector over an open surface S into a line integral of the vector along the contour C bounding the surface S . ◀

For the geometry shown in Fig. 3-24, **Stokes's theorem** states

$$\int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{s} = \oint_C \mathbf{B} \cdot d\mathbf{l}. \quad (3.107)$$

(Stokes's theorem)

Its validity follows from the definition of $\nabla \times \mathbf{B}$ given by Eq. (3.103). If $\nabla \times \mathbf{B} = 0$, the field $\mathbf{B}$ is said to be **conservative** or **irrotational** because its circulation, represented by the right-hand side of Eq. (3.107), is zero, irrespective of the contour chosen.

对于 x - y 平面上以原点为圆心、半径为 r 的圆形轮廓 C ，微分长度向量 $d\mathbf{l} = \hat{\boldsymbol{\phi}} r d\phi$ ， $\mathbf{B}$ 的循环为

Circulation $= \oint_C \mathbf{B} \cdot d\mathbf{l} = \int_0^{2\pi} \hat{\boldsymbol{\phi}} \frac{\mu_0 I}{2\pi r} \cdot \hat{\boldsymbol{\phi}} r d\phi = \mu_0 I. \quad (3.102)$

在这种情况下，流通量不为零。然而，如果轮廓 C 位于 x - z 或 y - z 平面上，则 $d\mathbf{l}$ 不会有一个 $\hat{\boldsymbol{\phi}}$ 分量，积分将产生零循环。显然， $\mathbf{B}$ 的循环取决于轮廓的选择及其遍历的方向。例如，为了描述龙卷风的环流，我们希望选择轮廓以使风场的环流最大，并且我们希望环流具有大小和方向，方向朝向龙卷风的涡旋。curl 运算符体现了这些属性。矢量场 $\mathbf{B}$ 的旋度，表示为旋度 $\mathbf{B}$ 或 $\nabla \times \mathbf{B}$ ，定义为

$$\nabla \times \mathbf{B} = \text{curl } \mathbf{B} = \lim_{\Delta s \rightarrow 0} \frac{1}{\Delta s} \left[\hat{\mathbf{n}} \oint_C \mathbf{B} \cdot d\mathbf{l} \right]_{\max}. \quad (3.103)$$

► 旋度 $\mathbf{B}$ 是每单位面积的 $\mathbf{B}$ 的循环，其中轮廓 C 的面积 Δs 被定向为使得循环最大。◀

卷曲 $\mathbf{B}$ 的方向为 $\hat{\mathbf{n}}$ ，即 Δs 的单位法线，根据右手定则定义，右手四指沿着轮廓方向 $d\mathbf{l}$ ，拇指指向 $\hat{\mathbf{n}}$ (图 3-24)。当我们使用符号 $\nabla \times \mathbf{B}$ 来表示旋度 $\mathbf{B}$ 时，它不应该被解释为 ∇ 和 $\mathbf{B}$ 的叉积。

对于在笛卡尔坐标中指定的向量 $\mathbf{B}$

$$\mathbf{B} = \hat{\mathbf{x}}B_x + \hat{\mathbf{y}}B_y + \hat{\mathbf{z}}B_z, \quad (3.104)$$

通过相当长的推导可以证明，方程。(3.103) 导致

$$\begin{aligned} \nabla \times \mathbf{B} &= \hat{\mathbf{x}} \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) + \hat{\mathbf{y}} \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) \\ &\quad + \hat{\mathbf{z}} \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right) \\ &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ B_x & B_y & B_z \end{vmatrix}. \end{aligned} \quad (3.105)$$

附录 C 中给出了本章考虑的三个正交坐标系的 $\nabla \times \mathbf{B}$ 的表达式。

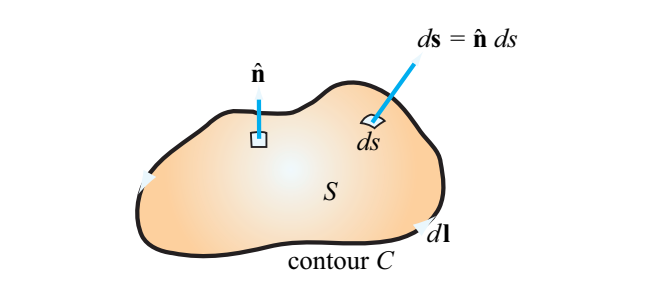


图3-24 当右手的其他四个手指跟随 $d\mathbf{l}$ 时，单位向量 $\hat{\mathbf{n}}$ 的方向是沿着拇指的。

3-6.1 涉及旋度的向量恒等式

对于任意两个向量 $\mathbf{A}$ 和 $\mathbf{B}$ 以及标量 V ，

(1) $\nabla \times (\mathbf{A} + \mathbf{B}) = \nabla \times \mathbf{A} + \nabla \times \mathbf{B}, \quad (3.106a)$

(2) $\nabla \cdot (\nabla \times \mathbf{A}) = 0, \quad (3.106b)$

(3) $\nabla \times (\nabla V) = 0. \quad (3.106c)$

3-6.2 斯托克斯定理

► 斯托克斯定理将向量在开放曲面 S 上的旋度的曲面积分转换为向量沿包围曲面 S 的轮廓 C 的线积分。◀

对于图 3-24 所示的几何形状，斯托克斯定理指出

$$\int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{s} = \oint_C \mathbf{B} \cdot d\mathbf{l}. \quad (3.107)$$

(Stokes's theorem)

它的有效性来自等式给出的 $\nabla \times \mathbf{B}$ 的定义。(3.103)。如果 $\nabla \times \mathbf{B} = 0$ ，则称域 B 为保守域或无旋域，因为它的循环由等式右侧表示。(3.107) 为零，无论选择什么轮廓。

Technology Brief 6:
X-Ray Computed Tomography

► The word **tomography** is derived from the Greek words **tome**, meaning section or slice, and **graphia**, meaning writing. ◀

Computed tomography, also known as **CT scan** or **CAT scan** (for computed axial tomography), refers to a technique capable of generating 3-D images of the X-ray attenuation (absorption) properties of an object. This is in contrast to the traditional, X-ray technique that produces only a 2-D profile of the object (Fig. TF6-1). CT was invented in 1972 by British electrical engineer **Godfrey Hounsfield** and independently by **Allan Cormack**, a South African-born American physicist. The two inventors shared the 1979 Nobel Prize in Physiology or



Figure TF6-1 2-D X-ray image.

Medicine. Among diagnostic imaging techniques, CT has the decided advantage in having the sensitivity to image body parts on a wide range of densities—from soft tissue to blood vessels and bones.

Principle of Operation

In the system shown in Fig. TF6-2, the X-ray source and detector array are contained inside a circular structure through which the patient is moved along a conveyor belt. A CAT scan technician can monitor the reconstructed images to ensure that they do not contain artifacts such as streaks or blurry sections caused by movement on the part of the patient during the measurement process.

A CT scanner uses an **X-ray source** with a narrow slit that generates a **fan-beam** that is wide enough to encompass the extent of the body but only a few millimeters in thickness (Fig. TF6-3(a)). Instead of recording the attenuated X-ray beam on film, it is captured by an array of some 700 **detectors**. The X-ray source and the detector array are mounted on a circular frame that rotates in steps of a fraction of a degree over a full 360° circle around the patient, each time recording an X-ray attenuation profile from a different angular perspective. Typically, 1,000 such profiles are recorded per each thin traverse slice of anatomy. In today's technology, this process is completed in less than 1 second. To image an entire part of the body, such as the chest or head, the process is repeated over multiple slices (layers), which typically takes about 10 seconds to complete.



Figure TF6-2 CT scanner.

技术简介 6: X 射线计算机断层扫描

► 断层扫描一词源自希腊语 **tome**，意思是截面或切片，而 **graphia**，意思是书写。◀

计算机断层扫描，也称为 CT 扫描或 CAT 扫描（计算机轴向断层扫描），是指能够生成物体 X 射线衰减（吸收）特性的 3D 图像的技术。这与传统的 X 射线技术形成鲜明对比，传统的 X 射线技术仅产生物体的二维轮廓（图 TF6-1）。CT 由英国电气工程师 Godfrey Hounsfield 于 1972 年发明，并由南非出生的美国物理学家 Allan Cormack 独立发明。两位发明者共同获得 1979 年诺贝尔生理学奖或

药品。在诊断成像技术中，CT 具有明显的优势，它对各种密度的身体部位（从软组织到血管和骨骼）的图像具有敏感性。

工作原理

在图 TF6-2 所示的系统中，X 射线源和探测器阵列包含在圆形结构内，患者沿着传送带移动通过该圆形结构。CAT 扫描技术人员可以监控重建图像，以确保它们不包含测量过程中因患者运动而造成的条纹或模糊部分等伪影。

CT 扫描仪使用带有窄缝的 X 射线源，产生扇形光束，其宽度足以包围整个身体，但厚度只有几毫米（图 TF6-3(a)）。衰减的 X 射线束不是在胶片上记录，而是由大约 700 个探测器组成的阵列捕获。X 射线源和探测器阵列安装在一个圆形框架上，该框架以零点几度的步长在围绕患者的完整 360° 圆上旋转，每次从不同的角度记录 X 射线衰减剖面。通常，每个薄横切解剖切片都会记录 1,000 个这样的剖面。在当今的技术中，这个过程不到 1 秒即可完成。为了对身体的整个部位（例如胸部或头部）进行成像，需要在多个切片（层）上重复该过程，通常需要大约 10 秒才能完成。



图 TF6-2 CT 扫描仪。

图 TF6-1 二维 X 射线图像。

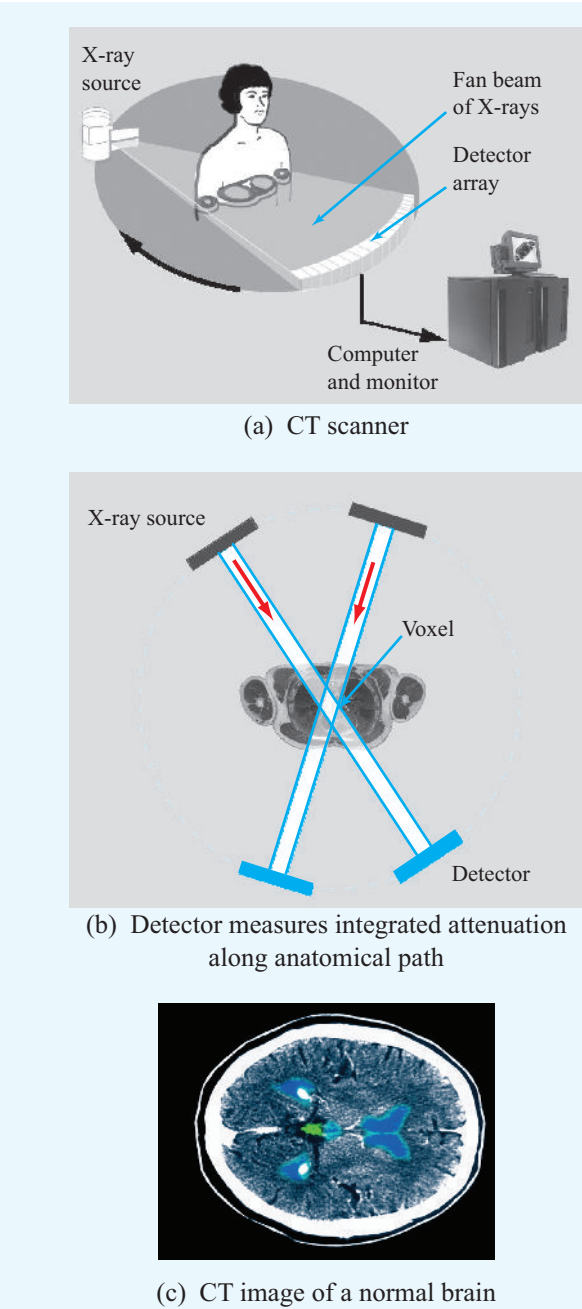


Figure TF6-3 Basic elements of a CT scanner.

Image Reconstruction

For each anatomical slice, the CT scanner generates on the order of 7×10^5 measurements (1,000 angular orientations $\times$ 700 detector channels). Each measurement represents the integrated path attenuation for the narrow beam between the X-ray source and the detector (Fig. TF6-3(b)), and each volume element (*voxel*) contributes to 1,000 such measurement beams.

► Commercial CT machines use a technique called **filtered back-projection** to “reconstruct” an image of the attenuation rate of each voxel in the anatomical slice and, by extension, for each voxel in the entire body organ. This is accomplished through the application of a sophisticated matrix inversion process. ◀

A sample CT image of the brain is shown in Fig. TF6-3(c).

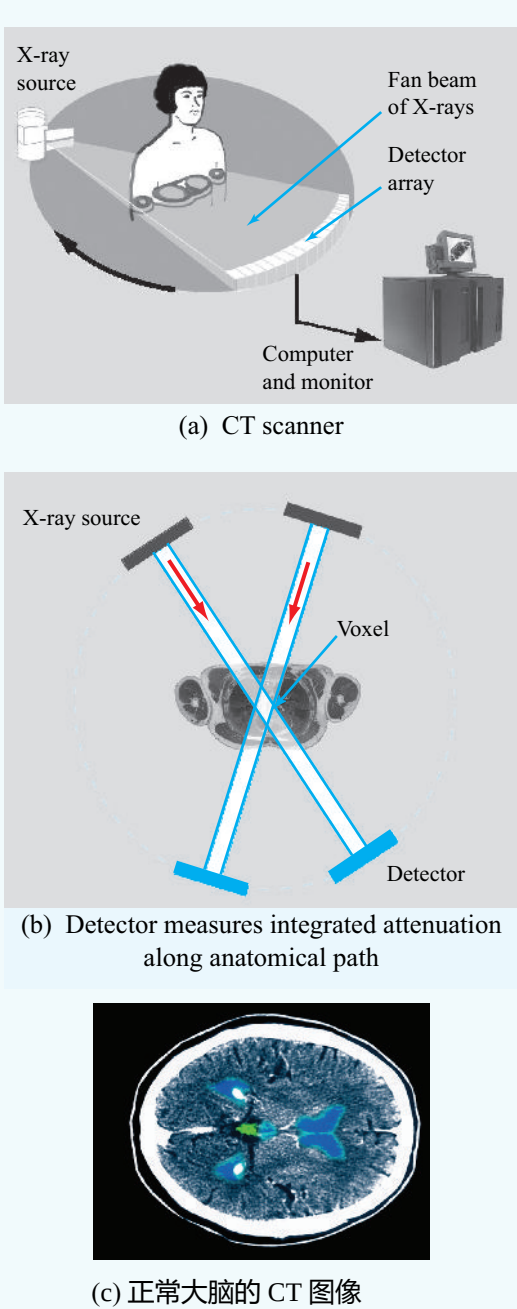


图 TF6-3 CT 扫描仪的基本元件。

图像重建

对于每个解剖切片，CT 扫描仪生成大约 7×10^5 测量值 (1,000 个角度方向 $\times$ 700 个探测器通道)。每个测量值代表X射线源和探测器之间窄光束的积分路径衰减 (图TF 6-3 (b))，每个体积元素 (体素) 贡献1,000个这样的测量光束。

► 商用 CT 机使用一种称为滤波反投影的技术来 “重建” 解剖切片中每个体素的衰减率图像，进而重建整个身体器官中每个体素的衰减率图像。这是通过应用复杂的矩阵求逆过程来实现的。 ◀

大脑 CT 图像样本如图 TF6-3(c) 所示。

3-6 CURL OF A VECTOR FIELD

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Example 3-14: Verification of Stokes's Theorem

For vector field $\mathbf{B} = \hat{\mathbf{z}} \cos \phi / r$, verify Stokes's theorem for a segment of a cylindrical surface defined by $r = 2$, $\pi/3 \leq \phi \leq \pi/2$, and $0 \leq z \leq 3$ (Fig. 3-25).

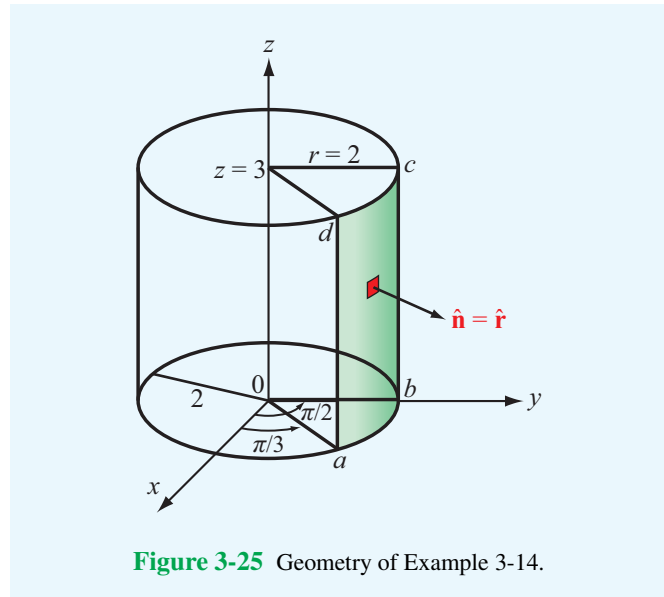


Figure 3-25 Geometry of Example 3-14.

Solution: Stokes's theorem states that

$$\int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{s} = \oint_C \mathbf{B} \cdot d\mathbf{l}.$$

Left-hand side: With $\mathbf{B}$ having only a component $B_z = \cos \phi / r$, use of the expression for $\nabla \times \mathbf{B}$ in cylindrical coordinates in Appendix C gives

$$\begin{aligned} \nabla \times \mathbf{B} &= \hat{\mathbf{r}} \left(\frac{1}{r} \frac{\partial B_z}{\partial \phi} - \frac{\partial B_\phi}{\partial z} \right) + \hat{\boldsymbol{\phi}} \left(\frac{\partial B_r}{\partial z} - \frac{\partial B_z}{\partial r} \right) \\ &\quad + \hat{\mathbf{z}} \frac{1}{r} \left(\frac{\partial}{\partial r} (r B_\phi) - \frac{\partial B_r}{\partial \phi} \right) \\ &= \hat{\mathbf{r}} \frac{1}{r} \frac{\partial}{\partial \phi} \left(\frac{\cos \phi}{r} \right) - \hat{\boldsymbol{\phi}} \frac{\partial}{\partial r} \left(\frac{\cos \phi}{r} \right) \\ &= -\hat{\mathbf{r}} \frac{\sin \phi}{r^2} + \hat{\boldsymbol{\phi}} \frac{\cos \phi}{r^2}. \end{aligned}$$

The integral of $\nabla \times \mathbf{B}$ over the specified surface S is

$$\begin{aligned} \int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{s} &= \int_{z=0}^3 \int_{\phi=\pi/3}^{\pi/2} \left(-\hat{\mathbf{r}} \frac{\sin \phi}{r^2} + \hat{\boldsymbol{\phi}} \frac{\cos \phi}{r^2} \right) \cdot \hat{\mathbf{r}} r d\phi dz \\ &= \int_0^3 \int_{\pi/3}^{\pi/2} -\frac{\sin \phi}{r} d\phi dz = -\frac{3}{2r} = -\frac{3}{4}. \end{aligned}$$

Right-hand side: The surface S is bounded by contour $C = abcd$ shown in Fig. 3-25. The direction of C is chosen so that it is compatible with the surface normal $\hat{\mathbf{r}}$ by the right-hand rule. Hence,

$$\begin{aligned} \oint_C \mathbf{B} \cdot d\mathbf{l} &= \int_a^b \mathbf{B}_{ab} \cdot d\mathbf{l} + \int_b^c \mathbf{B}_{bc} \cdot d\mathbf{l} \\ &\quad + \int_c^d \mathbf{B}_{cd} \cdot d\mathbf{l} + \int_d^a \mathbf{B}_{da} \cdot d\mathbf{l}, \end{aligned}$$

where $\mathbf{B}_{ab}$, $\mathbf{B}_{bc}$, $\mathbf{B}_{cd}$, and $\mathbf{B}_{da}$ are the field $\mathbf{B}$ along segments ab , bc , cd , and da , respectively. Over segment ab , the dot product of $\mathbf{B}_{ab} = \hat{\mathbf{z}} (\cos \phi) / 2$ and $d\mathbf{l} = \hat{\boldsymbol{\phi}} r d\phi$ is zero, and the same is true for segment cd . Over segment bc , $\phi = \pi/2$; hence, $\mathbf{B}_{bc} = \hat{\mathbf{z}} (\cos \pi/2) / 2 = 0$. For the last segment, $\mathbf{B}_{da} = \hat{\mathbf{z}} (\cos \pi/3) / 2 = \hat{\mathbf{z}} / 4$ and $d\mathbf{l} = \hat{\mathbf{z}} dz$. Hence,

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = \int_d^a \left(\hat{\mathbf{z}} \frac{1}{4} \right) \cdot \hat{\mathbf{z}} dz = \int_3^0 \frac{1}{4} dz = -\frac{3}{4},$$

which is the same as the result obtained by evaluating the left-hand side of Stokes's equation.

Exercise 3-18: Find $\nabla \times \mathbf{A}$ at $(2, 0, 3)$ in cylindrical coordinates for the vector field

$$\mathbf{A} = \hat{\mathbf{r}} 10e^{-2r} \cos \phi + \hat{\mathbf{z}} 10 \sin \phi.$$

Answer: (See [EM](#).)

$$\nabla \times \mathbf{A} = \left(\hat{\mathbf{r}} \frac{10 \cos \phi}{r} + \frac{\hat{\mathbf{z}} 10 e^{-2r}}{r} \sin \phi \right) \bigg|_{(2,0,3)} = \hat{\mathbf{r}} 5.$$

Exercise 3-19: Find $\nabla \times \mathbf{A}$ at $(3, \pi/6, 0)$ in spherical coordinates for the vector field $\mathbf{A} = \hat{\boldsymbol{\theta}} 12 \sin \theta$.

Answer: (See [EM](#).)

$$\nabla \times \mathbf{A} = \hat{\boldsymbol{\phi}} \frac{12 \sin \theta}{R} \bigg|_{(3,\pi/6,0)} = \hat{\boldsymbol{\phi}} 2.$$

3-6 CURL OF A VECTOR FIELD

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例 3-14: 斯托克斯定理的验证

对于矢量场 $\mathbf{B} = \hat{\mathbf{z}} \cos \phi / r$, 验证由 $r = 2$ 、 $\pi/3 \leq \phi \leq \pi/2$ 和 $0 \leq z \leq 3$ 定义的一段圆柱表面的斯托克斯定理 (图 3-25)。

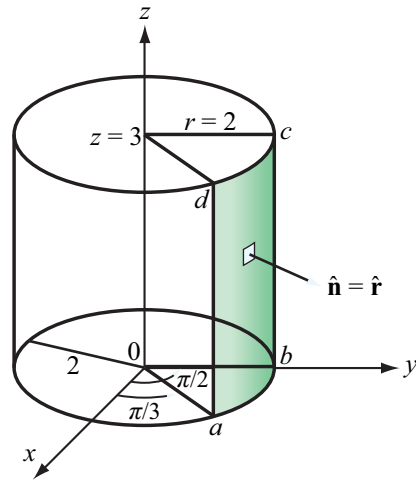


图 3-25 示例 3-14 的几何形状。

解: 斯托克斯定理指出

$$\int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{s} = \oint_C \mathbf{B} \cdot d\mathbf{l}.$$

左侧: 当 $\mathbf{B}$ 仅具有分量 $B_z = \cos \phi / r$ 时, 使用附录 C 中的柱坐标中的 $\nabla \times \mathbf{B}$ 表达式可得出

$$\begin{aligned} \nabla \times \mathbf{B} &= \hat{\mathbf{r}} \left(\frac{1}{r} \frac{\partial B_z}{\partial \phi} - \frac{\partial B_\phi}{\partial z} \right) + \hat{\boldsymbol{\phi}} \left(\frac{\partial B_r}{\partial z} - \frac{\partial B_z}{\partial r} \right) \\ &\quad + \hat{\mathbf{z}} \frac{1}{r} \left(\frac{\partial}{\partial r} (r B_\phi) - \frac{\partial B_r}{\partial \phi} \right) \\ &= \hat{\mathbf{r}} \frac{1}{r} \frac{\partial}{\partial \phi} \left(\frac{\cos \phi}{r} \right) - \hat{\boldsymbol{\phi}} \frac{\partial}{\partial r} \left(\frac{\cos \phi}{r} \right) \\ &= -\hat{\mathbf{r}} \frac{\sin \phi}{r^2} + \hat{\boldsymbol{\phi}} \frac{\cos \phi}{r^2}. \end{aligned}$$

$\nabla \times \mathbf{B}$ 在指定表面 S 上的积分为

$$\begin{aligned} \int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{s} &= \int_{z=0}^3 \int_{\phi=\pi/3}^{\pi/2} \left(-\hat{\mathbf{r}} \frac{\sin \phi}{r^2} + \hat{\boldsymbol{\phi}} \frac{\cos \phi}{r^2} \right) \cdot \hat{\mathbf{r}} r d\phi dz \\ &= \int_0^3 \int_{\pi/3}^{\pi/2} -\frac{\sin \phi}{r} d\phi dz = -\frac{3}{2r} = -\frac{3}{4}. \end{aligned}$$

右侧: 曲面 S 以轮廓 $C = abcd$ 为界, 如图 3-25 所示。选择 C 的方向, 使其与根据右手定则的表面法线兼容。因此,

$$\begin{aligned} \oint_C \mathbf{B} \cdot d\mathbf{l} &= \int_a^b \mathbf{B}_{ab} \cdot d\mathbf{l} + \int_b^c \mathbf{B}_{bc} \cdot d\mathbf{l} \\ &\quad + \int_c^d \mathbf{B}_{cd} \cdot d\mathbf{l} + \int_d^a \mathbf{B}_{da} \cdot d\mathbf{l}, \end{aligned}$$

其中 B_{ab} 、 B_{bc} 、 B_{cd} 和 B_{da} 分别是沿线段 ab 、 bc 、 cd 和 da 的域 $\mathbf{B}$ 。在线段 ab 上, $B_{ab} = \hat{\mathbf{z}} (\cos \phi) / 2$ 和 $d\mathbf{l} = \hat{\boldsymbol{\phi}} r d\phi$ 的点积为零, 线段 cd 也是如此。过 BC 段, $\phi = \pi/2$; 因此, $B_{bc} = \hat{\mathbf{z}} (\cos \pi/2) / 2 = 0$ 。对于最后一段, $B_{da} = \hat{\mathbf{z}} (\cos \pi/3) / 2 = \hat{\mathbf{z}} / 4$ 和 $d\mathbf{l} = \hat{\mathbf{z}} dz$ 。因此,

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = \int_d^a \left(\hat{\mathbf{z}} \frac{1}{4} \right) \cdot \hat{\mathbf{z}} dz = \int_3^0 \frac{1}{4} dz = -\frac{3}{4},$$

这与评估斯托克斯方程左侧得到的结果相同。

练习 3-18: 在向量场的柱坐标中求 $(2, 0, 3)$ 处的 $\nabla \times \mathbf{A}$

$$\mathbf{A} = \hat{\mathbf{r}} 10e^{-2r} \cos \phi + \hat{\mathbf{z}} 10 \sin \phi.$$

答案: (参见 [EM](#).)

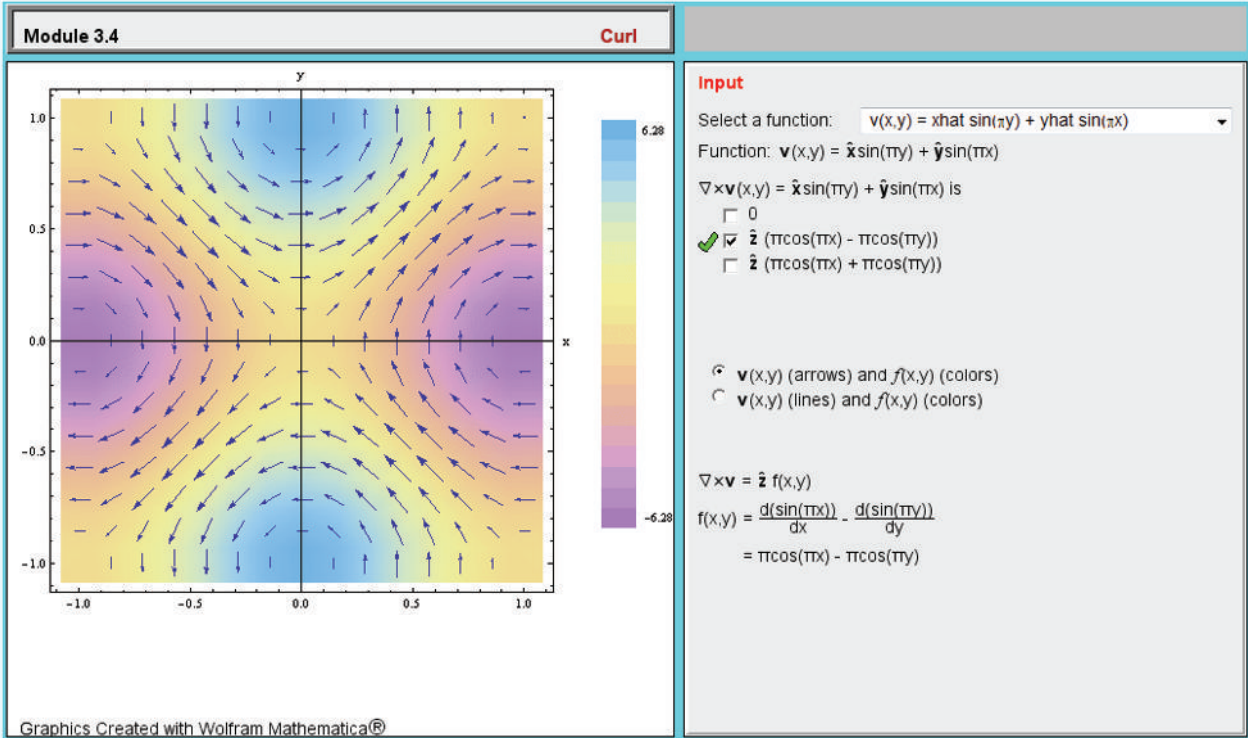
$$\nabla \times \mathbf{A} = \left(\hat{\mathbf{r}} \frac{10 \cos \phi}{r} + \frac{\hat{\mathbf{z}} 10 e^{-2r}}{r} \sin \phi \right) \bigg|_{(2,0,3)} = \hat{\mathbf{r}} 5.$$

练习 3-19: 在球坐标系中针对矢量场 $\mathbf{A} = \hat{\boldsymbol{\theta}} 12 \sin \theta$ 求 $(3, \pi/6, 0)$ 处的 $\nabla \times \mathbf{A}$ 。

Answer: (See [EM](#).)

$$\nabla \times \mathbf{A} = \hat{\boldsymbol{\phi}} \frac{12 \sin \theta}{R} \bigg|_{(3,\pi/6,0)} = \hat{\boldsymbol{\phi}} 2.$$

Module 3.4 Curl Select a vector $\mathbf{f}(x,y)$, evaluate its curl, and display both in the x - y plane.



3-7 Laplacian Operator

In later chapters, we sometimes deal with problems involving multiple combinations of operations on scalars and vectors. A frequently encountered combination is the divergence of the gradient of a scalar. For a scalar function V defined in Cartesian coordinates, its gradient is

$$\nabla V = \hat{\mathbf{x}} \frac{\partial V}{\partial x} + \hat{\mathbf{y}} \frac{\partial V}{\partial y} + \hat{\mathbf{z}} \frac{\partial V}{\partial z} = \hat{\mathbf{x}} A_x + \hat{\mathbf{y}} A_y + \hat{\mathbf{z}} A_z = \mathbf{A}, \tag{3.108}$$

where we defined a vector $\mathbf{A}$ with components $A_x = \partial V / \partial x$, $A_y = \partial V / \partial y$, and $A_z = \partial V / \partial z$. The divergence of ∇V is

$$\begin{aligned} \nabla \cdot (\nabla V) &= \nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \\ &= \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}. \end{aligned} \tag{3.109}$$

For convenience, $\nabla \cdot (\nabla V)$ is called the **Laplacian** of V and is denoted by $\nabla^2 V$ (the symbol ∇^2 is pronounced “del square”).

That is,

$$\nabla^2 V = \nabla \cdot (\nabla V) = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}. \tag{3.110}$$

As we can see from Eq. (3.110), the Laplacian of a scalar function is a scalar. Expressions for $\nabla^2 V$ in cylindrical and spherical coordinates are given in Appendix C.

The Laplacian of a scalar can be used to define the Laplacian of a vector. For a vector $\mathbf{E}$ specified in Cartesian coordinates as

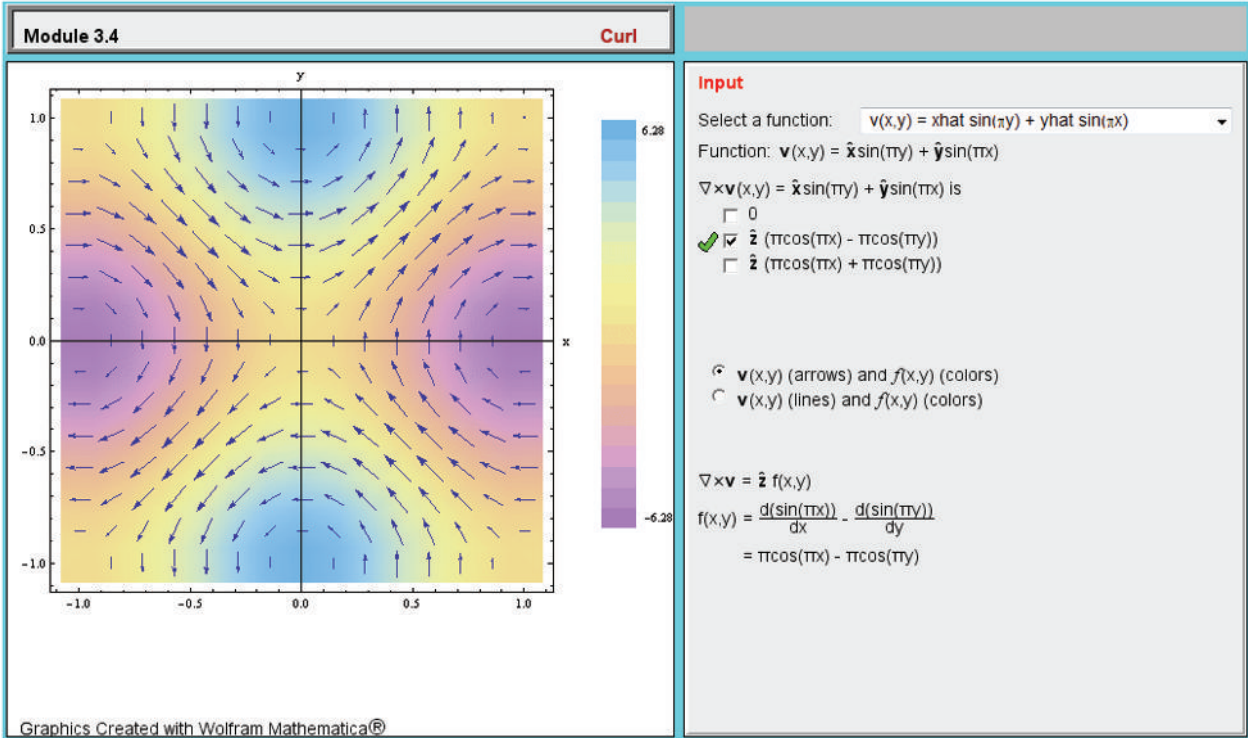
$$\mathbf{E} = \hat{\mathbf{x}} E_x + \hat{\mathbf{y}} E_y + \hat{\mathbf{z}} E_z, \tag{3.111}$$

the Laplacian of $\mathbf{E}$ is

$$\nabla^2 \mathbf{E} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \mathbf{E} = \hat{\mathbf{x}} \nabla^2 E_x + \hat{\mathbf{y}} \nabla^2 E_y + \hat{\mathbf{z}} \nabla^2 E_z. \tag{3.112}$$

Thus, in Cartesian coordinates the Laplacian of a vector is a vector whose components are equal to the Laplacians of the vector components. Through direct substitution, it can be

模块 3.4 旋度 选择一个向量 $\mathbf{f}(x, y)$ ，评估其旋度，并将两者显示在 x - y 平面中。



3-7 拉普拉斯算子

在后面的章节中，我们有时会处理涉及标量和向量运算的多种组合的问题。经常遇到的组合是标量梯度的散度。对于在笛卡尔坐标系中定义的标量函数 V ，其梯度为

$$\nabla V = \hat{\mathbf{x}} \frac{\partial V}{\partial x} + \hat{\mathbf{y}} \frac{\partial V}{\partial y} + \hat{\mathbf{z}} \frac{\partial V}{\partial z} = \hat{\mathbf{x}} A_x + \hat{\mathbf{y}} A_y + \hat{\mathbf{z}} A_z = \mathbf{A}, \tag{3.108}$$

其中我们定义了一个向量 $\mathbf{A}$ ，其分量为 $A_x = \partial V / \partial x$ 、 $A_y = \partial V / \partial y$ 和 $A_z = \partial V / \partial z$ 。 ∇V 的散度为

$$\begin{aligned} \nabla \cdot (\nabla V) &= \nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \\ &= \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}. \end{aligned} \tag{3.109}$$

为了方便起见， $\nabla \cdot (\nabla V)$ 被称为 V 的拉普拉斯算子，记为 $\nabla^2 V$ （符号 ∇^2 发音为 “del square”）。

那是，

$$\nabla^2 V = \nabla \cdot (\nabla V) = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}. \tag{3.110}$$

正如我们从等式中可以看到的那样。（3.110），标量函数的拉普拉斯算子是标量。附录 C 中给出了柱坐标和球坐标中 $\nabla^2 V$ 的表达式。

标量的拉普拉斯算子可用于定义向量的拉普拉斯算子。对于笛卡尔坐标中指定的向量 $\mathbf{E}$

$$\mathbf{E} = \hat{\mathbf{x}} E_x + \hat{\mathbf{y}} E_y + \hat{\mathbf{z}} E_z, \tag{3.111}$$

$\mathbf{E}$ 的拉普拉斯算子是

$$\nabla^2 \mathbf{E} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \mathbf{E} = \hat{\mathbf{x}} \nabla^2 E_x + \hat{\mathbf{y}} \nabla^2 E_y + \hat{\mathbf{z}} \nabla^2 E_z. \tag{3.112}$$

因此，在笛卡尔坐标中，向量的拉普拉斯算子是其分量等于向量分量的拉普拉斯算子的向量。通过直接替换，可以

shown that

$$\nabla^2 \mathbf{E} = \nabla(\nabla \cdot \mathbf{E}) - \nabla \times (\nabla \times \mathbf{E}).$$
(3.113)

Concept Question 3-11: What do the magnitude and direction of the gradient of a scalar quantity represent?

Concept Question 3-12: Prove the validity of Eq. (3.84c) in Cartesian coordinates.

Concept Question 3-13: What is the physical meaning of the divergence of a vector field?

Concept Question 3-14: If a vector field is solenoidal at a given point in space, does it necessarily follow that the vector field is zero at that point? Explain.

Concept Question 3-15: What is the meaning of the transformation provided by the divergence theorem?

Concept Question 3-16: How is the curl of a vector field at a point related to the circulation of the vector field?

Concept Question 3-17: What is the meaning of the transformation provided by Stokes’s theorem?

Concept Question 3-18: When is a vector field “conservative”?

Chapter 3 Summary

Concepts

- Vector algebra governs the laws of addition, subtraction, and multiplication of vectors, and vector calculus encompasses the laws of differentiation and integration of vectors.
- In a right-handed orthogonal coordinate system, the three base vectors are mutually perpendicular to each other at any point in space, and the cyclic relations governing the cross products of the base vectors obey the right-hand rule.
- The dot product of two vectors produces a scalar, whereas the cross product of two vectors produces another vector.
- A vector expressed in a given coordinate system can be expressed in another coordinate system through the use of transformation relations linking the two coordinate systems.
- The fundamental differential functions in vector calculus are the gradient, the divergence, and the curl.
- The gradient of a scalar function is a vector whose magnitude is equal to the maximum rate of increasing

- change of the scalar function per unit distance, and its direction is along the direction of maximum increase.
- The divergence of a vector field is a measure of the net outward flux per unit volume through a closed surface surrounding the unit volume.
- The divergence theorem transforms the volume integral of the divergence of a vector field into a surface integral of the field’s flux through a closed surface surrounding the volume.
- The curl of a vector field is a measure of the circulation of the vector field per unit area Δs , with the orientation of Δs chosen such that the circulation is maximum.
- Stokes’s theorem transforms the surface integral of the curl of a vector field into a line integral of the field over a contour that bounds the surface.
- The Laplacian of a scalar function is defined as the divergence of the gradient of that function.

表明

$$\nabla^2 \mathbf{E} = \nabla(\nabla \cdot \mathbf{E}) - \nabla \times (\nabla \times \mathbf{E}).$$
(3.113)

概念问题3-11：标量梯度的大小和方向代表什么？概念问题 3-12：证明方程的有效性。(3.84c) 在笛卡尔坐标中。概念问题3-13：矢量场散度的物理意义是什么？

Concept Question 3-14: If a vector field is solenoidal at a given point in space, does it necessarily follow that the vector field is zero at that point? Explain.

Concept Question 3-15: What is the meaning of the transformation provided by the divergence theorem?

Concept Question 3-16: How is the curl of a vector field at a point related to the circulation of the vector field?

Concept Question 3-17: What is the meaning of the transformation provided by Stokes’s theorem?

Concept Question 3-18: When is a vector field “conservative”?

第三章小结

概念

- 向量代数控制向量的加法、减法和乘法定律，向量微积分包含向量的微分和积分定律。
- 在右手正交坐标系中，三个基向量在空间中的任意点都相互垂直，并且控制基向量叉积的循环关系遵循右手定则。
- 两个向量的点积产生一个标量，而两个向量的叉积产生另一个向量。
- 通过使用连接两个坐标系的变换关系，可以在给定坐标系中表达的向量在另一个坐标系中表达。
- 矢量微积分中的基本微分函数是梯度、散度和旋度。
- 标量函数的梯度是一个向量，其大小等于最大增加率

- 标量函数每单位距离的变化，其方向是沿着增量最大的方向。
- 矢量场的散度是通过单位体积周围的封闭表面的每单位体积的净向外通量的度量。
- 散度定理将矢量场散度的体积积分转换为通过围绕体积的闭合表面的场通量的表面积分。
- 矢量场的旋度是每单位面积 Δs 矢量场循环的度量，选择 Δs 的方向以使循环最大。
- 斯托克斯定理将矢量场旋度的表面积分转换为该场在限定该表面的轮廓上的线积分。
- 标量函数的拉普拉斯算子定义为该函数梯度的散度。

Mathematical and Physical Models

Distance Between Two Points

$$d = [(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]^{1/2}$$

$$d = [r_2^2 + r_1^2 - 2r_1r_2 \cos(\phi_2 - \phi_1) + (z_2 - z_1)^2]^{1/2}$$

$$d = \{R_2^2 + R_1^2 - 2R_1R_2[\cos \theta_2 \cos \theta_1 + \sin \theta_1 \sin \theta_2 \cos(\phi_2 - \phi_1)]\}^{1/2}$$

Coordinate Systems Table 3-1

Coordinate Transformations Table 3-2

Vector Products

$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta_{AB}$$

$$\mathbf{A} \times \mathbf{B} = \hat{\mathbf{n}} AB \sin \theta_{AB}$$

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$$

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

Divergence Theorem

$$\int_v \nabla \cdot \mathbf{E} \, d\mathcal{V} = \oint_S \mathbf{E} \cdot d\mathbf{s}$$

Vector Operators

$$\nabla T = \hat{\mathbf{x}} \frac{\partial T}{\partial x} + \hat{\mathbf{y}} \frac{\partial T}{\partial y} + \hat{\mathbf{z}} \frac{\partial T}{\partial z}$$

$$\nabla \cdot \mathbf{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}$$

$$\nabla \times \mathbf{B} = \hat{\mathbf{x}} \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) + \hat{\mathbf{y}} \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) + \hat{\mathbf{z}} \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right)$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

(see Appendix C for cylindrical and spherical coordinates)

Stokes’s Theorem

$$\int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{s} = \oint_C \mathbf{B} \cdot d\mathbf{l}$$

Important Terms

Provide definitions or explain the meaning of the following terms:

azimuth angle	divergence operator	radial distance r
base vectors	divergence theorem	range R
Cartesian coordinate system	divergenceless	right-hand rule
circulation of a vector	dot product	scalar product
conservative field	field lines	scalar quantity
cross product	flux density	simple product
curl operator	flux lines	solenoidal field
cylindrical coordinate system	gradient operator	spherical coordinate system
differential area vector	irrotational field	Stokes’s theorem
differential length vector	Laplacian operator	vector product
differential volume	magnitude	vector quantity
directional derivative	orthogonal coordinate system	unit vector
distance vector	position vector	zenith angle

PROBLEMS

Section 3-1: Vector Algebra

*3.1 Vector $\mathbf{A}$ starts at point $(1, -1, -3)$ and ends at point $(2, -1, 0)$. Find a unit vector in the direction of $\mathbf{A}$.

* Answer(s) available in Appendix E.

3.2 Given vectors $\mathbf{A} = \hat{\mathbf{x}}2 - \hat{\mathbf{y}}3 + \hat{\mathbf{z}}$, $\mathbf{B} = \hat{\mathbf{x}}2 - \hat{\mathbf{y}} + \hat{\mathbf{z}}3$, and $\mathbf{C} = \hat{\mathbf{x}}4 + \hat{\mathbf{y}}2 - \hat{\mathbf{z}}2$, show that $\mathbf{C}$ is perpendicular to both $\mathbf{A}$ and $\mathbf{B}$.

*3.3 In Cartesian coordinates, the three corners of a triangle are $P_1 = (0, 4, 4)$, $P_2 = (4, -4, 4)$, and $P_3 = (2, 2, -4)$. Find the area of the triangle.

数学与物理模型

两点间距离

$$d = [(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]^{1/2}$$

$$d = [r_2^2 + r_1^2 - 2r_1r_2 \cos(\phi_2 - \phi_1) + (z_2 - z_1)^2]^{1/2}$$

$$d = \{R_2^2 + R_1^2 - 2R_1R_2[\cos \theta_2 \cos \theta_1 + \sin \theta_1 \sin \theta_2 \cos(\phi_2 - \phi_1)]\}^{1/2}$$

坐标系 表3-1

坐标变换 表3-2

矢量积运算

$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta_{AB}$$

$$\mathbf{A} \times \mathbf{B} = \hat{\mathbf{n}} AB \sin \theta_{AB}$$

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$$

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

散度定理

$$\int_v \nabla \cdot \mathbf{E} \, d\mathcal{V} = \oint_S \mathbf{E} \cdot d\mathbf{s}$$

重要术语

请给出下列术语的定义或解释其含义。

方位角	散度算子	径向距离 r
基矢量	散度定理	球径 R
直角坐标系	无散性	右手定则
矢量的环量	点积	标量积
保守场	场线	标量
叉积	通量密度	标量与矢量的乘积
旋度算子	通量线	管形场（无散场）
圆柱坐标系	梯度算子	球坐标系
有向面元	无旋场	斯托克斯定理
有向线元	拉普拉斯算子	矢量积
体积元	模（大小）	矢量
方向导数	正交坐标系	单位矢量
两点间的位移矢量	位置矢量	天顶角

问题

第 3-1 节：向量代数

*3.1 向量 $\mathbf{A}$ 从点 $(1, -1, -3)$ 开始，到点 $(2, -1, 0)$ 结束。求 $\mathbf{A}$ 方向的单位向量。

* 标记的习题在附录 E 提供答案。

矢量微分算子

$$\nabla T = \hat{\mathbf{x}} \frac{\partial T}{\partial x} + \hat{\mathbf{y}} \frac{\partial T}{\partial y} + \hat{\mathbf{z}} \frac{\partial T}{\partial z}$$

$$\nabla \cdot \mathbf{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}$$

$$\nabla \times \mathbf{B} = \hat{\mathbf{x}} \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) + \hat{\mathbf{y}} \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) + \hat{\mathbf{z}} \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right)$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

圆柱坐标与球坐标下的表达式见附录 C。

斯托克斯定理

$$\int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{s} = \oint_C \mathbf{B} \cdot d\mathbf{l}$$

3.2 给定向量 $\mathbf{A} = \hat{\mathbf{x}}2 - \hat{\mathbf{y}}3 + \hat{\mathbf{z}}$ 、 $\mathbf{B} = \hat{\mathbf{x}}2 - \hat{\mathbf{y}} + \hat{\mathbf{z}}3$ 和 $\mathbf{C} = \hat{\mathbf{x}}4 + \hat{\mathbf{y}}2 - \hat{\mathbf{z}}2$ ，表明 $\mathbf{C}$ 垂直于 $\mathbf{A}$ 和 $\mathbf{B}$ 。

*3.3 在笛卡尔坐标系中，三角形的三个角为 $P_1 = (0, 4, 4)$ ， $P_2 = (4, -4, 4)$ 和 $P_3 = (2, 2, -4)$ 。求三角形的面积。

- 3.4** Given $\mathbf{A} = \hat{x}2 - \hat{y}3 + \hat{z}1$ and $\mathbf{B} = \hat{x}B_x + \hat{y}2 + \hat{z}B_z$:
- (a) Find B_x and B_z if $\mathbf{A}$ is parallel to $\mathbf{B}$.
 - (b) Find a relation between B_x and B_z if $\mathbf{A}$ is perpendicular to $\mathbf{B}$.

3.5 Given vectors $\mathbf{A} = \hat{x} + \hat{y}2 - \hat{z}3$, $\mathbf{B} = \hat{x}2 - \hat{y}4$, and $\mathbf{C} = \hat{y}2 - \hat{z}4$, find the following:

- (a)** A and $\hat{a}$
- The component of $\mathbf{B}$ along $\mathbf{C}$
- θ_{AC}
- $\mathbf{A} \times \mathbf{C}$
- $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$
- $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$
- $\hat{x} \times \mathbf{B}$
- $(\mathbf{A} \times \hat{y}) \cdot \hat{z}$

3.6 Given vectors $\mathbf{A} = \hat{x}2 - \hat{y} + \hat{z}3$ and $\mathbf{B} = \hat{x}3 - \hat{z}2$, find a vector $\mathbf{C}$ whose magnitude is 9 and whose direction is perpendicular to both $\mathbf{A}$ and $\mathbf{B}$.

3.7 Given $\mathbf{A} = \hat{x}(x + 2y) - \hat{y}(y + 3z) + \hat{z}(3x - y)$, determine a unit vector parallel to $\mathbf{A}$ at point $P = (1, -1, 2)$.

- 3.8** By expansion in Cartesian coordinates, prove:
- (a) The relation for the scalar triple product given by Eq. (3.29).
 - (b) The relation for the vector triple product given by Eq. (3.33).

3.9 Find an expression for the unit vector directed toward the origin from an arbitrary point on the line described by $x = 1$ and $z = -2$.

3.10 Find an expression for the unit vector directed toward the point P located on the z axis at a height h above the x - y plane from an arbitrary point $Q = (x, y, -5)$ in the plane $z = -5$.

3.11 Find a unit vector parallel to either direction of the line described by

$$2x + z = 4$$

3.12 Two lines in the x - y plane are described by the following expressions:

$$\begin{array}{ll} \text{Line 1} & x + 2y = -6 \\ \text{Line 2} & 3x + 4y = 8 \end{array}$$

Use vector algebra to find the smaller angle between the lines at their intersection point.

3.13 A given line is described by

$$x + 2y = 4.$$

Vector $\mathbf{A}$ starts at the origin and ends at point P on the line such that $\mathbf{A}$ is orthogonal to the line. Find an expression for $\mathbf{A}$.

3.14 Show the following given two vectors $\mathbf{A}$ and $\mathbf{B}$:

- (a) The vector $\mathbf{C}$ defined as the vector component of $\mathbf{B}$ in the direction of $\mathbf{A}$ is given by

$$\mathbf{C} = \hat{\mathbf{a}}(\mathbf{B} \cdot \hat{\mathbf{a}}) = \frac{\mathbf{A}(\mathbf{B} \cdot \mathbf{A})}{|\mathbf{A}|^2},$$

where $\hat{\mathbf{a}}$ is the unit vector of $\mathbf{A}$.

- (b) The vector $\mathbf{D}$ defined as the vector component of $\mathbf{B}$ perpendicular to $\mathbf{A}$ is given by

$$\mathbf{D} = \mathbf{B} - \frac{\mathbf{A}(\mathbf{B} \cdot \mathbf{A})}{|\mathbf{A}|^2}$$

3.15 A certain plane is described by

$$2x + 3y + 4z = 16$$

Find the unit vector normal to the surface in the direction away from the origin.

3.16 Given $\mathbf{B} = \hat{x}(z - 3y) + \hat{y}(2x - 3z) - \hat{z}(x + y)$, find a unit vector parallel to $\mathbf{B}$ at point $P = (1, 0, -1)$.

3.17 Find a vector $\mathbf{G}$ whose magnitude is 4 and whose direction is perpendicular to both vectors $\mathbf{E}$ and $\mathbf{F}$, where $\mathbf{E} = \hat{x} + \hat{y}2 - \hat{z}2$ and $\mathbf{F} = \hat{y}3 - \hat{z}6$.

3.18 A given line is described by the equation:

$$y = x - 1.$$

Vector $\mathbf{A}$ starts at point $P_1 = (0, 2)$ and ends at point P_2 on the line, at which $\mathbf{A}$ is orthogonal to the line. Find an expression for $\mathbf{A}$.

3.19 Vector field $\mathbf{E}$ is given by

$$\mathbf{E} = \hat{\mathbf{r}}5R \cos \theta - \hat{\boldsymbol{\theta}}\frac{12}{R} \sin \theta \cos \phi + \hat{\boldsymbol{\phi}}3 \sin \phi.$$

Determine the component of $\mathbf{E}$ tangential to the spherical surface $R = 2$ at point $P = (2, 30^\circ, 60^\circ)$.

3.20 When sketching or demonstrating the spatial variation of a vector field, we often use arrows, as in **Fig. P3.20**, wherein the length of the arrow is made to be proportional to the strength of the field and the direction of the arrow is the same as that of the field's. The sketch shown in **Fig. P3.20**, which represents the vector field $\mathbf{E} = \hat{\mathbf{r}}r$, consists of arrows pointing radially away from the origin and their lengths increasing linearly in proportion to their distance away from the origin.

- 3.4 给定 $\mathbf{A} = \hat{x}2 - \hat{y}3 + \hat{z}1$ 和 $\mathbf{B} = \hat{x}B_x + \hat{y}2 + \hat{z}B_z$:
- (a) 如果 $\mathbf{A}$ 与 $\mathbf{B}$ 平行, 求 B_x 和 B_z 。
 - (b) 如果 $\mathbf{A}$ 垂直于 $\mathbf{B}$, 求 B_x 和 B_z 之间的关系。

3.5 给定向量 $\mathbf{A} = \hat{x} + \hat{y}2 - \hat{z}3$ 、 $\mathbf{B} = \hat{x}2 - \hat{y}4$ 和 $\mathbf{C} = \hat{y}2 - \hat{z}4$, 求出以下结果:

- (a) A 和 $\hat{a}$
- (b) $\mathbf{B}$ 沿 $\mathbf{C}$ 的分量
- (c) θ_{AC}
- (d) $\mathbf{A} \times \mathbf{C}$
- (e) $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$
- (f) $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$
- (g) $\hat{x} \times \mathbf{B}$
- (h) $(\mathbf{A} \times \hat{y}) \cdot \hat{z}$

3.6 给定向量 $\mathbf{A} = \hat{x}2 - \hat{y} + \hat{z}3$ 和 $\mathbf{B} = \hat{x}3 - \hat{z}2$, 找到一个向量 $\mathbf{C}$, 其大小为 9, 方向垂直于 $\mathbf{A}$ 和 $\mathbf{B}$ 。

3.7 给定 $\mathbf{A} = \hat{x}(x + 2y) - \hat{y}(y + 3z) + \hat{z}(3x - y)$, 确定在点 $P = (1, -1, 2)$ 处与 $\mathbf{A}$ 平行的单位向量。

3.8 通过笛卡尔坐标展开, 证明:

- (a) 等式给出的标量三重积的关系。(3.29)。
- (b) 由方程给出的向量三重积的关系。(3.33)。

3.9 找出从 $x = 1$ 和 $z = -2$ 描述的直线上的任意点指向原点的单位向量的表达式。

3.10 找到从平面 $z = -5$ 中的任意点 $Q = (x, y, -5)$ 指向位于 x - y 平面上方高度 h 的 z 轴上的点 P 的单位向量的表达式。

3.11 求一个与下式描述的线的任一方向平行的单位向量

$$2x + z = 4$$

3.12 x - y 平面中的两条直线由以下表达式描述:

$$\begin{array}{ll} \text{Line 1} & x + 2y = -6 \\ \text{Line 2} & 3x + 4y = 8 \end{array}$$

使用向量代数找到线交点处的较小角度。

3.13 给定的线由下式描述

$$x + 2y = 4.$$

矢量 $\mathbf{A}$ 从原点开始, 到线上的点 P 结束, 使得 $\mathbf{A}$ 与该线正交。找出 $\mathbf{A}$ 的表达式。

3.14 显示以下给定的两个向量 $\mathbf{A}$ 和 $\mathbf{B}$:

- (a) 向量 $\mathbf{C}$ 定义为 $\mathbf{B}$ 在 $\mathbf{A}$ 方向上的向量分量, 由下式给出

$$\mathbf{C} = \hat{\mathbf{a}}(\mathbf{B} \cdot \hat{\mathbf{a}}) = \frac{\mathbf{A}(\mathbf{B} \cdot \mathbf{A})}{|\mathbf{A}|^2},$$

其中 $\hat{\mathbf{a}}$ 是 $\mathbf{A}$ 的单位向量。

- (b) 向量 $\mathbf{D}$ 定义为垂直于 $\mathbf{A}$ 的 $\mathbf{B}$ 的向量分量, 由下式给出

$$\mathbf{D} = \mathbf{B} - \frac{\mathbf{A}(\mathbf{B} \cdot \mathbf{A})}{|\mathbf{A}|^2}$$

3.15 某个平面被描述为

$$2x + 3y + 4z = 16$$

求远离原点方向上垂直于表面的单位向量。

3.16 给定 $\mathbf{B} = \hat{x}(z - 3y) + \hat{y}(2x - 3z) - \hat{z}(x + y)$, 求在点 $P = (1, 0, -1)$ 处平行于 $\mathbf{B}$ 的单位向量。

3.17 求一个向量 $\mathbf{G}$, 其大小为 4, 方向垂直于向量 $\mathbf{E}$ 和 $\mathbf{F}$, 其中 $\mathbf{E} = \hat{x} + \hat{y}2 - \hat{z}2$ 和 $\mathbf{F} = \hat{y}3 - \hat{z}6$ 。

3.18 给定的直线由以下方程描述:

$$y = x - 1.$$

矢量 $\mathbf{A}$ 从线上的点 $P_1 = (0, 2)$ 开始, 到点 P_2 结束, 在该点 $\mathbf{A}$ 与线正交。找出 $\mathbf{A}$ 的表达式。

3.19 向量场 $\mathbf{E}$ 由下式给出

$$\mathbf{E} = \hat{\mathbf{r}}5R \cos \theta - \hat{\boldsymbol{\theta}}\frac{12}{R} \sin \theta \cos \phi + \hat{\boldsymbol{\phi}}3 \sin \phi.$$

确定 $\mathbf{E}$ 在点 $P = (2, 30^\circ, 60^\circ)$ 处与球面 $R = 2$ 相切的分量。

3.20 当描绘或演示矢量场的空间变化时, 我们经常使用箭头, 如图 P3.20 所示, 其中箭头的长度与场的强度成正比, 并且箭头的方向与场的方向相同。图 P3.20 所示的草图表示矢量场 $\mathbf{E} = \hat{\mathbf{r}}r$, 由远离原点的径向箭头组成, 它们的长度与远离原点的距离成正比线性增加。

Using this arrow representation, sketch each of the following vector fields:

- (a) $\mathbf{E}_1 = -\hat{x}y$
- (b) $\mathbf{E}_2 = \hat{y}x$
- (c) $\mathbf{E}_3 = \hat{x}x + \hat{y}y$
- (d) $\mathbf{E}_4 = \hat{x}x + \hat{y}2y$
- (e) $\mathbf{E}_5 = \hat{\phi}r$
- (f) $\mathbf{E}_6 = \hat{r} \sin \phi$

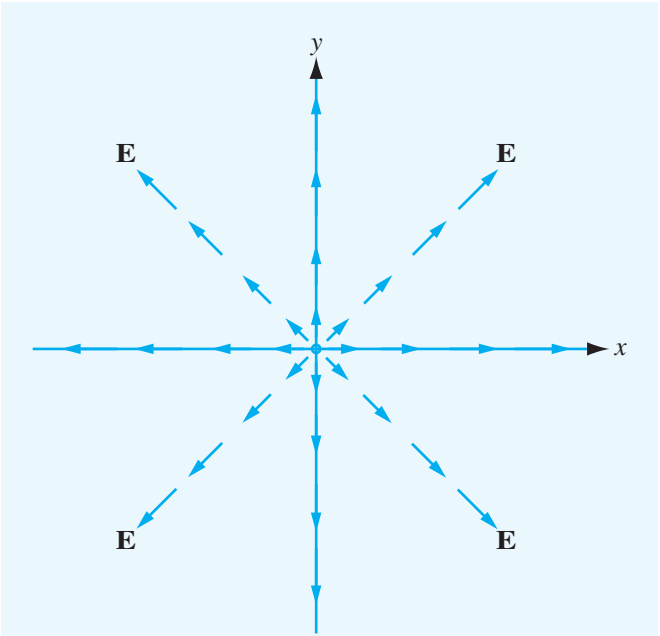


Figure P3.20 Arrow representation for vector field $\mathbf{E} = \hat{r}r$ (Problem 3.20).

3.21 Use arrows to sketch each of the following vector fields:

- (a) $\mathbf{E}_1 = \hat{x}x - \hat{y}y$
- (b) $\mathbf{E}_2 = -\hat{\phi}$
- (c) $\mathbf{E}_3 = \hat{y} (1/x)$
- (d) $\mathbf{E}_4 = \hat{r} \cos \phi$

Sections 3-2 and 3-3: Coordinate Systems

3.22 Convert the coordinates of the following points from Cartesian to cylindrical and spherical coordinates:

- * (a) $P_1 = (1, 2, 0)$
- (b) $P_2 = (0, 0, 2)$
- (c) $P_3 = (1, 1, 3)$

- * (d) $P_4 = (-2, 2, -2)$

3.23 Convert the coordinates of the following points from cylindrical to Cartesian coordinates:

- (a) $P_1 = (2, \pi/4, -3)$
- (b) $P_2 = (3, 0, -2)$
- (c) $P_3 = (4, \pi, 5)$

3.24 Convert the coordinates of the following points from spherical to cylindrical coordinates:

- * (a) $P_1 = (5, 0, 0)$
- (b) $P_2 = (5, 0, \pi)$
- (c) $P_3 = (3, \pi/2, 0)$

3.25 Use the appropriate expression for the differential surface area ds to determine the area of each of the following surfaces:

- (a) $r = 3; 0 \leq \phi \leq \pi/3; -2 \leq z \leq 2$
- (b) $2 \leq r \leq 5; \pi/2 \leq \phi \leq \pi; z = 0$
- * (c) $2 \leq r \leq 5; \phi = \pi/4; -2 \leq z \leq 2$
- (d) $R = 2; 0 \leq \theta \leq \pi/3; 0 \leq \phi \leq \pi$
- (e) $0 \leq R \leq 5; \theta = \pi/3; 0 \leq \phi \leq 2\pi$

Also sketch the outline of each surface.

3.26 Find the volumes described by the following:

- * (a) $2 \leq r \leq 5; \pi/2 \leq \phi \leq \pi; 0 \leq z \leq 2$
- (b) $0 \leq R \leq 5; 0 \leq \theta \leq \pi/3; 0 \leq \phi \leq 2\pi$

Also sketch the outline of each volume.

3.27 A section of a sphere is described by $0 \leq R \leq 2, 0 \leq \theta \leq 90^\circ$, and $30^\circ \leq \phi \leq 90^\circ$. Find the following:

- (a) The surface area of the spherical section.
- (b) The enclosed volume.

Also sketch the outline of the section.

3.28 A vector field is given in cylindrical coordinates by

$$\mathbf{E} = \hat{r}r \cos \phi + \hat{\phi}r \sin \phi + \hat{z}z^2.$$

Point $P = (2, \pi, 3)$ is located on the surface of the cylinder described by $r = 2$. At point P , find:

- (a) The vector component of $\mathbf{E}$ perpendicular to the cylinder.
- (b) The vector component of $\mathbf{E}$ tangential to the cylinder.

3.29 At a given point in space, vectors $\mathbf{A}$ and $\mathbf{B}$ are given in spherical coordinates by

$$\mathbf{A} = \hat{\mathbf{R}}4 + \hat{\boldsymbol{\theta}}2 - \hat{\boldsymbol{\phi}},$$

$$\mathbf{B} = -\hat{\mathbf{R}}2 + \hat{\boldsymbol{\phi}}3.$$

使用此箭头表示， 绘制以下每个向量场的草图：

- (a) $\mathbf{E}_1 = -\hat{x}y$ (b) $\mathbf{E}_2 = \hat{y}x$ (c) $\mathbf{E}_3 = \hat{x}x + \hat{y}y$ (d) $\mathbf{E}_4 = \hat{x}x + \hat{y}2y$ (e) $\mathbf{E}_5 = \hat{\phi}r$ (f) $\mathbf{E}_6 = \hat{r} \sin \phi$

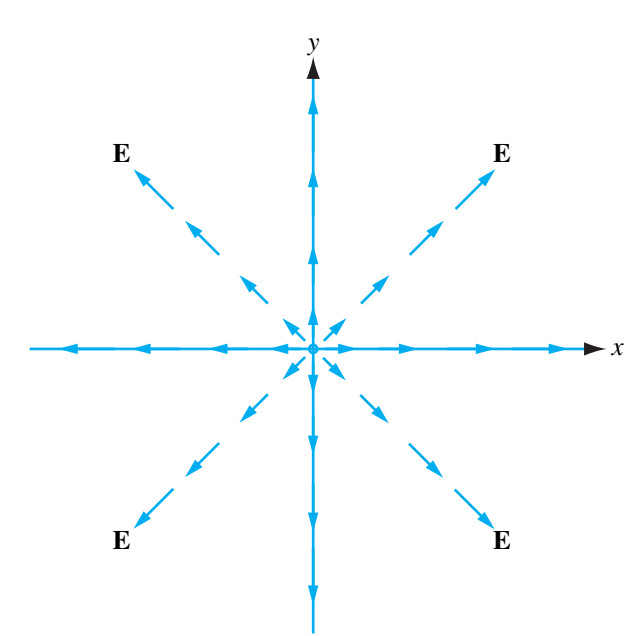


图 P3.20 矢量场 $\mathbf{E} = \hat{r}r$ 的箭头表示（问题 3.20）。

3.21 用箭头画出以下每个向量场：

- (a) $\mathbf{E}_1 = \hat{x}x - \hat{y}y$
- (b) $\mathbf{E}_2 = -\hat{\phi}$ (c) $\mathbf{E}_3 = \hat{y} (1/x)$ (d) $\mathbf{E}_4 = \hat{r} \cos \phi$

第 3-2 节和第 3-3 节：坐标系

3.22 将下列点的坐标从笛卡尔坐标转换为柱坐标和球坐标：

- * (a) $P_1 = (1, 2, 0)$ (b) $P_2 = (0, 0, 2)$ (c) $P_3 = (1, 1, 3)$
- () () ()

- * (d) $P_4 = (-2, 2, -2)$

3.23 将下列点的坐标从圆柱坐标转换为笛卡尔坐标：

- (a) $P_1 = (2, \pi/4, -3)$ (b) $P_2 = (3, 0, -2)$ (c) $P_3 = (4, \pi, 5)$
- () () ()

3.24 将下列点的坐标从球坐标转换为柱坐标：

- * (a) $P_1 = (5, 0, 0)$ (b) $P_2 = (5, 0, \pi)$ (c) $P_3 = (3, \pi/2, 0)$
- () () ()

3.25 使用微分表面积 ds 的适当表达式来确定以下每个表面的面积：

- (a) $r = 3; 0 \leq \phi \leq \pi/3; -2 \leq z \leq 2$ (
- b) $2 \leq r \leq 5; \pi/2 \leq \phi \leq \pi; z = 0$

- * (c) $2 \leq r \leq 5; \phi = \pi/4; -2 \leq z \leq 2$ (
- d) $R = 2; 0 \leq \theta \leq \pi/3; 0 \leq \phi \leq \pi$ e) $0 \leq R \leq 5; \theta = \pi/3; 0 \leq \phi \leq 2\pi$

还勾勒出每个表面的轮廓。

3.26 找到下列描述的卷：

- * (a) $2 \leq r \leq 5; \pi/2 \leq \phi \leq \pi; 0 \leq z \leq 2$ (
- b) $0 \leq R \leq 5; 0 \leq \theta \leq \pi/3; 0 \leq \phi \leq 2\pi$

还勾勒出每卷的轮廓。

3.27 球体的截面由 $0 \leq R \leq 2, 0 \leq \theta \leq 90^\circ$ 和 $30^\circ \leq \phi \leq 90^\circ$ 描述。找到以下内容：

- (a) 球形截面的表面积。
- (b) 封闭体积。

还勾勒出该部分的轮廓。

3.28 向量场在柱坐标系中由下式给出

$$\mathbf{E} = \hat{r}r \cos \phi + \hat{\phi}r \sin \phi + \hat{z}z^2.$$

点 $P = (2, \pi, 3)$ 位于 $r = 2$ 描述的圆柱体表面上。在点 P 处，找到：

- (a) $\mathbf{E}$ 垂直于圆柱体的矢量分量。
- (b) $\mathbf{E}$ 与圆柱体相切的矢量分量。

3.29 在空间中的给定点，向量 $\mathbf{A}$ 和 $\mathbf{B}$ 在球坐标中给出：

$$\mathbf{A} = \hat{\mathbf{R}}4 + \hat{\boldsymbol{\theta}}2 - \hat{\boldsymbol{\phi}},$$

$$\mathbf{B} = -\hat{\mathbf{R}}2 + \hat{\boldsymbol{\phi}}3.$$

Find:

- (a) The scalar component, or projection, of $\mathbf{B}$ in the direction of $\mathbf{A}$.
- (b) The vector component of $\mathbf{B}$ in the direction of $\mathbf{A}$.
- (c) The vector component of $\mathbf{B}$ perpendicular to $\mathbf{A}$.

*3.30 Given vectors

$$\mathbf{A} = \hat{\mathbf{r}}(\cos \phi + 3z) - \hat{\boldsymbol{\phi}}(2r + 4 \sin \phi) + \hat{\mathbf{z}}(r - 2z),$$

$$\mathbf{B} = -\hat{\mathbf{r}} \sin \phi + \hat{\mathbf{z}} \cos \phi,$$

find

- (a) θ_{AB} at $(2, \pi/2, 0)$
- (b) A unit vector perpendicular to both $\mathbf{A}$ and $\mathbf{B}$ at $(2, \pi/3, 1)$

3.31 Find the distance between the following pairs of points:

- (a) $P_1 = (1, 2, 3)$ and $P_2 = (-2, -3, -2)$ in Cartesian coordinates.
- (b) $P_3 = (1, \pi/4, 3)$ and $P_4 = (3, \pi/4, 4)$ in cylindrical coordinates.
- (c) $P_5 = (4, \pi/2, 0)$ and $P_6 = (3, \pi, 0)$ in spherical coordinates.

3.32 Determine the distance between the following pairs of points:

- (a) $P_1 = (1, 1, 2)$ and $P_2 = (0, 2, 3)$
- (b) $P_3 = (2, \pi/3, 1)$ and $P_4 = (4, \pi/2, 3)$
- (c) $P_5 = (3, \pi, \pi/2)$ and $P_6 = (4, \pi/2, \pi)$

3.33 Transform the vector

$$\mathbf{A} = \hat{\mathbf{R}} \sin^2 \theta \cos \phi + \hat{\boldsymbol{\theta}} \cos^2 \phi - \hat{\boldsymbol{\phi}} \sin \phi$$

into cylindrical coordinates and then evaluate it at $P = (2, \pi/2, \pi/2)$.

3.34 Transform the following vectors into cylindrical coordinates and then evaluate them at the indicated points:

- (a) $\mathbf{A} = \hat{\mathbf{x}}(x + y)$ at $P_1 = (1, 2, 3)$
- (b) $\mathbf{B} = \hat{\mathbf{x}}(y - x) + \hat{\mathbf{y}}(x - y)$ at $P_2 = (1, 0, 2)$

* (c) $\mathbf{C} = \hat{\mathbf{x}}y^2/(x^2 + y^2) - \hat{\mathbf{y}}x^2/(x^2 + y^2) + \hat{\mathbf{z}}4$ at $P_3 = (1, -1, 2)$

(d) $\mathbf{D} = \hat{\mathbf{R}} \sin \theta + \hat{\boldsymbol{\theta}} \cos \theta + \hat{\boldsymbol{\phi}} \cos^2 \phi$ at $P_4 = (2, \pi/2, \pi/4)$

* (e) $\mathbf{E} = \hat{\mathbf{R}} \cos \phi + \hat{\boldsymbol{\theta}} \sin \phi + \hat{\boldsymbol{\phi}} \sin^2 \theta$ at $P_5 = (3, \pi/2, \pi)$

3.35 Transform the following vectors into spherical coordinates and then evaluate them at the indicated points:

- (a) $\mathbf{A} = \hat{\mathbf{x}}y^2 + \hat{\mathbf{y}}xz + \hat{\mathbf{z}}4$ at $P_1 = (1, -1, 2)$

(b) $\mathbf{B} = \hat{\mathbf{y}}(x^2 + y^2 + z^2) - \hat{\mathbf{z}}(x^2 + y^2)$ at $P_2 = (-1, 0, 2)$

* (c) $\mathbf{C} = \hat{\mathbf{r}} \cos \phi - \hat{\boldsymbol{\phi}} \sin \phi + \hat{\mathbf{z}} \cos \phi \sin \phi$ at $P_3 = (2, \pi/4, 2)$

(d) $\mathbf{D} = \hat{\mathbf{x}}y^2/(x^2 + y^2) - \hat{\mathbf{y}}x^2/(x^2 + y^2) + \hat{\mathbf{z}}4$ at $P_4 = (1, -1, 2)$

Sections 3-4 to 3-7: Gradient, Divergence, and Curl Operators

3.36 Find the gradient of the following scalar functions:

- (a) $T = 3/(x^2 + z^2)$
- (b) $V = xy^2z^4$
- (c) $U = z \cos \phi / (1 + r^2)$
- (d) $W = e^{-R} \sin \theta$
- (e) $S = 4x^2e^{-z} + y^3$
- (f) $N = r^2 \cos^2 \phi$
- (g) $M = R \cos \theta \sin \phi$

3.37 For each of the following scalar fields, obtain an analytical solution for ∇T and generate a corresponding arrow representation.

- (a) $T = 10 + x$, for $-10 \leq x \leq 10$
- (b) $T = x^2$, for $-10 \leq x \leq 10$
- (b) $T = 100 + xy$, for $-10 \leq x \leq 10$
- (c) $T = x^2y^2$, for $-10 \leq x, y \leq 10$
- (e) $T = 20 + x + y$, for $-10 \leq x, y \leq 10$
- (f) $T = 1 + \sin(\pi x/3)$, for $-10 \leq x \leq 10$
- (g) $T = 1 + \cos(\pi x/3)$, for $-10 \leq x \leq 10$
- (h) $T = 15 + r \cos \phi$, for $\begin{cases} 0 \leq r \leq 10 \\ 0 \leq \phi \leq 2\pi. \end{cases}$
- (i) $T = 15 + r \cos^2 \phi$, for $\begin{cases} 0 \leq r \leq 10 \\ 0 \leq \phi \leq 2\pi. \end{cases}$

*3.38 The gradient of a scalar function T is given by

$$\nabla T = \hat{\mathbf{z}}e^{-3z}$$

If $T = 10$ at $z = 0$, find $T(z)$.

3.39 Follow a procedure similar to that leading to Eq. (3.82) to derive the expression given by Eq. (3.83) for ∇ in spherical coordinates.

*3.40 For the scalar function $V = xy^2 - z^2$, determine its directional derivative along the direction of vector $\mathbf{A} = (\hat{\mathbf{x}} - \hat{\mathbf{y}}\mathbf{z})$ and then evaluate it at $P = (1, -1, 4)$.

3.41 Evaluate the line integral of $\mathbf{E} = \hat{\mathbf{x}}x - \hat{\mathbf{y}}y$ along the segment P_1 to P_2 of the circular path shown in Fig. P3.41.

寻找:

- (a) B 在 A 方向上的标量分量或投影。
- (b) B 在 A 方向上的矢量分量。
- (c) B 垂直于 A 的矢量分量。

*3.30 给定向量

$$\mathbf{A} = \hat{\mathbf{r}}(\cos \phi + 3z) - \hat{\boldsymbol{\phi}}(2r + 4 \sin \phi) + \hat{\mathbf{z}}(r - 2z),$$

$$\mathbf{B} = -\hat{\mathbf{r}} \sin \phi + \hat{\mathbf{z}} \cos \phi,$$

找到

- (a) $(2, \pi/2, 0)$ 处的 θ_{AB} (b) $(2, \pi/3, 1)$ 处垂直于 A 和 B 的单位向量

3.31 求下列两点之间的距离:

- (a) 笛卡尔坐标中的 $P_1 = (1, 2, 3)$ 和 $P_2 = (-2, -3, -2)$ 。
- (b) 柱坐标中的 $P_3 = (1, \pi/4, 3)$ 和 $P_4 = (3, \pi/4, 4)$ 。
- (c) 球坐标中的 $P_5 = (4, \pi/2, 0)$ 和 $P_6 = (3, \pi, 0)$ 。

3.32 确定下列两点之间的距离:

- (a) $P_1 = (1, 1, 2)$ 和 $P_2 = (0, 2, 3)$
- (b) $P_3 = (2, \pi/3, 1)$ 和 $P_4 = (4, \pi/2, 3)$
- (c) $P_5 = (3, \pi, \pi/2)$ 和 $P_6 = (4, \pi/2, \pi)$

3.33 变换向量

$$\mathbf{A} = \hat{\mathbf{R}} \sin^2 \theta \cos \phi + \hat{\boldsymbol{\theta}} \cos^2 \phi - \hat{\boldsymbol{\phi}} \sin \phi$$

转换为柱坐标, 然后在 $P = (2, \pi/2, \pi/2)$ 处对其进行评估。

3.34 将下列向量变换为柱坐标, 然后在指定点处对它们求值:

- (a) $\mathbf{A} = \hat{\mathbf{x}}(x + y)$ at $P_1 = (1, 2, 3)$
- (b) $\mathbf{B} = \hat{\mathbf{x}}(y - x) + \hat{\mathbf{y}}(x - y)$ at $P_2 = (1, 0, 2)$

* (c) $\mathbf{C} = \hat{\mathbf{x}}y^2/(x^2 + y^2) - \hat{\mathbf{y}}x^2/(x^2 + y^2) + \hat{\mathbf{z}}4$ at $P_3 = (1, -1, 2)$

(d) $\mathbf{D} = \hat{\mathbf{R}} \sin \theta + \hat{\boldsymbol{\theta}} \cos \theta + \hat{\boldsymbol{\phi}} \cos^2 \phi$ at $P_4 = (2, \pi/2, \pi/4)$

* (e) $\mathbf{E} = \hat{\mathbf{R}} \cos \phi + \hat{\boldsymbol{\theta}} \sin \phi + \hat{\boldsymbol{\phi}} \sin^2 \theta$ at $P_5 = (3, \pi/2, \pi)$

3.35 将下列向量变换为球坐标, 然后在指定点对其求值:

- (a) $\mathbf{A} = \hat{\mathbf{x}}y^2 + \hat{\mathbf{y}}xz + \hat{\mathbf{z}}4$ at $P_1 = (1, -1, 2)$

(b) $\mathbf{B} = \hat{\mathbf{y}}(x^2 + y^2 + z^2) - \hat{\mathbf{z}}(x^2 + y^2)$ at $P_2 = (-1, 0, 2)$

* (c) $\mathbf{C} = \hat{\mathbf{r}} \cos \phi - \hat{\boldsymbol{\phi}} \sin \phi + \hat{\mathbf{z}} \cos \phi \sin \phi$ at $P_3 = (2, \pi/4, 2)$

(d) $\mathbf{D} = \hat{\mathbf{x}}y^2/(x^2 + y^2) - \hat{\mathbf{y}}x^2/(x^2 + y^2) + \hat{\mathbf{z}}4$ at $P_4 = (1, -1, 2)$

第 3-4 节到 3-7 节: 梯度、散度和卷曲运算符

3.36 求下列标量函数的梯度:

- (a) $T = 3/(x^2 + z^2)$ (b) $V = xy^2z^4$ (c) $U = z \cos \phi / (1 + r^2)$ (d) $W = e^{-R}$

3.37 对于以下每个标量场, 获得 ∇T 的解析解并生成相应的箭头表示。

- (a) $T = 10 + x$, 对于 $-10 \leq x \leq 10$
- (b) $T = x^2$, 对于 $-10 \leq x \leq 10$
- (b) $T = 100 + xy$, 对于 $-10 \leq x \leq 10$
- (c) $T = x^2y^2$, 对于 $-10 \leq x, y \leq 10$
- (e) $T = 20 + x + y$, 对于 $-10 \leq x, y \leq 10$
- (f) $T = 1 + \sin(\pi x/3)$, 对于 $-10 \leq x \leq 10$
- (g) $T = 1 + \cos(\pi x/3)$, for $-10 \leq x \leq 10$

- (h) $T = 15 + r \cos \phi$, 对于 $\begin{cases} 0 \leq r \leq 10 \\ 0 \leq \phi \leq 2\pi. \end{cases}$
- (i) $T = 15 + r \cos^2 \phi$, 对于 $\begin{cases} 0 \leq r \leq 10 \\ 0 \leq \phi \leq 2\pi. \end{cases}$

3.38 标量函数 T 的梯度由下式给出

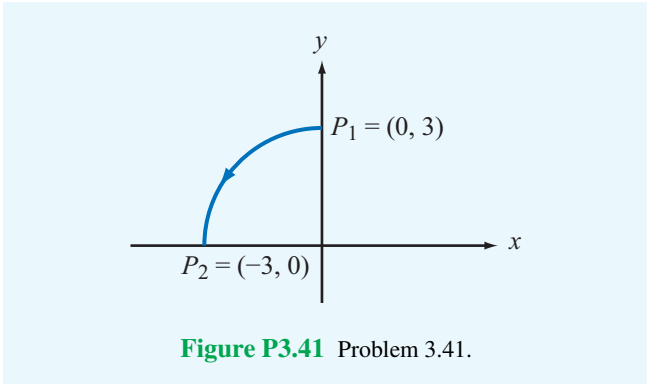
$$\nabla T = \hat{\mathbf{z}}e^{-3z}$$

如果 $T = 10$ 在 $z = 0$ 处, 求 $T(z)$ 。

3.39 遵循类似于公式 3 的程序。 (3.82) 推导出式 (3.82) 给出的表达式。 (3.83) 对于球坐标中的 ∇ 。

*3.40 对于标量函数 $V = xy^2 - z^2$, 确定其沿向量 $\mathbf{A} = (\hat{\mathbf{x}} - \hat{\mathbf{y}}\mathbf{z})$ 方向的方向导数, 然后在 $P = (1, -1, 4)$ 处对其求值。

3.41 计算 $\mathbf{E} = \hat{\mathbf{x}}x - \hat{\mathbf{y}}y$ 沿图 P3.41 所示圆形路径的线段 P_1 至 P_2 的线积分。



(b) $\mathbf{A} = -\hat{x} \sin 2y + \hat{y} \cos 2x$, for $-\pi \leq x, y \leq \pi$

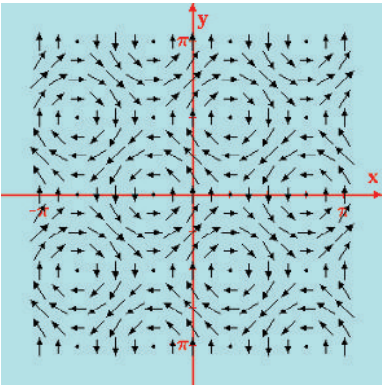


Figure P3.44(b)

(c) $\mathbf{A} = -\hat{x}xy + \hat{y}y^2$, for $-10 \leq x, y \leq 10$

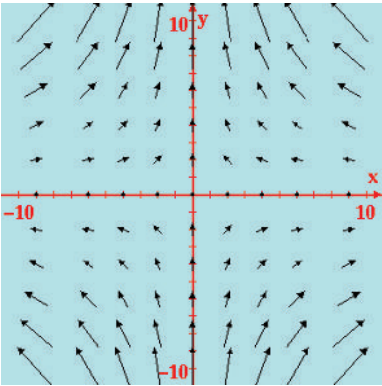


Figure P3.44(c)

(d) $\mathbf{A} = -\hat{x} \cos x + \hat{y} \sin y$, for $-\pi \leq x, y \leq \pi$

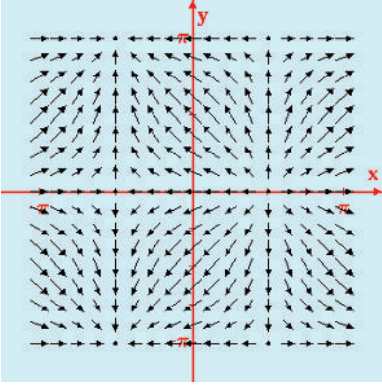


Figure P3.44(d)

3.42 For the scalar function $T = \frac{1}{2} e^{-r/5} \cos \phi$, determine its directional derivative along the radial direction $\hat{\mathbf{r}}$ and then evaluate it at $P = (2, \pi/4, 3)$.

***3.43** For the scalar function $U = \frac{1}{R} \sin^2 \theta$, determine its directional derivative along the range direction $\hat{\mathbf{R}}$ and then evaluate it at $P = (5, \pi/4, \pi/2)$.

3.44 Each of the following vector fields is displayed in **Fig. P3.44** in the form of a vector representation. Determine $\nabla \cdot \mathbf{A}$ analytically and then compare the result with your expectations on the basis of the displayed pattern.

(a) $\mathbf{A} = -\hat{x} \cos x \sin y + \hat{y} \sin x \cos y$, for $-\pi \leq x, y \leq \pi$

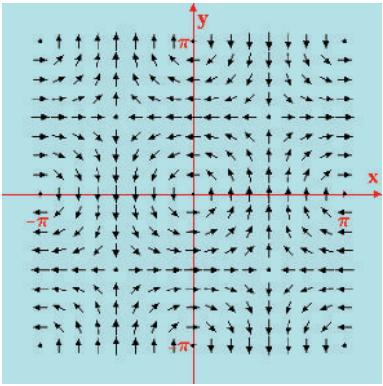


Figure P3.44(a)

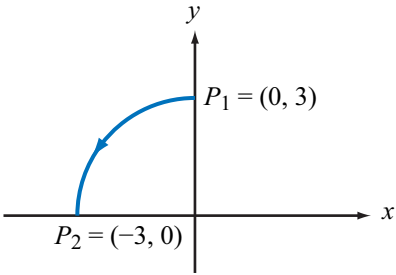
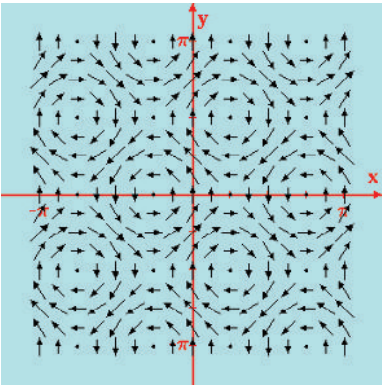


图 P3.41 问题 3.41。

(b) $\mathbf{A} = -\hat{x} \sin 2y + \hat{y} \cos 2x$, 对于 $-\pi \leq x, y \leq \pi$



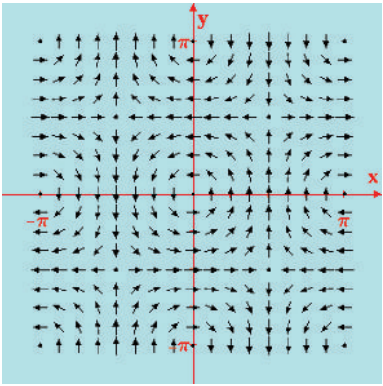
图P3.44(b)

3.42 对于标量函数 $T = \frac{1}{2} e^{-r/5} \cos \phi$ ，确定其沿径向方向 $\hat{\mathbf{r}}$ 的方向导数，然后在 $P = (2, \pi/4, 3)$ 处对其求值。

***3.43** 对于标量函数 $U = \frac{1}{R} \sin^2 \theta$ ，确定其沿范围方向 $\hat{\mathbf{R}}$ 的方向导数，然后在 $P = (5, \pi/4, \pi/2)$ 处对其求值。

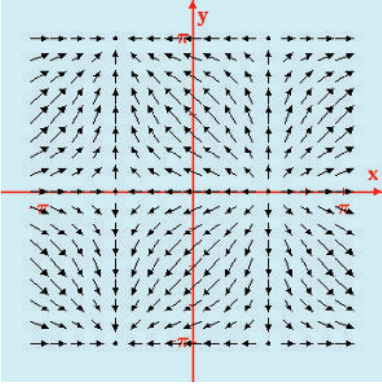
3.44 以下每个向量场都以向量表示的形式显示在图 P3.44 中。通过分析确定 $\nabla \cdot \mathbf{A}$ ，然后根据显示的模式将结果与您的期望进行比较。

(a) $\mathbf{A} = -\hat{x} \cos x \sin y + \hat{y} \sin x \cos y$, 对于 $-\pi \leq x, y \leq \pi$



图P3.44(a)

(d) $\mathbf{A} = -\hat{x} \cos x + \hat{y} \sin y$, 对于 $-\pi \leq x, y \leq \pi$



图P3.44(d)

(e) $\mathbf{A} = \hat{\mathbf{x}}x$, for $-10 \leq x \leq 10$

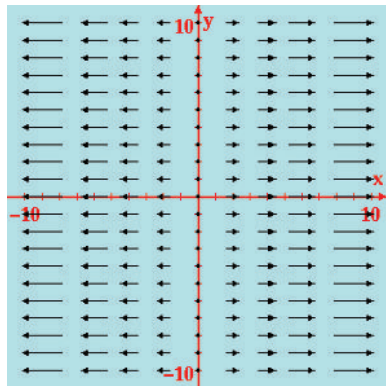


Figure P3.44(e)

(f) $\mathbf{A} = \hat{\mathbf{x}}xy^2$, for $-10 \leq x, y \leq 10$

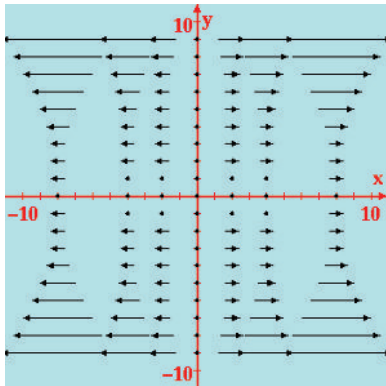


Figure P3.44(f)

(g) $\mathbf{A} = \hat{\mathbf{x}}xy^2 + \hat{\mathbf{y}}x^2y$, for $-10 \leq x, y \leq 10$

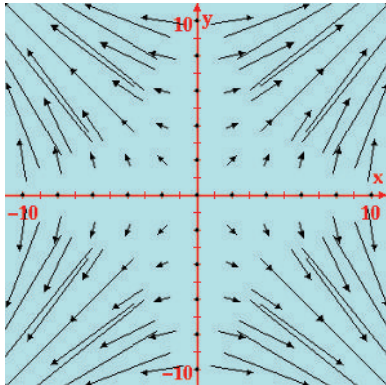


Figure P3.44(g)

(h) $\mathbf{A} = \hat{\mathbf{x}}\sin\left(\frac{\pi x}{10}\right) + \hat{\mathbf{y}}\sin\left(\frac{\pi y}{10}\right)$, for $-10 \leq x, y \leq 10$

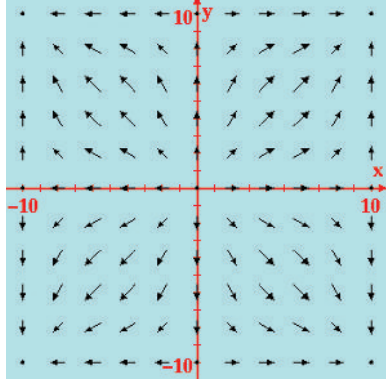


Figure P3.44(h)

(i) $\mathbf{A} = \hat{\mathbf{r}}r + \hat{\boldsymbol{\phi}}r\cos\phi$, for $\begin{cases} 0 \leq r \leq 10 \\ 0 \leq \phi \leq 2\pi \end{cases}$

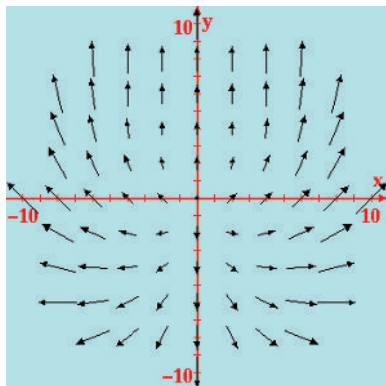


Figure P3.44(i)

(j) $\mathbf{A} = \hat{\mathbf{r}}r^2 + \hat{\boldsymbol{\phi}}r^2\sin\phi$, for $\begin{cases} 0 \leq r \leq 10 \\ 0 \leq \phi \leq 2\pi \end{cases}$

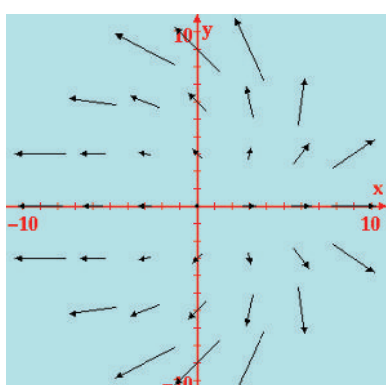
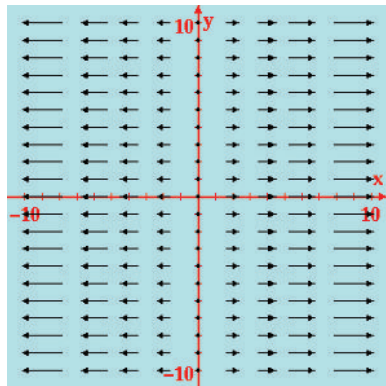


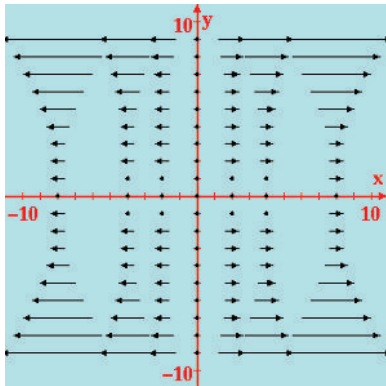
Figure P3.44(j)

(e) $\mathbf{A} = \hat{\mathbf{x}}x$, 对于 $-10 \leq x \leq 10$



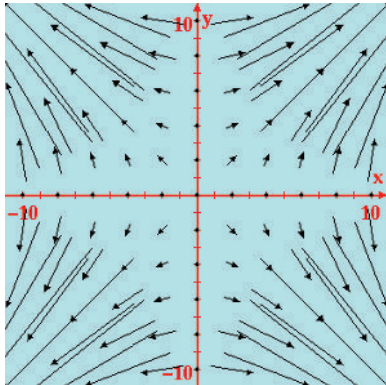
图P3.44(e)

(f) $\mathbf{A} = \hat{\mathbf{x}}xy^2$, 对于 $-10 \leq x, y \leq 10$



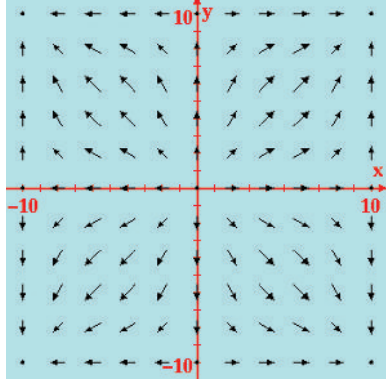
图P3.44(f)

(g) $\mathbf{A} = \hat{\mathbf{x}}xy^2 + \hat{\mathbf{y}}x^2y$, 对于 $-10 \leq x, y \leq 10$



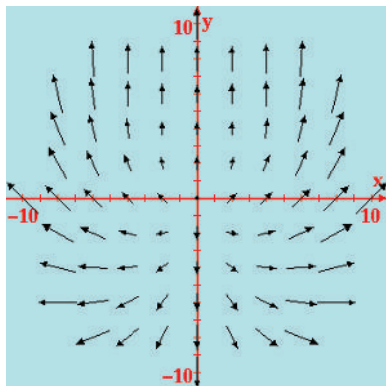
图P3.44(g)

(h) $\mathbf{A} = \hat{\mathbf{x}}\sin\left(\frac{\pi x}{10}\right) + \hat{\mathbf{y}}\sin\left(\frac{\pi y}{10}\right)$, 对于 $-10 \leq x, y \leq 10$



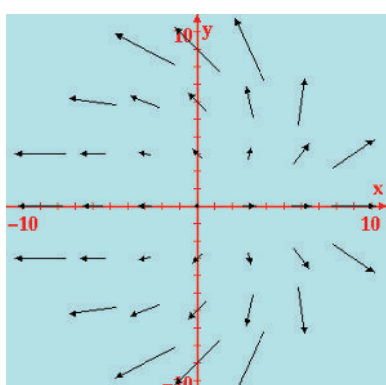
图P3.44(h)

(i) $\mathbf{A} = \hat{\mathbf{r}}r + \hat{\boldsymbol{\phi}}r\cos\phi$, 对于 $\begin{cases} 0 \leq r \leq 10 \\ 0 \leq \phi \leq 2\pi \end{cases}$



图P3.44(i)

(j) $\mathbf{A} = \hat{\mathbf{r}}r^2 + \hat{\boldsymbol{\phi}}r^2\sin\phi$, 对于 $\begin{cases} 0 \leq r \leq 10 \\ 0 \leq \phi \leq 2\pi \end{cases}$



图P3.44(j)

3.45 The scalar function V is given by

$$V = 2\sin x + xy^2 + ye^{2z}.$$

Determine ∇V and then evaluate it at $P(0, 1, 2)$.

3.46 The scalar function V is given by

$$V = \frac{2z}{x^2 + y^2}.$$

- (a) Determine ∇V in Cartesian coordinates.
- (b) Convert the result of part (a) from Cartesian to cylindrical coordinates.
- (c) Convert the expression for V into cylindrical coordinates and then determine ∇V in those coordinates. Compare the results of parts (b) and (c).

***3.47** Vector field $\mathbf{E}$ is characterized by the following properties: (a) $\mathbf{E}$ points along $\hat{\mathbf{r}}$; (b) the magnitude of $\mathbf{E}$ is a function of only the distance from the origin; (c) $\mathbf{E}$ vanishes at the origin; and (d) $\nabla \cdot \mathbf{E} = 12$, everywhere. Find an expression for $\mathbf{E}$ that satisfies these properties.

3.48 For the vector field $\mathbf{E} = \hat{\mathbf{x}}xz - \hat{\mathbf{y}}yz^2 - \hat{\mathbf{z}}xy$, verify the divergence theorem by computing

- (a) the total outward flux flowing through the surface of a cube centered at the origin and with sides equal to 2 units each and parallel to the Cartesian axes, and
- (b) the integral of $\nabla \cdot \mathbf{E}$ over the cube's volume.

3.49 For the vector field $\mathbf{E} = \hat{\mathbf{r}}10e^{-r} - \hat{\mathbf{z}}3z$, verify the divergence theorem for the cylindrical region enclosed by $r = 2$, $z = 0$, and $z = 4$.

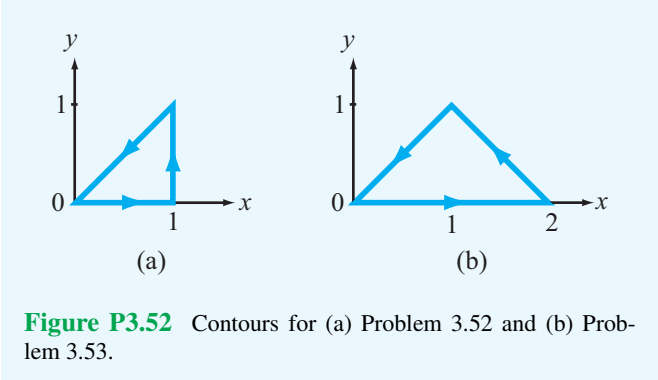
***3.50** A vector field $\mathbf{D} = \hat{\mathbf{r}}r^3$ exists in the region between two concentric cylindrical surfaces defined by $r = 1$ and $r = 2$ with both cylinders extending between $z = 0$ and $z = 5$. Verify the divergence theorem by evaluating the following:

- (a) $\oint_S \mathbf{D} \cdot d\mathbf{s}$
- (b) $\int_V \nabla \cdot \mathbf{D} \, dV$

3.51 For the vector field $\mathbf{D} = \hat{\mathbf{R}}3R^2$, evaluate both sides of the divergence theorem for the region enclosed between the spherical shells defined by $R = 1$ and $R = 2$.

3.52 For the vector field $\mathbf{E} = \hat{\mathbf{x}}xy - \hat{\mathbf{y}}(x^2 + 2y^2)$, calculate

- (a) $\oint_C \mathbf{E} \cdot d\mathbf{l}$ around the triangular contour shown in Fig. P3.52(a).
- (b) $\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{s}$ over the area of the triangle.



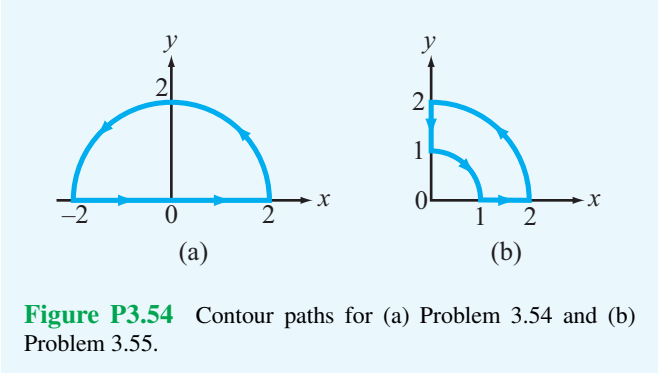
3.53 Repeat Problem 3.52 for the contour shown in Fig. P3.52(b).

3.54 Verify Stokes's theorem for the vector field

$$\mathbf{B} = (\hat{\mathbf{r}}r \cos \phi + \hat{\phi} \sin \phi)$$

by evaluating

- (a) $\oint_C \mathbf{B} \cdot d\mathbf{l}$ over the semicircular contour shown in Fig. P3.54(a), and
- (b) $\int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{s}$ over the surface of the semicircle.



3.55 Repeat Problem 3.54 for the contour shown in Fig. P3.54(b).

3.56 Verify Stokes's theorem for the vector field

$$\mathbf{A} = \hat{\mathbf{R}} \cos \theta + \hat{\phi} \sin \theta$$

by evaluating it on the hemisphere of unit radius.

3.45 标量函数 V 由下式给出

$$V = 2\sin x + xy^2 + ye^{2z}.$$

确定 ∇V ，然后在 $P(0, 1, 2)$ 处对其进行评估。

3.46 标量函数 V 由下式给出

$$V = \frac{2z}{x^2 + y^2}.$$

- (a) 确定笛卡尔坐标中的 ∇V 。
- (b) 将(a)部分的结果从笛卡尔坐标转换为柱坐标。(c) 将 V 的表达式转换为柱坐标，然后在该坐标中确定 ∇V 。比较 (b) 和 (c) 部分的结果。

***3.47** 矢量场 $\mathbf{E}$ 具有以下性质：(a) $\mathbf{E}$ 沿 $\hat{\mathbf{r}}$ 指向；(b) $\mathbf{E}$ 的大小仅是距原点距离的函数；(c) $\mathbf{E}$ 在原点消失；(d) $\nabla \cdot \mathbf{E} = 12$ ，无处不在。找到满足这些属性的 $\mathbf{E}$ 表达式。

3.48 对于向量场 $\mathbf{E} = \hat{\mathbf{x}}xz - \hat{\mathbf{y}}yz^2 - \hat{\mathbf{z}}xy$ ，通过计算验证散度定理

(a) 流过以原点为中心、每条边等于 2 个单位且平行于笛卡尔轴的立方体表面的总向外通量，以及 (b) $\nabla \cdot \mathbf{E}$ 对立方体体积的积分。

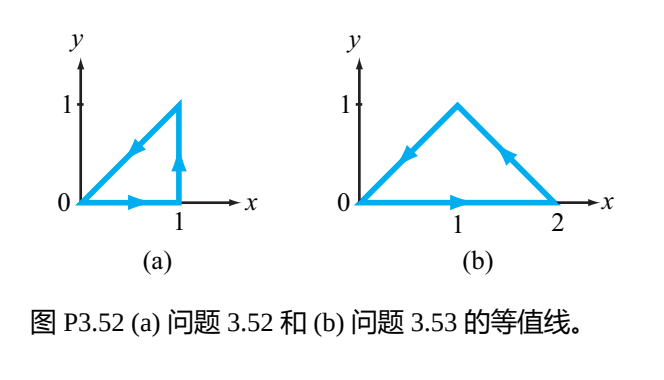
3.49 对于向量场 $\mathbf{E} = \hat{\mathbf{r}}10e^{-r} - \hat{\mathbf{z}}3z$ ，验证 $r = 2$ 、 $z = 0$ 和 $z = 4$ 围成的圆柱区域的散度定理。

***3.50** 矢量场 $\mathbf{D} = \hat{\mathbf{r}}r^3$ 存在于由 $r = 1$ 和 $r = 2$ 定义的两个同心圆柱面之间的区域中，两个圆柱面都在 $z = 0$ 和 $z = 5$ 之间延伸。通过评估以下内容来验证散度定理：

- (a) $\oint_S \mathbf{D} \cdot d\mathbf{s}$
- (b) $\int_V \nabla \cdot \mathbf{D} \, dV$

3.51 对于向量场 $\mathbf{D} = \hat{\mathbf{R}}3R^2$ ，计算由 $R = 1$ 和 $R = 2$ 定义的球壳之间包围的区域的散度定理两边。

3.52 对于向量场 $\mathbf{E} = \hat{\mathbf{x}}xy - \hat{\mathbf{y}}(x^2 + 2y^2)$ ，计算图 P3.52(a) 所示三角形轮廓周围的 (a) $\oint_C \mathbf{E} \cdot d\mathbf{l}$ 。(b) 三角形面积上的 $\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{s}$ 。



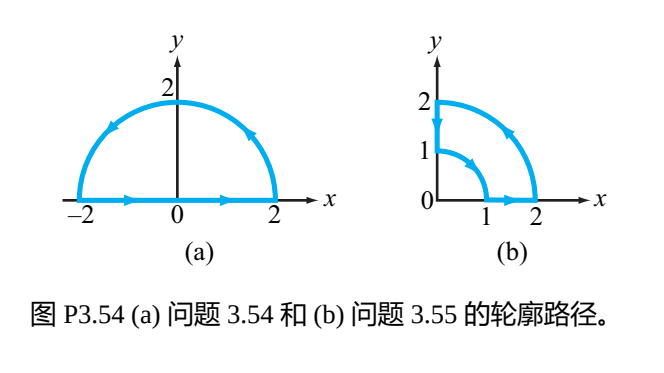
3.53 对图 P3.52(b) 所示的轮廓重复问题 3.52。

3.54 验证向量场的斯托克斯定理

$$\mathbf{B} = (\hat{\mathbf{r}}r \cos \phi + \hat{\phi} \sin \phi)$$

通过评估

(a)图P3.54(a)所示半圆轮廓上的 $\oint_C \mathbf{B} \cdot d\mathbf{l}$ ，以及(b)半圆表面上的 $\int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{s}$ 。



3.55 对图 P3.54(b) 所示的轮廓重复问题 3.54。

3.56 验证向量场的斯托克斯定理

$$\mathbf{A} = \hat{\mathbf{R}} \cos \theta + \hat{\phi} \sin \theta$$

通过在单位半径的半球上对其进行评估。

3.57 Verify Stokes’s theorem for the vector field

$$\mathbf{B} = (\hat{\mathbf{r}} \cos \phi + \hat{\boldsymbol{\phi}} \sin \phi)$$

by evaluating:

- (a) $\oint_C \mathbf{B} \cdot d\boldsymbol{\ell}$ over the path comprising a quarter section of a circle, as shown in Fig. P3.57, and
- (b) $\int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{s}$ over the surface of the quarter section.

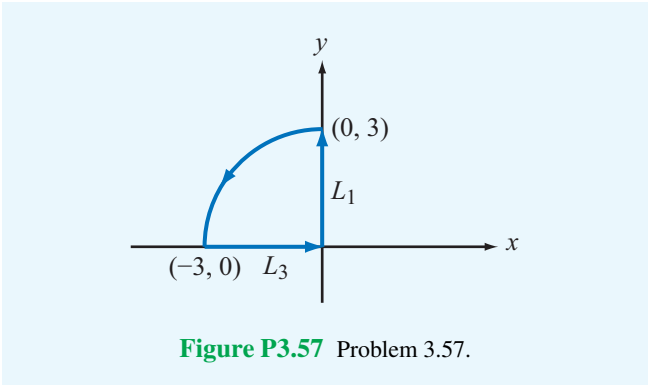


Figure P3.57 Problem 3.57.

3.58 Determine if each of the following vector fields is solenoidal, conservative, or both:

- (a) $\mathbf{A} = \hat{\mathbf{x}}x^2 - \hat{\mathbf{y}}2xy$
- (b) $\mathbf{B} = \hat{\mathbf{x}}x^2 - \hat{\mathbf{y}}y^2 + \hat{\mathbf{z}}2z$
- (c) $\mathbf{C} = \hat{\mathbf{r}}(\sin \phi)/r^2 + \hat{\boldsymbol{\phi}}(\cos \phi)/r^2$
- (d) $\mathbf{D} = \hat{\mathbf{R}}/R$
- (e) $\mathbf{E} = \hat{\mathbf{r}}(3 - \frac{r}{1+r}) + \hat{\mathbf{z}}z$
- (f) $\mathbf{F} = (\hat{\mathbf{x}}y + \hat{\mathbf{y}}x)/(x^2 + y^2)$
- (g) $\mathbf{G} = \hat{\mathbf{x}}(x^2 + z^2) - \hat{\mathbf{y}}(y^2 + x^2) - \hat{\mathbf{z}}(y^2 + z^2)$
- (h) $\mathbf{H} = \hat{\mathbf{R}}(Re^{-R})$

3.59 Find the Laplacian of the following scalar functions:

- (a) $V = 4xy^2z^3$
- (b) $V = xy + yz + zx$
- (c) $V = 3/(x^2 + y^2)$
- (d) $V = 5e^{-r} \cos \phi$
- (e) $V = 10e^{-R} \sin \theta$

3.60 Find the Laplacian of the following scalar functions:

- (a) $V_1 = 10r^3 \sin 2\phi$
- (b) $V_2 = (2/R^2) \cos \theta \sin \phi$

3.57 验证向量场的斯托克斯定理

$$\mathbf{B} = (\hat{\mathbf{r}} \cos \phi + \hat{\boldsymbol{\phi}} \sin \phi)$$

通过评估：

- (a) 在包含四分之一圆截面的路径上的 $\oint_C \mathbf{B} \cdot d\boldsymbol{\ell}$, 如图 P3.57 所示, 以及 (b) 在四分之一截面的表面上的 $\int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{s}$ 。

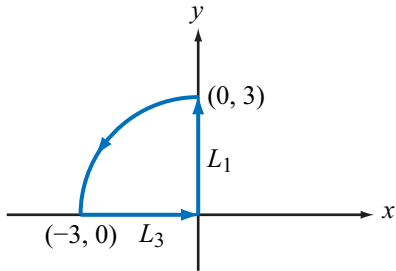


图 P3.57 问题 3.57。

3.58 确定以下每个矢量场是螺线管矢量场、保守矢量场还是两者兼而有之：

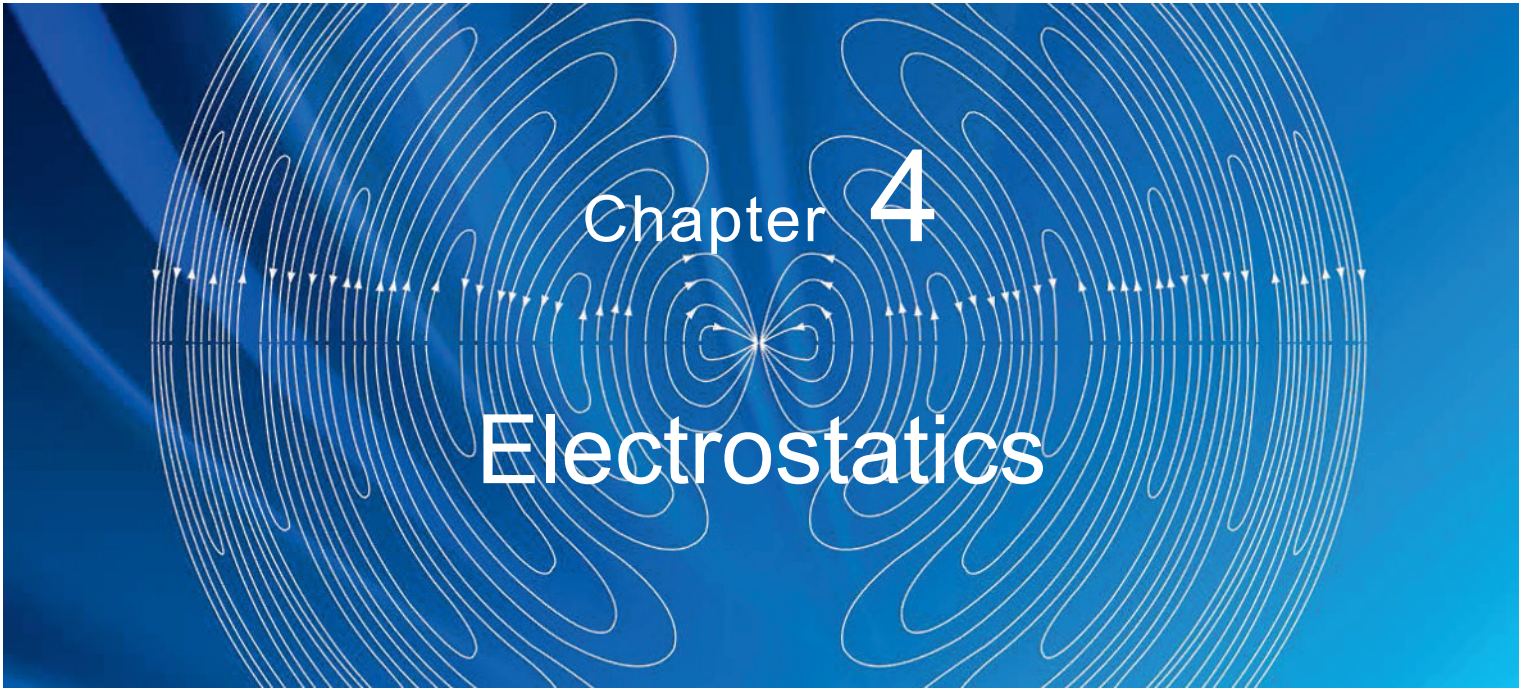
- (a) $\mathbf{A} = \hat{\mathbf{x}}x^2 - \hat{\mathbf{y}}2xy$
- (b) $\mathbf{B} = \hat{\mathbf{x}}x^2 - \hat{\mathbf{y}}y^2 + \hat{\mathbf{z}}2z$
- (c) $\mathbf{C} = \hat{\mathbf{r}}(\sin \phi)/r^2 + \hat{\boldsymbol{\phi}}(\cos \phi)/r^2$
- (d) $\mathbf{D} = \hat{\mathbf{R}}/R$
- (e) $\mathbf{E} = \hat{\mathbf{r}}(3 - \frac{r}{1+r}) + \hat{\mathbf{z}}z$
- (f) $\mathbf{F} = (\hat{\mathbf{x}}y + \hat{\mathbf{y}}x)/(x^2 + y^2)$
- (g) $\mathbf{G} = \hat{\mathbf{x}}(x^2 + z^2) - \hat{\mathbf{y}}(y^2 + x^2) - \hat{\mathbf{z}}(y^2 + z^2)$
- (h) $\mathbf{H} = \hat{\mathbf{R}}(Re^{-R})$

3.59 求下列标量函数的拉普拉斯算子：

- (a) $V = 4xy^2z^3$
- (b) $V = xy + yz + zx$
- (c) $V = 3/(x^2 + y^2)$
- (d) $V = 5e^{-r} \cos \phi$
- (e) $V = 10e^{-R} \sin \theta$

3.60 求下列标量函数的拉普拉斯算子：

- (a) $V_1 = 10r^3 \sin 2\phi$
- (b) $V_2 = (2/R^2) \cos \theta \sin \phi$



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Objectives

Upon learning the material presented in this chapter, you should be able to:

1. Evaluate the electric field and electric potential due to any distribution of electric charges.
2. Apply Gauss’s law.
3. Calculate the resistance R of any shaped object given the electric field at every point in its volume.
4. Describe the operational principles of resistive and capacitive sensors.
5. Calculate the capacitance of two-conductor configurations.



章节内容

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目标

学习本章介绍的材料后, 您应该能够:

1. 评估由于电荷分布而产生的电场和电势。
2. 应用高斯定律。
3. 在给定其体积内每个点的电场的情况下, 计算任何形状物体的电阻 R 。
4. 描述电阻式和电容式传感器的工作原理。
5. 计算双导体配置的电容。

4-1 Maxwell’s Equations

The modern theory of electromagnetism is based on a set of four fundamental relations known as *Maxwell’s equations*:

$$\nabla \cdot \mathbf{D} = \rho_v,$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t},$$

$$\nabla \cdot \mathbf{B} = 0,$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}.$$

(4.1a)

(4.1b)

(4.1c)

(4.1d)

Here **E** and **D** are the *electric field intensity* and *flux density* interrelated by **D** = ε**E** where ε is the electrical permittivity; **H** and **B** are *magnetic field intensity* and *flux density* inter-related by **B** = μ**H** where μ is the magnetic permeability; ρ_v is the electric charge density per unit volume; and **J** is the current density per unit area. The fields and fluxes **E**, **D**, **B**, **H** were introduced in Section 1-3, and ρ_v and **J** will be discussed in Section 4-2. Maxwell’s equations hold in any material, including free space (vacuum). In general, all of the above quantities may depend on spatial location and time *t*. In the interest of readability, we will not, however, explicitly reference these dependencies (as in **E**(*x,y,z,t*)) except when the context calls for it. By formulating these equations, published in a classic treatise in 1873, James Clerk Maxwell established the first unified theory of electricity and magnetism. His equations are deduced from experimental observations reported by Coulomb, Gauss, Ampère, Faraday, and others; they not only encapsulate the connection between the electric field and electric charge and between the magnetic field and electric current but also capture the bilateral coupling between electric and magnetic fields and fluxes. Together with some auxiliary relations, Maxwell’s equations comprise the fundamental tenets of electromagnetic theory.

► Under *static* conditions, none of the quantities appearing in Maxwell’s equations are functions of time (i.e., ∂/∂*t* = 0). *This happens when all charges are permanently fixed in space. If they move, they do so at a steady rate so that ρ_v and **J** are constant in time.* ◀

Under these circumstances, the time derivatives of **B** and **D** in Eqs. (4.1b) and (4.1d) vanish, and Maxwell’s equations reduce to the following pairs.

Electrostatics

$$\nabla \cdot \mathbf{D} = \rho_v,$$

$$\nabla \times \mathbf{E} = 0.$$

(4.2a)

(4.2b)

Magnetostatics

$$\nabla \cdot \mathbf{B} = 0,$$

$$\nabla \times \mathbf{H} = \mathbf{J}.$$

(4.3a)

(4.3b)

Maxwell’s four equations separate into two uncoupled pairs with the first pair involving only the electric field and flux **E** and **D** and the second pair involving only the magnetic field and flux **H** and **B**.

► Electric and magnetic fields become *decoupled* under static conditions. ◀

This allows us to study electricity and magnetism as two distinct and separate phenomena as long as the spatial distributions of charge and current flow remain constant in time. We refer to the study of electric and magnetic phenomena under static conditions as *electrostatics* and *magnetostatics*, respectively. Electrostatics is the subject of the present chapter, and we learn about magnetostatics in Chapter 5. The experience gained through studying electrostatic and magnetostatic phenomena will prove invaluable in tackling the more involved material in subsequent chapters that deal with time-varying fields, charge densities, and currents.

We study electrostatics not only as a prelude to the study of time-varying fields but also because it is an important field in its own right. Many electronic devices and systems are based on the principles of electrostatics. They include x-ray machines, oscilloscopes, ink-jet electrostatic printers, liquid crystal displays, copy machines, micro-electromechanical switches and accelerometers, and many solid-state–based control devices. Electrostatic principles also guide the design of medical diagnostic sensors, such as the electrocardiogram, which records the heart’s pumping pattern, and the electroencephalogram, which records brain activity, as well as the development of numerous industrial applications.

4-2 Charge and Current Distributions

In electromagnetics, we encounter various forms of electric charge distributions. When put in motion, these charge distributions constitute current distributions. Charges and currents may be distributed over a volume of space, across a surface, or along a line.

4-1 麦克斯韦方程组

现代电磁理论基于一组称为麦克斯韦方程组的四个基本关系：

$$\nabla \cdot \mathbf{D} = \rho_v,$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t},$$

$$\nabla \cdot \mathbf{B} = 0,$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}.$$

(4.1a)

(4.1b)

(4.1c)

(4.1d)

这里，E 为电场强度，D 为电通量密度，二者满足 D = εE，ε 为介电率；H 为磁场强度，B 为磁通密度，二者满足 B = μH，μ 为磁导率；ρ_v 为体电荷密度，J 为电流密度。E、D、B、H 已在第1-3节介绍，ρ_v 和 J 将在第4-2节讨论。麦克斯韦方程组适用于包括自由空间（真空）在内的各类介质。一般而言，上述量均可依赖空间位置与时间 t；为使记号简洁，除上下文需要外，不显式写出 E(x,y,z,t) 等函数的自变量。麦克斯韦在1873年出版的经典著作中系统阐述了这些方程，建立了统一的电磁理论。这些方程综合了库仑、高斯、安培、法拉第等人的实验成果，不仅描述电荷与电场、电流与磁场之间的联系，也描述电场与磁场之间的双向耦合。麦克斯韦方程组及相应的辅助关系共同构成电磁理论的基础。

► 在静态条件下，麦克斯韦方程组中出现的所有量都不是时间的函数（即 ∂/∂t = 0）。当所有电荷永久固定在空间中时，就会发生这种情况。如果它们移动，它们会以稳定的速率移动，因此 ρ_v 和 J 在时间上恒定。 ◀

在这种情况下，方程中B和D的时间导数。（4.1b）和（4.1d）消失，麦克斯韦方程组简化为以下对。

静电学

$$\nabla \cdot \mathbf{D} = \rho_v,$$

$$\nabla \times \mathbf{E} = 0.$$

(4.2a)

(4.2b)

静磁学

$$\nabla \cdot \mathbf{B} = 0,$$

$$\nabla \times \mathbf{H} = \mathbf{J}.$$

(4.3a)

(4.3b)

麦克斯韦的四个方程分成两对非耦合方程，第一对仅涉及电场以及磁通 E 和 D，第二对仅涉及磁场以及磁通 H 和 B。

► 电场和磁场在静态条件下会解耦。 ◀

只要电荷和电流的空间分布随时间保持恒定，这使我们能够将电和磁作为两种不同且独立的现象来研究。我们将静态条件下的电磁现象的研究分别称为静电学和静磁学。静电学是本章的主题，我们在第五章中学习静磁学。通过研究静电和静磁现象获得的经验对于处理后续章节中涉及时变场、电荷密度和电流的更复杂的材料将证明是非常宝贵的。

我们研究静电不仅是作为研究时变领域的前奏，而且还因为它本身就是一个重要的领域。许多电子设备和系统都是基于静电原理。它们包括 X 射线机、示波器、喷墨静电打印机、液晶显示器、复印机、微机电开关和加速度计以及许多固态控制设备。静电原理还指导医疗诊断传感器的设计，例如记录心脏泵血模式的心电图和记录大脑活动的脑电图，以及众多工业应用的开发。

4-2 电荷和电流分布

在电磁学中，我们遇到各种形式的电荷分布。当运动时，这些电荷分布就构成了电流分布。电荷和电流可以分布在一定体积的空间、整个表面或沿着一条线。

4-2.1 Charge Densities

At the atomic scale, the charge distribution in a material is discrete, meaning that charge exists only where electrons and nuclei are and nowhere else. In electromagnetics, we usually are interested in studying phenomena at a much larger scale, typically three or more orders of magnitude greater than the spacing between adjacent atoms. At such a macroscopic scale, we can disregard the discontinuous nature of the charge distribution and treat the net charge contained in an elemental volume $\Delta\mathcal{V}$ as if it were uniformly distributed within. Accordingly, we define the **volume charge density** ρ_v as

$$\rho_v = \lim_{\Delta v \rightarrow 0} \frac{\Delta q}{\Delta \mathcal{V}} = \frac{dq}{d\mathcal{V}} \quad (\text{C/m}^3), \quad (4.4)$$

where Δq is the charge contained in $\Delta\mathcal{V}$. In general, ρ_v depends on spatial location (x, y, z) and t ; thus, $\rho_v = \rho_v(x, y, z, t)$. Physically, ρ_v represents the average charge per unit volume for a volume $\Delta\mathcal{V}$ centered at (x, y, z) with $\Delta\mathcal{V}$ being large enough to contain a large number of atoms, yet it is small enough to be regarded as a point at the macroscopic scale under consideration. The variation of ρ_v with spatial location is called its **spatial distribution** (or simply its **distribution**). The total charge contained in volume $\mathcal{V}$ is

$$Q = \int_{\mathcal{V}} \rho_v d\mathcal{V} \quad (\text{C}). \quad (4.5)$$

In some cases, particularly when dealing with conductors, electric charge may be distributed across the surface of a material, where the quantity of interest is the **surface charge density** ρ_s , which is defined as

$$\rho_s = \lim_{\Delta s \rightarrow 0} \frac{\Delta q}{\Delta s} = \frac{dq}{ds} \quad (\text{C/m}^2), \quad (4.6)$$

where Δq is the charge present across an elemental surface area Δs . Similarly, if the charge is, for all practical purposes, confined to a line, which need not be straight, we characterize its distribution in terms of the **line charge density** ρ_ℓ , defined as

$$\rho_\ell = \lim_{\Delta l \rightarrow 0} \frac{\Delta q}{\Delta l} = \frac{dq}{dl} \quad (\text{C/m}). \quad (4.7)$$

Example 4-1: Line Charge Distribution

Calculate the total charge Q contained in a cylindrical tube oriented along the z axis, as shown in Fig. 4-1(a). The line charge density is $\rho_\ell = 2z$, where z is the distance in meters from the bottom end of the tube. The tube length is 10 cm.

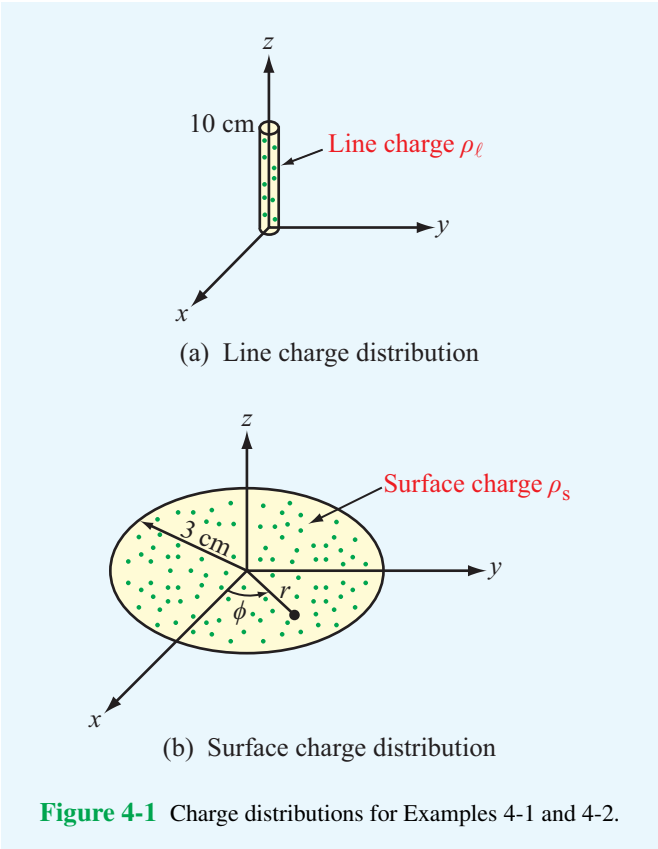


Figure 4-1 Charge distributions for Examples 4-1 and 4-2.

Solution: The total charge Q is

$$Q = \int_0^{0.1} \rho_\ell dz = \int_0^{0.1} 2z dz = z^2 \Big|_0^{0.1} = 10^{-2} \text{ C}.$$

Example 4-2: Surface Charge Distribution

The circular disk of electric charge shown in Fig. 4-1(b) is characterized by an azimuthally symmetric surface charge density that increases linearly with r from zero at the center to 6 C/m^2 at $r = 3 \text{ cm}$. Find the total charge present on the disk surface.

Solution: Since ρ_s is symmetrical with respect to the azimuth angle ϕ , it depends only on r and is given by

$$\rho_s = \frac{6r}{3 \times 10^{-2}} = 2 \times 10^2 r \quad (\text{C/m}^2),$$

where r is in meters. In polar coordinates, an elemental area is $ds = r dr d\phi$, and for the disk shown in Fig. 4-1(b), the

4-2.1 电荷密度

在原子尺度上，材料中的电荷分布是离散的，这意味着电荷仅存在于电子和原子核所在的地方，而不存在于其他地方。在电磁学中，我们通常有兴趣研究更大尺度的现象，通常比相邻原子之间的间距大三个或更多数量级。在这样的宏观尺度上，我们可以忽略电荷分布的不连续性质，并将元素体积 $\Delta\mathcal{V}$ 中包含的净电荷视为均匀分布。因此，我们将体积电荷密度 ρ_v 定义为

$$\rho_v = \lim_{\Delta v \rightarrow 0} \frac{\Delta q}{\Delta \mathcal{V}} = \frac{dq}{d\mathcal{V}} \quad (\text{C/m}^3), \quad (4.4)$$

其中 Δq 是 $\Delta\mathcal{V}$ 中包含的电荷。一般来说， ρ_v 取决于空间位置 (x, y, z) 和 t ；因此， $\rho_v = \rho_v(x, y, z, t)$ 。物理上， ρ_v 表示以 (x, y, z) 为中心的体积 $\Delta\mathcal{V}$ 的每单位体积的平均电荷，其中 $\Delta\mathcal{V}$ 大到足以包含大量原子，但又小到足以被视为所考虑的宏观尺度上的一个点。 ρ_v 随空间位置的变化称为其空间分布（或简称其分布）。体积 $\mathcal{V}$ 中包含的总电荷为

$$Q = \int_{\mathcal{V}} \rho_v d\mathcal{V} \quad (\text{C}). \quad (4.5)$$

在某些情况下，特别是在处理导体时，电荷可能分布在材料的表面，其中感兴趣的量是表面电荷密度 ρ_s ，其定义为

$$\rho_s = \lim_{\Delta s \rightarrow 0} \frac{\Delta q}{\Delta s} = \frac{dq}{ds} \quad (\text{C/m}^2), \quad (4.6)$$

其中 Δq 是元素表面积 Δs 上存在的电荷。类似地，如果出于所有实际目的，电荷被限制在一条线上，而这条线不必是直的，我们就用线电荷密度 ρ_ℓ 来表征其分布，定义为

$$\rho_\ell = \lim_{\Delta l \rightarrow 0} \frac{\Delta q}{\Delta l} = \frac{dq}{dl} \quad (\text{C/m}). \quad (4.7)$$

例 4-1：线路电荷分布

计算沿 z 轴定向的圆柱管中所含的总电荷 Q ，如图 4-1 (a) 所示。线电荷密度为 $\rho_\ell = 2z$ ，其中 z 是距管底端的距离（以米为单位）。管子长度为 10 厘米。

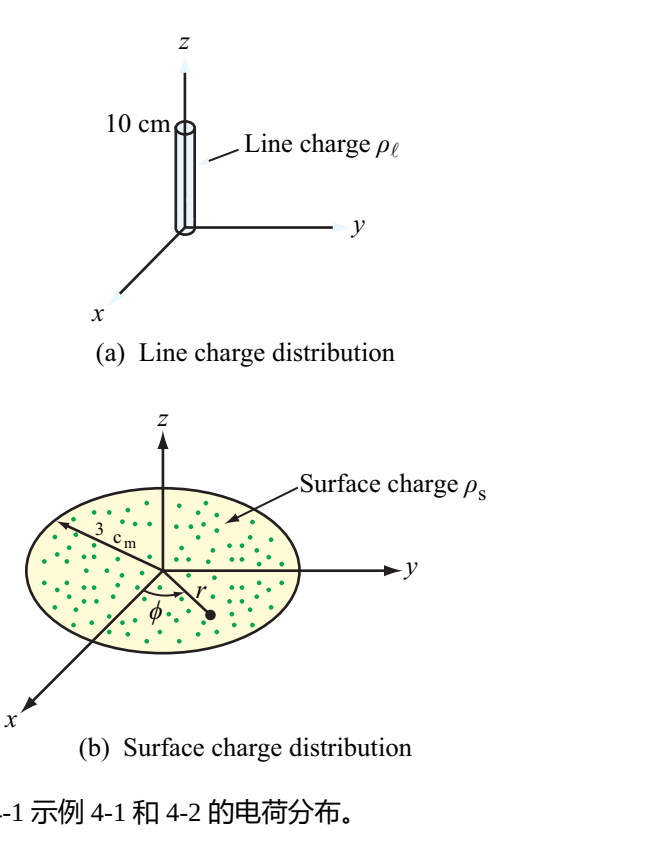


图 4-1 示例 4-1 和 4-2 的电荷分布。

解：总电荷 Q 为

$$Q = \int_0^{0.1} \rho_\ell dz = \int_0^{0.1} 2z dz = z^2 \Big|_0^{0.1} = 10^{-2} \text{ C}.$$

例 4-2：表面电荷分布

图 4-1(b) 所示的电荷圆盘的特征是方位对称的表面电荷密度，随着 r 从中心的 0 线性增加到 $r = 3 \text{ cm}$ 处的 6 C/m^2 。找出磁盘表面上存在的总电荷。

解：由于 ρ_s 相对于方位角 ϕ 对称，因此它仅取决于 r ，由下式给出

$$\rho_s = \frac{6r}{3 \times 10^{-2}} = 2 \times 10^2 r \quad (\text{C/m}^2),$$

其中 r 的单位是米。在极坐标中，一个基本区域为 $ds = r dr d\phi$ ，对于图 4-1(b) 所示的圆盘，

4-2 CHARGE AND CURRENT DISTRIBUTIONS

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limits of integration are from 0 to 2π (rad) for ϕ and from 0 to 3×10^{-2} m for r . Hence,

$$Q = \int_S \rho_s ds = \int_{\phi=0}^{2\pi} \int_{r=0}^{3 \times 10^{-2}} (2 \times 10^2 r) r dr d\phi$$
$$= 2\pi \times 2 \times 10^2 \frac{r^3}{3} \Big|_0^{3 \times 10^{-2}} = 11.31 \text{ (mC)}.$$

Exercise 4-1: A square plate residing in the x - y plane is situated in the space defined by $-3 \text{ m} \leq x \leq 3 \text{ m}$ and $-3 \text{ m} \leq y \leq 3 \text{ m}$. Find the total charge on the plate if the surface charge density is $\rho_s = 4y^2$ ($\mu\text{C}/\text{m}^2$).

Answer: $Q = 0.432$ (mC). (See [EM](#).)

Exercise 4-2: A thick spherical shell centered at the origin extends between $R = 2 \text{ cm}$ and $R = 3 \text{ cm}$. If the volume charge density is $\rho_v = 3R \times 10^{-4}$ (C/m^3), find the total charge contained in the shell.

Answer: $Q = 0.61$ (nC). (See [EM](#).)

4-2.2 Current Density

Consider a tube with volume charge density ρ_v (Fig. 4-2(a)). The charges in the tube move with velocity $\mathbf{u}$ along the tube axis. Over a period Δt , the charges move a distance $\Delta l = u \Delta t$. The amount of charge that crosses the tube's cross-sectional surface $\Delta s'$ in time Δt is therefore

$$\Delta q' = \rho_v \Delta V = \rho_v \Delta l \Delta s' = \rho_v u \Delta s' \Delta t. \quad (4.8)$$

Now consider the more general case where the charges are flowing through a surface Δs with normal $\hat{\mathbf{n}}$ not necessarily parallel to $\mathbf{u}$ (Fig. 4-2(b)). In this case, the amount of charge Δq flowing through Δs is

$$\Delta q = \rho_v \mathbf{u} \cdot \Delta \mathbf{s} \Delta t, \quad (4.9)$$

where $\Delta \mathbf{s} = \hat{\mathbf{n}} \Delta s$ and the corresponding total current flowing in the tube is

$$\Delta I = \frac{\Delta q}{\Delta t} = \rho_v \mathbf{u} \cdot \Delta \mathbf{s} = \mathbf{J} \cdot \Delta \mathbf{s}. \quad (4.10)$$

Here

$$\mathbf{J} = \rho_v \mathbf{u} \quad (\text{A}/\text{m}^2) \quad (4.11)$$

is defined as the **current density** in ampere per square meter. Generalizing to an arbitrary surface S , the total current flowing

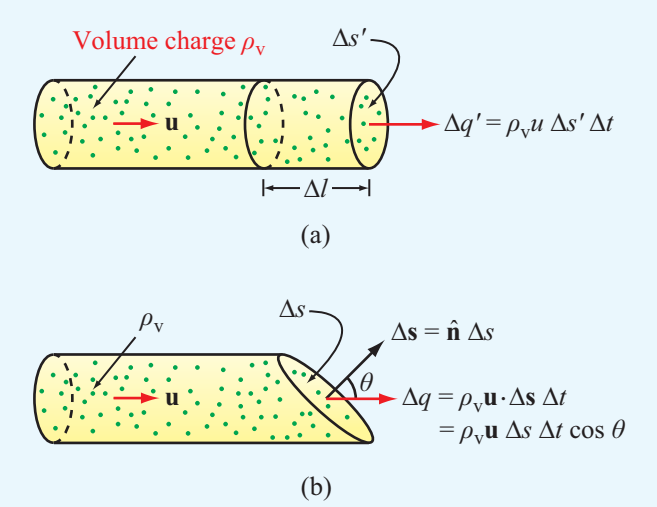


Figure 4-2 Charges with velocity $\mathbf{u}$ moving through a cross section $\Delta s'$ in (a) and Δs in (b).

through it is

$$I = \int_S \mathbf{J} \cdot d\mathbf{s} \quad (\text{A}). \quad (4.12)$$

► When a current is due to the actual movement of electrically charged matter, it is called a **convection current**, and $\mathbf{J}$ is called a **convection current density**. ◀

A wind-driven charged cloud, for example, gives rise to a convection current. In some cases, the charged matter constituting the convection current consists solely of charged particles, such as the electron beam of a scanning electron microscope or the ion beam of a plasma propulsion system.

When a current is due to the movement of charged particles relative to their host material, $\mathbf{J}$ is called a **conduction current density**. In a metal wire, for example, there are equal amounts of positive charges (in atomic nuclei) and negative charges (in the electron shells of the atoms). None of the positive charges and few of the negative charges can move; only those electrons in the outermost electron shells of the atoms can be pushed from one atom to the next if a voltage is applied across the ends of the wire.

► This movement of electrons from atom to atom constitutes a **conduction current**. The electrons that emerge from the wire are not necessarily the same electrons that entered the wire at the other end. ◀

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ϕ 的积分限制为 0 到 2π (rad), r 的积分限制为 0 到 3×10^{-2} m。因此,

$$Q = \int_S \rho_s ds = \int_{\phi=0}^{2\pi} \int_{r=0}^{3 \times 10^{-2}} (2 \times 10^2 r) r dr d\phi$$
$$= 2\pi \times 2 \times 10^2 \frac{r^3}{3} \Big|_0^{3 \times 10^{-2}} = 11.31 \text{ (mC)}.$$

练习 4-1: 位于 x - y 平面上的方形板位于由 $-3 \text{ m} \leq x \leq 3 \text{ m}$ 和 $-3 \text{ m} \leq y \leq 3 \text{ m}$ 定义的空间中。如果表面电荷密度为 $\rho_s = 4y^2$ ($\mu\text{C}/\text{m}^2$), 求极板上的总电荷。

答案: $Q = 0.432$ (mC)。 (参见 [EM](#)。)

练习 4-2: 以原点为中心的厚球壳在 $R = 2 \text{ cm}$ 和 $R = 3 \text{ cm}$ 之间延伸。如果体积电荷密度为 $\rho_v = 3R \times 10^{-4}$ (C/m^3), 求壳层中包含的总电荷。

答案: $Q = 0.61$ (nC)。 (参见 [EM](#)。)

4-2.2 电流密度

考虑具有体积电荷密度 ρ_v (图 1 的管子。 4-2(a))。管内的电荷沿管轴以速度 $\mathbf{u}$ 移动。在 Δt 期间, 电荷移动距离 $\Delta l = u \Delta t$ 。因此, 在时间 Δt 内穿过管横截面 $\Delta s'$ 的电荷量为

$$\Delta q' = \rho_v \Delta V = \rho_v \Delta l \Delta s' = \rho_v u \Delta s' \Delta t. \quad (4.8)$$

现在考虑更一般的情况, 其中电荷流过表面 Δs , 其法线 $\hat{\mathbf{n}}$ 不一定平行于 $\mathbf{u}$ (图 4-2(b))。在这种情况下, 流过 Δs 的电荷量 Δq 为

$$\Delta q = \rho_v \mathbf{u} \cdot \Delta \mathbf{s} \Delta t, \quad (4.9)$$

其中 $\Delta \mathbf{s} = \hat{\mathbf{n}} \Delta s$ 和相应的管中流动的总电流为

$$\Delta I = \frac{\Delta q}{\Delta t} = \rho_v \mathbf{u} \cdot \Delta \mathbf{s} = \mathbf{J} \cdot \Delta \mathbf{s}. \quad (4.10)$$

这里

$$\mathbf{J} = \rho_v \mathbf{u} \quad (\text{A}/\text{m}^2) \quad (4.11)$$

定义为电流密度, 单位为安培每平方米。推广到任意表面 S , 流过的总电流

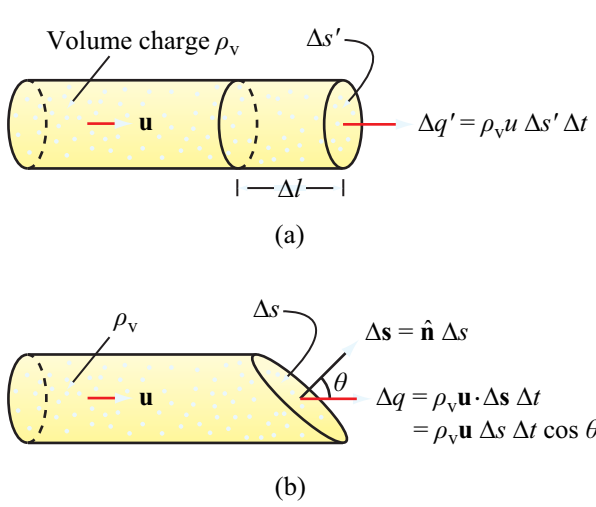


图 4-2 电荷以速度 $\mathbf{u}$ 穿过 (a) 中的横截面 $\Delta s'$ 和 (b) 中的 Δs 横截面。

通过它是

$$I = \int_S \mathbf{J} \cdot d\mathbf{s} \quad (\text{A}). \quad (4.12)$$

► 当电流是由于带电物质的实际运动而产生时, 称为对流电流, $\mathbf{J}$ 称为对流电流密度。 ◀

例如, 风驱动的带电云会产生对流。在一些情况下, 构成对流的带电物质仅由带电粒子组成, 例如扫描电子显微镜的电子束或等离子体推进系统的离子束。

当电流是由于带电粒子相对于其主体材料的运动而产生时, $\mathbf{J}$ 称为传导电流密度。例如, 在金属线中, 有等量的正电荷 (在原子核中) 和负电荷 (在原子的电子壳层中)。正电荷和负电荷都不能移动; 如果在导线两端施加电压, 则只有原子最外层电子壳层中的电子才能从一个原子推到下一个原子。

► 电子从一个原子到另一个原子的运动构成了传导电流。从导线中出现的电子不一定与在另一端进入导线的电子相同。 ◀

Conduction current, which is discussed in more detail in Section 4-6, obeys Ohm’s law, whereas convection current does not.

Concept Question 4-1: What happens to Maxwell’s equations under static conditions?

Concept Question 4-2: How is the current density $\mathbf{J}$ related to the volume charge density ρ_v ?

Concept Question 4-3: What is the difference between convection and conduction currents?

4-3 Coulomb’s Law

One of the primary goals of this chapter is to develop dexterity in applying the expressions for the *electric field intensity* $\mathbf{E}$ and associated *electric flux density* $\mathbf{D}$ induced by a specified distribution of charge. Our discussion will be limited to electrostatic fields induced by stationary charge densities.

We begin by reviewing the expression for the electric field introduced in Section 1-3.2 on the basis of the results of Coulomb’s experiments on the electrical force between charged bodies. *Coulomb’s law*, which was first introduced for electrical charges in air and later generalized to material media, implies that:

- (1) An isolated charge q induces an electric field $\mathbf{E}$ at every point in space, and at any specific point P , $\mathbf{E}$ is given by

$$\mathbf{E} = \hat{\mathbf{R}} \frac{q}{4\pi\epsilon R^2} \quad (\text{V/m}), \quad (4.13)$$

where $\hat{\mathbf{R}}$ is a unit vector pointing from q to P (Fig. 4-3), R is the distance between them, and ϵ is the electrical permittivity of the medium containing the observation point P .

- (2) In the presence of an electric field $\mathbf{E}$ at a given point in space that may be due to a single charge or a distribution of charges, the force acting on a test charge q' when placed at P is

$$\mathbf{F} = q'\mathbf{E} \quad (\text{N}). \quad (4.14)$$

With $\mathbf{F}$ measured in newtons (N) and q' in coulombs (C), the unit of $\mathbf{E}$ is (N/C), which will be shown later in Section 4-5 to be the same as volt per meter (V/m).

For a material with electrical permittivity ϵ , the electric field quantities $\mathbf{D}$ and $\mathbf{E}$ are related by

$$\mathbf{D} = \epsilon\mathbf{E} \quad (4.15)$$

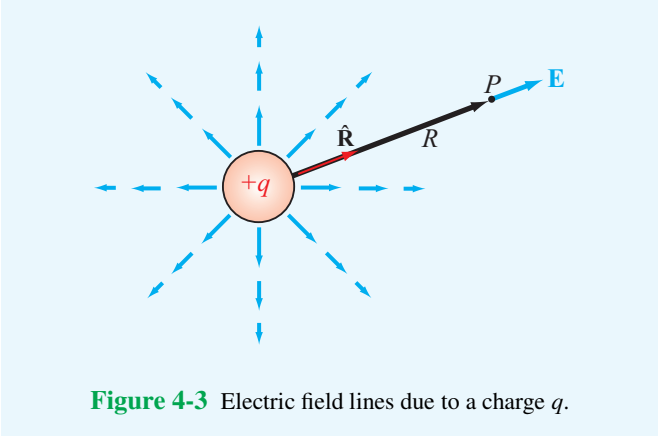


Figure 4-3 Electric field lines due to a charge q .

with
$$\epsilon = \epsilon_r \epsilon_0, \quad (4.16)$$

where

$$\epsilon_0 = 8.85 \times 10^{-12} \approx (1/36\pi) \times 10^{-9} \quad (\text{F/m})$$

is the electrical permittivity of free space and $\epsilon_r = \epsilon/\epsilon_0$ is called the *relative permittivity* (or *dielectric constant*) of the material. For most materials and under a wide range of conditions, ϵ is independent of both the magnitude and direction of $\mathbf{E}$ [as implied by Eq. (4.15)].

► If ϵ is independent of the magnitude of $\mathbf{E}$, then the material is said to be *linear* because $\mathbf{D}$ and $\mathbf{E}$ are related linearly, and if it is independent of the direction of $\mathbf{E}$, the material is said to be *isotropic*. ◀

Materials usually do not exhibit nonlinear permittivity behavior except when the amplitude of $\mathbf{E}$ is very high (at levels approaching dielectric breakdown conditions discussed later in Section 4-7), and anisotropy is present only in certain materials with peculiar crystalline structures. Hence, except for unique materials under very special circumstances, the quantities $\mathbf{D}$ and $\mathbf{E}$ are effectively redundant; for a material with known ϵ , knowledge of either $\mathbf{D}$ or $\mathbf{E}$ is sufficient to specify the other in that material.

4-3.1 Electric Field Due to Multiple Point Charges

The expression given by Eq. (4.13) for the field $\mathbf{E}$ due to a single point charge can be extended to multiple charges. We begin by considering two point charges, q_1 and q_2 , with position vectors $\mathbf{R}_1$ and $\mathbf{R}_2$ (measured from the origin in

传导电流（将在第 4-6 节中详细讨论）遵守欧姆定律，而对流电流则不然。

Concept Question 4-1: What happens to Maxwell’s equations under static conditions?

Concept Question 4-2: How is the current density $\mathbf{J}$ related to the volume charge density ρ_v ?

Concept Question 4-3: What is the difference between convection and conduction currents?

4-3 库仑定律

本章的主要目标之一是提高应用特定电荷分布引起的电场强度 $\mathbf{E}$ 和相关电通量密度 $\mathbf{D}$ 表达式的灵活性。我们的讨论将仅限于由静止电荷密度引起的静电场。

我们首先根据库仑关于带电体之间电力的实验结果回顾 1-3.2 节中介绍的电场表达式。库仑定律首先针对空气中的电荷引入，后来推广到材料介质，它意味着：

- (1) 孤立电荷 q 在空间中的每一点感应出一个电场 $\mathbf{E}$ ，并且在任何特定点 P ， $\mathbf{E}$ 由下式给出

$$\mathbf{E} = \hat{\mathbf{R}} \frac{q}{4\pi\epsilon R^2} \quad (\text{V/m}), \quad (4.13)$$

其中 $\hat{\mathbf{R}}$ 是从 q 指向 P 的单位向量（图4-3）， R 是它们之间的距离， ϵ 是包含观察点 P 的介质的介电常数。

- (2) 当空间中给定点存在电场 $\mathbf{E}$ （可能是由于单个电荷或电荷分布）时，当放置在 P 处时，作用在测试电荷 q' 上的力为

$$\mathbf{F} = q'\mathbf{E} \quad (\text{N}). \quad (4.14)$$

$\mathbf{F}$ 的单位为牛顿 (N)， q' 的单位为库仑 (C)， $\mathbf{E}$ 的单位为 (N/C)，稍后将在第 4-5 节中显示，它与伏每米 (V/m) 相同。

对于介电常数为 ϵ 的材料，电场量 $\mathbf{D}$ 和 $\mathbf{E}$ 的关系为

$$\mathbf{D} = \epsilon\mathbf{E} \quad (4.15)$$

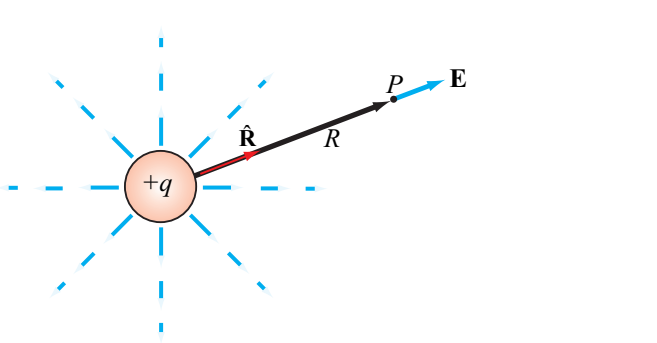


图 4-3 电荷 q 产生的电场线。

和
$$\epsilon = \epsilon_r \epsilon_0, \quad (4.16)$$

在哪里

$$\epsilon_0 = 8.85 \times 10^{-12} \approx (1/36\pi) \times 10^{-9} \quad (\text{F/m})$$

是自由空间的介电常数， $\epsilon_r = \epsilon/\epsilon_0$ 称为材料的相对介电常数（或介电常数）。对于大多数材料和在各种条件下， ϵ 与 $\mathbf{E}$ 的大小和方向无关[如方程 1 所暗示的那样]。（4.15）]。

► 如果 ϵ 与 $\mathbf{E}$ 的大小无关，则该材料被称为线性，因为 $\mathbf{D}$ 和 $\mathbf{E}$ 线性相关；如果它与 $\mathbf{E}$ 的方向无关，则该材料被称为各向同性。 ◀

材料通常不会表现出非线性介电常数行为，除非 $\mathbf{E}$ 的幅度非常高（接近后面第 4-7 节讨论的介电击穿条件的水平），并且各向异性仅存在于某些具有特殊晶体结构的材料中。因此，除了非常特殊情况下的独特材料外，数量 $\mathbf{D}$ 和 $\mathbf{E}$ 实际上是多余的；对于已知 ϵ 的材料， $\mathbf{D}$ 或 $\mathbf{E}$ 的知识足以指定该材料中的另一个。

4-3.1 多点电荷产生的电场

方程给出的表达式。（4.13）对于场 $\mathbf{E}$ ，由于单点电荷可以扩展到多个电荷。我们首先考虑两个点电荷 q_1 和 q_2 ，位置向量 $\mathbf{R}_1$ 和 $\mathbf{R}_2$ （从原点测量

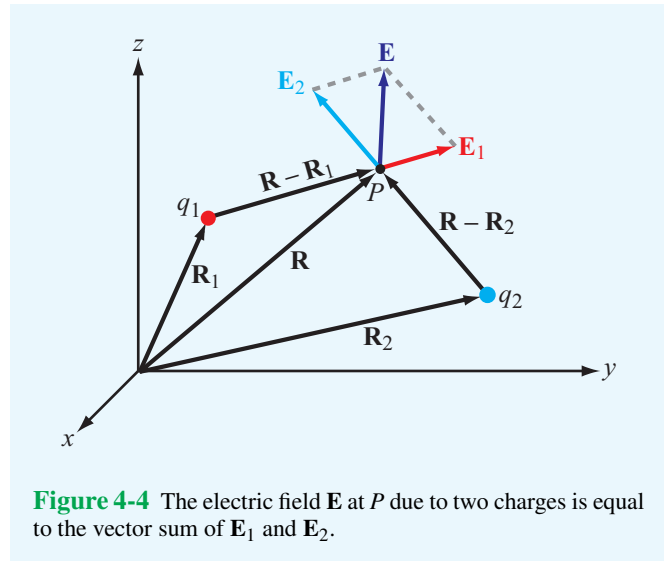


Figure 4-4 The electric field $\mathbf{E}$ at P due to two charges is equal to the vector sum of $\mathbf{E}_1$ and $\mathbf{E}_2$.

Fig. 4-4). The electric field $\mathbf{E}$ is to be evaluated at a point P with position vector $\mathbf{R}$. At P , the electric field $\mathbf{E}_1$ due to q_1 alone is given by Eq. (4.13) with $\mathbf{R}$, which is the distance between q_1 and P , replaced with $|\mathbf{R} - \mathbf{R}_1|$ and the unit vector $\hat{\mathbf{R}}$ replaced with $(\mathbf{R} - \mathbf{R}_1)/|\mathbf{R} - \mathbf{R}_1|$. Thus,

$$\mathbf{E}_1 = \frac{q_1(\mathbf{R} - \mathbf{R}_1)}{4\pi\epsilon|\mathbf{R} - \mathbf{R}_1|^3} \quad (\text{V/m}). \quad (4.17a)$$

Similarly, the electric field at P due to q_2 alone is

$$\mathbf{E}_2 = \frac{q_2(\mathbf{R} - \mathbf{R}_2)}{4\pi\epsilon|\mathbf{R} - \mathbf{R}_2|^3} \quad (\text{V/m}). \quad (4.17b)$$

► The electric field obeys the principle of linear superposition. ◀

Hence, the total electric field $\mathbf{E}$ at P due to q_1 and q_2 together is determined as

$$\begin{aligned} \mathbf{E} &= \mathbf{E}_1 + \mathbf{E}_2 \\ &= \frac{1}{4\pi\epsilon} \left[\frac{q_1(\mathbf{R} - \mathbf{R}_1)}{|\mathbf{R} - \mathbf{R}_1|^3} + \frac{q_2(\mathbf{R} - \mathbf{R}_2)}{|\mathbf{R} - \mathbf{R}_2|^3} \right]. \end{aligned} \quad (4.18)$$

Generalizing the preceding result to the case of N point charges, the electric field $\mathbf{E}$ at point P with position vector $\mathbf{R}$ due to charges $q_1, q_2, \dots, q_N$ located at points with position vectors $\mathbf{R}_1, \mathbf{R}_2, \dots, \mathbf{R}_N$ equals the vector sum of the electric fields induced by all the individual charges. Thus,

$$\mathbf{E} = \frac{1}{4\pi\epsilon} \sum_{i=1}^N \frac{q_i(\mathbf{R} - \mathbf{R}_i)}{|\mathbf{R} - \mathbf{R}_i|^3} \quad (\text{V/m}). \quad (4.19)$$

Example 4-3: Electric Field Due to Two Point Charges

Two point charges with

$$q_1 = 2 \times 10^{-5} \text{ C}$$

and

$$q_2 = -4 \times 10^{-5} \text{ C}$$

are located in free space at points with Cartesian coordinates $(1, 3, -1)$ and $(-3, 1, -2)$, respectively. Find (a) the electric field $\mathbf{E}$ at $(3, 1, -2)$ and (b) the force on a $8 \times 10^{-5} \text{ C}$ charge located at that point. All distances are in meters.

Solution: (a) From Eq. (4.18), the electric field $\mathbf{E}$ with $\epsilon = \epsilon_0$ (free space) is

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \left[q_1 \frac{(\mathbf{R} - \mathbf{R}_1)}{|\mathbf{R} - \mathbf{R}_1|^3} + q_2 \frac{(\mathbf{R} - \mathbf{R}_2)}{|\mathbf{R} - \mathbf{R}_2|^3} \right] \quad (\text{V/m}).$$

The vectors $\mathbf{R}_1$, $\mathbf{R}_2$, and $\mathbf{R}$ are

$$\begin{aligned} \mathbf{R}_1 &= \hat{\mathbf{x}} + \hat{\mathbf{y}}3 - \hat{\mathbf{z}}, \\ \mathbf{R}_2 &= -\hat{\mathbf{x}}3 + \hat{\mathbf{y}} - \hat{\mathbf{z}}2, \\ \mathbf{R} &= \hat{\mathbf{x}}3 + \hat{\mathbf{y}} - \hat{\mathbf{z}}2. \end{aligned}$$

Hence,

$$\begin{aligned} \mathbf{E} &= \frac{1}{4\pi\epsilon_0} \left[\frac{2(\hat{\mathbf{x}}2 - \hat{\mathbf{y}}2 - \hat{\mathbf{z}})}{27} - \frac{4(\hat{\mathbf{x}}6)}{216} \right] \times 10^{-5} \\ &= \frac{\hat{\mathbf{x}} - \hat{\mathbf{y}}4 - \hat{\mathbf{z}}2}{108\pi\epsilon_0} \times 10^{-5} \quad (\text{V/m}). \end{aligned}$$

(b) The force on q_3 is

$$\begin{aligned} \mathbf{F} = q_3\mathbf{E} &= 8 \times 10^{-5} \times \frac{\hat{\mathbf{x}} - \hat{\mathbf{y}}4 - \hat{\mathbf{z}}2}{108\pi\epsilon_0} \times 10^{-5} \\ &= \frac{\hat{\mathbf{x}}2 - \hat{\mathbf{y}}8 - \hat{\mathbf{z}}4}{27\pi\epsilon_0} \times 10^{-10} \quad (\text{N}). \end{aligned}$$

Exercise 4-3: Four charges of $10 \mu\text{C}$ each are located in free space at points with Cartesian coordinates $(-3, 0, 0)$, $(3, 0, 0)$, $(0, -3, 0)$, and $(0, 3, 0)$. Find the force on a $20 \mu\text{C}$ charge located at $(0, 0, 4)$. All distances are in meters.

Answer: $\mathbf{F} = \hat{\mathbf{z}}0.23 \text{ N}$. (See 4.19.)

Exercise 4-4: Two identical charges are located on the x axis at $x = 3$ and $x = 7$. At what point in space is the net electric field zero?

Answer: At point $(5, 0, 0)$. (See 4.19.)

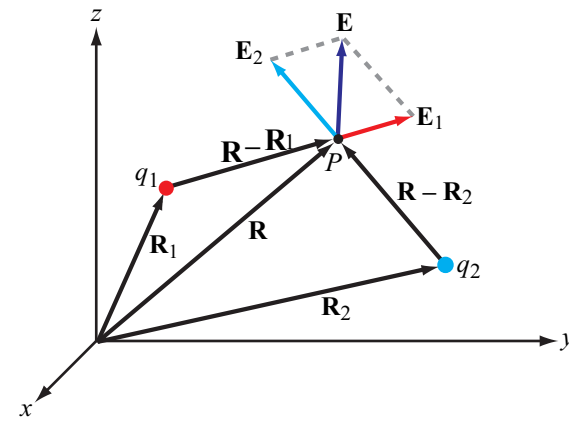


图 4-4 P 处两个电荷产生的电场 $\mathbf{E}$ 等于 $\mathbf{E}_1$ 和 $\mathbf{E}_2$ 的矢量和。

图4-4)。电场 $\mathbf{E}$ 将在具有位置矢量 $\mathbf{R}$ 的点 P 处进行评估。在 P 处，仅由 q_1 产生的电场 $\mathbf{E}_1$ 由方程 1 给出。(4.13) 将 $\mathbf{R}$ (即 q_1 和 P 之间的距离) 替换为 $|\mathbf{R} - \mathbf{R}_1|$ ，单位向量 $\hat{\mathbf{R}}$ 替换为 $(\mathbf{R} - \mathbf{R}_1)/|\mathbf{R} - \mathbf{R}_1|$ 。因此，

$$\mathbf{E}_1 = \frac{q_1(\mathbf{R} - \mathbf{R}_1)}{4\pi\epsilon|\mathbf{R} - \mathbf{R}_1|^3} \quad (\text{V/m}). \quad (4.17a)$$

类似地，仅由 q_2 产生的 P 处电场为

$$\mathbf{E}_2 = \frac{q_2(\mathbf{R} - \mathbf{R}_2)}{4\pi\epsilon|\mathbf{R} - \mathbf{R}_2|^3} \quad (\text{V/m}). \quad (4.17b)$$

► 电场遵循线性叠加原理位置。◀

因此，由 q_1 和 q_2 共同产生的 P 处的总电场 $\mathbf{E}$ 确定为

$$\begin{aligned} \mathbf{E} &= \mathbf{E}_1 + \mathbf{E}_2 \\ &= \frac{1}{4\pi\epsilon} \left[\frac{q_1(\mathbf{R} - \mathbf{R}_1)}{|\mathbf{R} - \mathbf{R}_1|^3} + \frac{q_2(\mathbf{R} - \mathbf{R}_2)}{|\mathbf{R} - \mathbf{R}_2|^3} \right]. \end{aligned} \quad (4.18)$$

将前面的结果推广到 N 个点电荷的情况，由于位于位置矢量 $\mathbf{R}_1, \mathbf{R}_2, \dots, \mathbf{R}_N$ 的点上的电荷 $q_1, q_2, \dots, q_N$ ，在位置矢量 $\mathbf{R}$ 的 P 点处的电场 $\mathbf{E}$ 等于所有单个电荷感应的电场的矢量和。因此，

$$\mathbf{E} = \frac{1}{4\pi\epsilon} \sum_{i=1}^N \frac{q_i(\mathbf{R} - \mathbf{R}_i)}{|\mathbf{R} - \mathbf{R}_i|^3} \quad (\text{V/m}). \quad (4.19)$$

例 4-3：两点电荷产生的电场

两点电荷

$$q_1 = 2 \times 10^{-5} \text{ C}$$

和

$$q_2 = -4 \times 10^{-5} \text{ C}$$

分别位于自由空间中具有笛卡尔坐标 $(1, 3, -1)$ 和 $(-3, 1, -2)$ 的点处。求 (a) $(3, 1, -2)$ 处的电场 $\mathbf{E}$ 和 (b) 位于该点的 $8 \times 10^{-5} \text{ C}$ 电荷上的力。所有距离均以米为单位。

解：(a) 根据方程。(4.18)， $\epsilon = \epsilon_0$ (自由空间) 的电场 $\mathbf{E}$ 为

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \left[q_1 \frac{(\mathbf{R} - \mathbf{R}_1)}{|\mathbf{R} - \mathbf{R}_1|^3} + q_2 \frac{(\mathbf{R} - \mathbf{R}_2)}{|\mathbf{R} - \mathbf{R}_2|^3} \right] \quad (\text{V/m}).$$

向量 $\mathbf{R}_1$ 、 $\mathbf{R}_2$ 和 $\mathbf{R}$ 是

$$\begin{aligned} \mathbf{R}_1 &= \hat{\mathbf{x}} + \hat{\mathbf{y}}3 - \hat{\mathbf{z}}, \\ \mathbf{R}_2 &= -\hat{\mathbf{x}}3 + \hat{\mathbf{y}} - \hat{\mathbf{z}}2, \\ \mathbf{R} &= \hat{\mathbf{x}}3 + \hat{\mathbf{y}} - \hat{\mathbf{z}}2. \end{aligned}$$

因此，

$$\begin{aligned} \mathbf{E} &= \frac{1}{4\pi\epsilon_0} \left[\frac{2(\hat{\mathbf{x}}2 - \hat{\mathbf{y}}2 - \hat{\mathbf{z}})}{27} - \frac{4(\hat{\mathbf{x}}6)}{216} \right] \times 10^{-5} \\ &= \frac{\hat{\mathbf{x}} - \hat{\mathbf{y}}4 - \hat{\mathbf{z}}2}{108\pi\epsilon_0} \times 10^{-5} \quad (\text{V/m}). \end{aligned}$$

(b) q_3 上的力为

$$\begin{aligned} \mathbf{F} = q_3\mathbf{E} &= 8 \times 10^{-5} \times \frac{\hat{\mathbf{x}} - \hat{\mathbf{y}}4 - \hat{\mathbf{z}}2}{108\pi\epsilon_0} \times 10^{-5} \\ &= \frac{\hat{\mathbf{x}}2 - \hat{\mathbf{y}}8 - \hat{\mathbf{z}}4}{27\pi\epsilon_0} \times 10^{-10} \quad (\text{N}). \end{aligned}$$

练习 4-3：四个 $10 \mu\text{C}$ 的电荷位于自由空间中笛卡尔坐标为 $(-3, 0, 0)$ 、 $(3, 0, 0)$ 、 $(0, -3, 0)$ 和 $(0, 3, 0)$ 的点处。求 $(0, 0, 4)$ 处 $20 \mu\text{C}$ 电荷所受的力。所有距离均以米为单位。

Answer: $\mathbf{F} = \hat{\mathbf{z}}0.23 \text{ N}$. (See 4.19.)

练习4-4：两个相同的电荷位于 x 轴上的 $x = 3$ 和 $x = 7$ 处。空间中的什么点净电场为零？

答案：在点 $(5, 0, 0)$ 。(参见 4.19.)

Exercise 4-5: In a hydrogen atom, the electron and proton are separated by an average distance of 5.3×10^{-11} m. Find the magnitude of the electrical force F_e between the two particles, and compare it with the gravitational force F_g between them.

Answer: $F_e = 8.2 \times 10^{-8}$ N, and $F_g = 3.6 \times 10^{-47}$ N. (See )

4-3.2 Electric Field Due to a Charge Distribution

We now extend the results obtained for the field due to discrete point charges to continuous charge distributions. Consider a volume $\mathcal{V}'$ that contains a distribution of electric charge with volume charge density ρ_v , which may vary spatially within $\mathcal{V}'$ (**Fig. 4-5**). The differential electric field at a point P due to a differential amount of charge $dq = \rho_v d\mathcal{V}'$ contained in a differential volume $d\mathcal{V}'$ is

$$d\mathbf{E} = \hat{\mathbf{R}}' \frac{dq}{4\pi\epsilon R'^2} = \hat{\mathbf{R}}' \frac{\rho_v d\mathcal{V}'}{4\pi\epsilon R'^2}, \quad (4.20)$$

where $\mathbf{R}'$ is the vector from the differential volume $d\mathcal{V}'$ to point P . Applying the principle of linear superposition, the total electric field $\mathbf{E}$ is obtained by integrating the fields due to all differential charges in $\mathcal{V}'$. Thus,

$$\mathbf{E} = \int_{\mathcal{V}'} d\mathbf{E} = \frac{1}{4\pi\epsilon} \int_{\mathcal{V}'} \hat{\mathbf{R}}' \frac{\rho_v d\mathcal{V}'}{R'^2}. \quad (4.21a)$$

(volume distribution)

It is important to note that, in general, both R' and $\hat{\mathbf{R}}'$ vary as a function of position over the integration volume $\mathcal{V}'$.

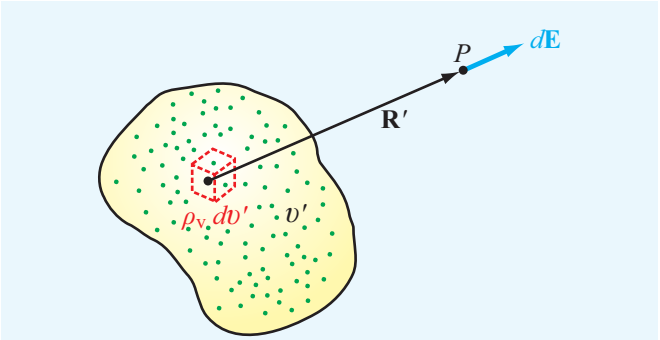


Figure 4-5 Electric field due to a volume charge distribution.

If the charge is distributed across a surface S' with surface charge density ρ_s , then $dq = \rho_s ds'$, and if it is distributed along a line l' with a line charge density ρ_ℓ , then $dq = \rho_\ell dl'$. Accordingly, the electric fields due to surface and line charge distributions are

$$\mathbf{E} = \frac{1}{4\pi\epsilon} \int_{S'} \hat{\mathbf{R}}' \frac{\rho_s ds'}{R'^2}, \quad (4.21b)$$

(surface distribution)

$$\mathbf{E} = \frac{1}{4\pi\epsilon} \int_{l'} \hat{\mathbf{R}}' \frac{\rho_\ell dl'}{R'^2}. \quad (4.21c)$$

(line distribution)

Example 4-4: Electric Field of a Ring of Charge

A ring of charge of radius b is characterized by a uniform line charge density of positive polarity ρ_ℓ . The ring resides in free space and is positioned in the x - y plane, as shown in **Fig. 4-6**. Determine the electric field intensity $\mathbf{E}$ at a point $P = (0, 0, h)$ along the axis of the ring at a distance h from its center.

Solution: We start by considering the electric field generated by a differential ring segment with cylindrical coordinates $(b, \phi, 0)$ in **Fig. 4-6(a)**. The segment has length $dl = b d\phi$ and contains charge $dq = \rho_\ell dl = \rho_\ell b d\phi$. The distance vector $\mathbf{R}'_1$ from segment 1 to point $P = (0, 0, h)$ is

$$\mathbf{R}'_1 = -\hat{\mathbf{r}}b + \hat{\mathbf{z}}h,$$

from which it follows that

$$R'_1 = |\mathbf{R}'_1| = \sqrt{b^2 + h^2}, \quad \hat{\mathbf{R}}'_1 = \frac{\mathbf{R}'_1}{|\mathbf{R}'_1|} = \frac{-\hat{\mathbf{r}}b + \hat{\mathbf{z}}h}{\sqrt{b^2 + h^2}}.$$

The electric field at $P = (0, 0, h)$ due to the charge in segment 1 therefore is

$$d\mathbf{E}_1 = \frac{1}{4\pi\epsilon_0} \hat{\mathbf{R}}'_1 \frac{\rho_\ell dl}{R'^2_1} = \frac{\rho_\ell b}{4\pi\epsilon_0} \frac{(-\hat{\mathbf{r}}b + \hat{\mathbf{z}}h)}{(b^2 + h^2)^{3/2}} d\phi.$$

The field $d\mathbf{E}_1$ has component dE_{1r} along $-\hat{\mathbf{r}}$ and component dE_{1z} along $\hat{\mathbf{z}}$. From symmetry considerations, the field $d\mathbf{E}_2$ generated by differential segment 2 in **Fig. 4-6(b)**, which is located diametrically opposite to segment 1, is identical to $d\mathbf{E}_1$ except that the $\hat{\mathbf{r}}$ component of $d\mathbf{E}_2$ is opposite that

练习 4-5：在氢原子中，电子和质子的平均距离为 5.3×10^{-11} m。求两个粒子之间的电力 F_e 的大小，并将其与它们之间的重力 F_g 进行比较。

答案： $F_e = 8.2 \times 10^{-8}$ N，和 $F_g = 3.6 \times 10^{-47}$ N。（参见 。）

4-3.2 电荷分布产生的电场

现在，我们将由于离散点电荷而获得的结果扩展到连续电荷分布。考虑一个体积 $\mathcal{V}'$ ，它包含具有体积电荷密度 ρ_v 的电荷分布，其可能在 $\mathcal{V}'$ （内随空间变化。4-5）。由于微分体积 $d\mathcal{V}'$ 中包含的微分电荷量 $dq = \rho_v d\mathcal{V}'$ ，在点 P 处的微分电场为

$$d\mathbf{E} = \hat{\mathbf{R}}' \frac{dq}{4\pi\epsilon R'^2} = \hat{\mathbf{R}}' \frac{\rho_v d\mathcal{V}'}{4\pi\epsilon R'^2}, \quad (4.20)$$

其中 $\mathbf{R}'$ 是微分体积 $d\mathcal{V}'$ 到点 P 的矢量。应用线性叠加原理，对 $\mathcal{V}'$ 中所有微分电荷产生的电场进行积分，得到总电场 $\mathbf{E}$ 。因此，

$$\mathbf{E} = \int_{\mathcal{V}'} d\mathbf{E} = \frac{1}{4\pi\epsilon} \int_{\mathcal{V}'} \hat{\mathbf{R}}' \frac{\rho_v d\mathcal{V}'}{R'^2}. \quad (4.21a)$$

(volume distribution)

值得注意的是，一般来说， R' 和 $\hat{\mathbf{R}}'$ 都随着积分体积 $\mathcal{V}'$ 上的位置而变化。

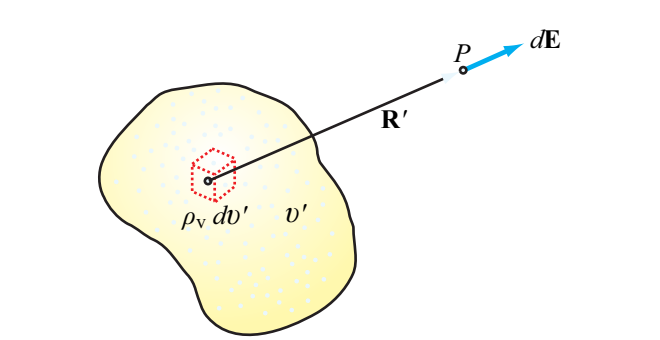


图 4-5 体积电荷分布产生的电场。

如果电荷以表面电荷密度 ρ_s 分布在表面 S' 上，则为 $dq = \rho_s ds'$ ；如果电荷以线电荷密度 ρ_ℓ 沿线 l' 分布，则为 $dq = \rho_\ell dl'$ 。因此，由于表面和线电荷分布而产生的电场为

$$\mathbf{E} = \frac{1}{4\pi\epsilon} \int_{S'} \hat{\mathbf{R}}' \frac{\rho_s ds'}{R'^2}, \quad (4.21b)$$

(表面分布)

$$\mathbf{E} = \frac{1}{4\pi\epsilon} \int_{l'} \hat{\mathbf{R}}' \frac{\rho_\ell dl'}{R'^2}. \quad (4.21c)$$

(line distribution)

例 4-4：电荷环的电场

半径为 b 的电荷环的特征是均匀的正极性线电荷密度 ρ_ℓ 。该环位于自由空间中，位于 x - y 平面中，如图 4-6 所示。确定沿环轴距环中心距离 h 的点 $P = (0, 0, h)$ 处的电场强度 $\mathbf{E}$ 。

解：我们首先考虑图4-6 (a) 中柱坐标 $(b, \phi, 0)$ 的微分环段产生的电场。该段的长度为 $dl = b d\phi$ ，并包含电荷 $dq = \rho_\ell dl = \rho_\ell b d\phi$ 。从线段 1 到点 $P = (0, 0, h)$ 的距离向量 $\mathbf{R}'_1$ 为

$$\mathbf{R}'_1 = -\hat{\mathbf{r}}b + \hat{\mathbf{z}}h,$$

由此可见

$$R'_1 = |\mathbf{R}'_1| = \sqrt{b^2 + h^2}, \quad \hat{\mathbf{R}}'_1 = \frac{\mathbf{R}'_1}{|\mathbf{R}'_1|} = \frac{-\hat{\mathbf{r}}b + \hat{\mathbf{z}}h}{\sqrt{b^2 + h^2}}.$$

因此，由于段 1 中的电荷， $P = (0, 0, h)$ 处的电场为

$$d\mathbf{E}_1 = \frac{1}{4\pi\epsilon_0} \hat{\mathbf{R}}'_1 \frac{\rho_\ell dl}{R'^2_1} = \frac{\rho_\ell b}{4\pi\epsilon_0} \frac{(-\hat{\mathbf{r}}b + \hat{\mathbf{z}}h)}{(b^2 + h^2)^{3/2}} d\phi.$$

域 $d\mathbf{E}_1$ 具有沿 $-\hat{\mathbf{r}}$ 的分量 dE_{1r} 和沿 $\hat{\mathbf{z}}$ 的分量 dE_{1z} 。从对称性考虑，图 4-6(b) 中与线段 1 完全相反的微分线段 2 生成的场 $d\mathbf{E}_2$ 与 $d\mathbf{E}_1$ 相同，只是 $d\mathbf{E}_2$ 的 $\hat{\mathbf{r}}$ 分量与

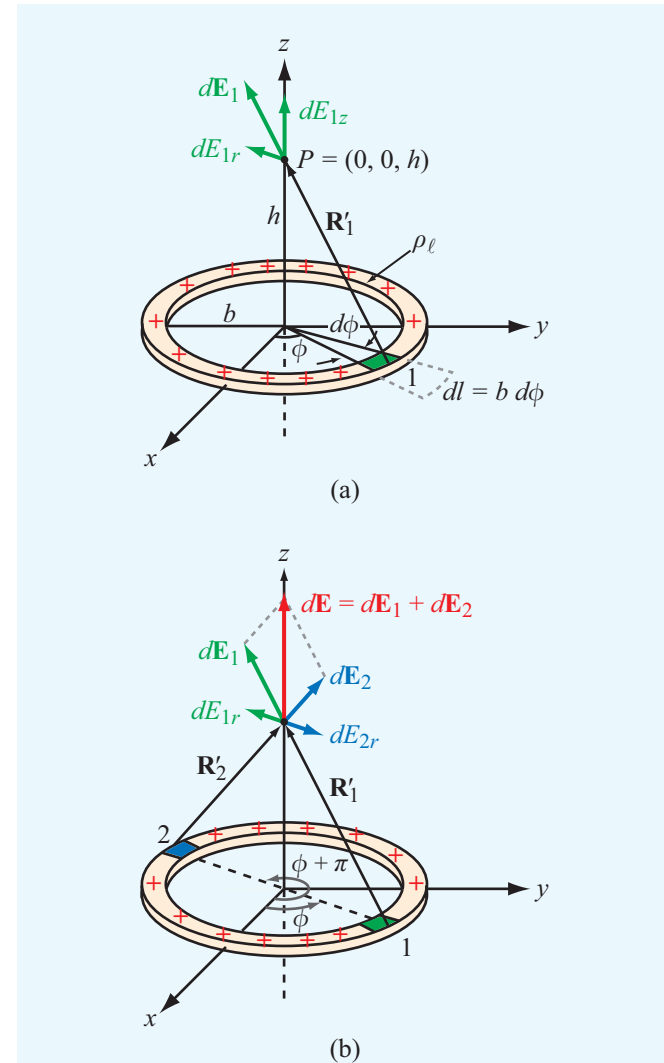


Figure 4-6 Ring of charge with line density ρ_ℓ . (a) The field $d\mathbf{E}_1$ due to infinitesimal segment 1 and (b) the fields $d\mathbf{E}_1$ and $d\mathbf{E}_2$ due to segments at diametrically opposite locations (Example 4-4).

of $d\mathbf{E}_1$. Hence, the $\hat{\mathbf{r}}$ components in the sum cancel and the $\hat{\mathbf{z}}$ contributions add. The sum of the two contributions is

$$d\mathbf{E} = d\mathbf{E}_1 + d\mathbf{E}_2 = \hat{\mathbf{z}} \frac{\rho_\ell b h}{2\pi\epsilon_0} \frac{d\phi}{(b^2 + h^2)^{3/2}}. \quad (4.22)$$

Since for every ring segment in the semicircle defined over the azimuthal range $0 \leq \phi \leq \pi$ (the right-hand half of the circular ring) there is a corresponding segment located diametrically

opposite at $(\phi + \pi)$, we can obtain the total field generated by the ring by integrating Eq. (4.22) over a semicircle as

$$\begin{aligned} \mathbf{E} &= \hat{\mathbf{z}} \frac{\rho_\ell b h}{2\pi\epsilon_0 (b^2 + h^2)^{3/2}} \int_0^\pi d\phi \\ &= \hat{\mathbf{z}} \frac{\rho_\ell b h}{2\epsilon_0 (b^2 + h^2)^{3/2}} = \hat{\mathbf{z}} \frac{h}{4\pi\epsilon_0 (b^2 + h^2)^{3/2}} Q, \end{aligned} \quad (4.23)$$

where $Q = 2\pi b \rho_\ell$ is the total charge on the ring.

Example 4-5: Electric Field of a Circular Disk of Charge

Find the electric field at point P with Cartesian coordinates $(0, 0, h)$ due to a circular disk of radius a and uniform charge density ρ_s residing in the x - y plane (Fig. 4-7). Also, evaluate $\mathbf{E}$ due to an infinite sheet of charge density ρ_s by letting $a \rightarrow \infty$.

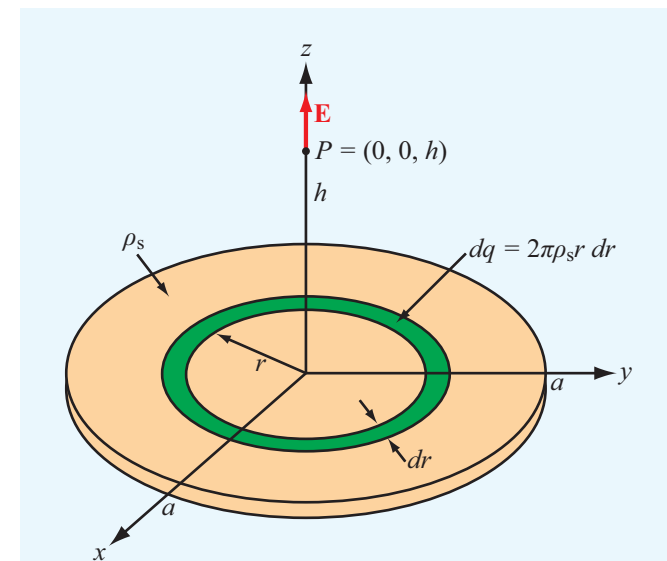


Figure 4-7 Circular disk of charge with surface charge density ρ_s . The electric field at $P = (0, 0, h)$ points along the z direction (Example 4-5).

Solution: Building on the expression obtained in Example 4-4 for the on-axis electric field due to a circular ring of charge, we can determine the field due to the circular disk by treating the disk as a set of concentric rings. A ring of radius r and width dr has an area $ds = 2\pi r dr$ and contains charge $dq = \rho_s ds = 2\pi \rho_s r dr$. Upon using this expression in

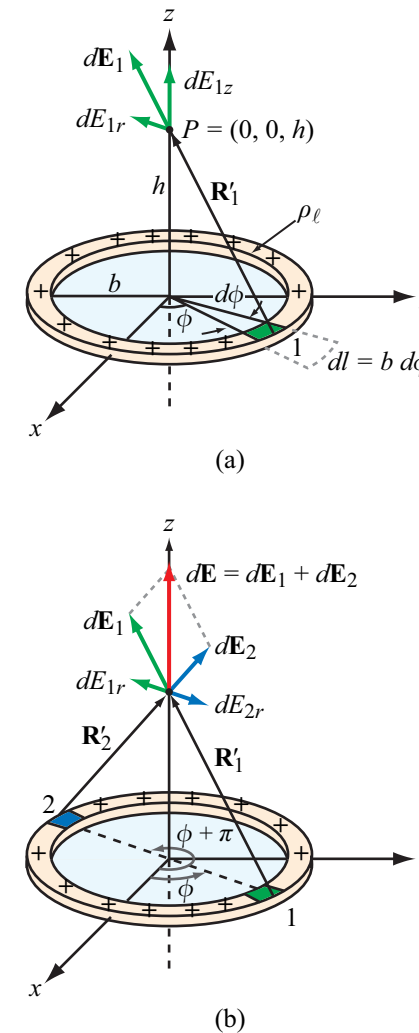


图 4-6 具有线密度 ρ_ℓ 的电荷环。(a) 由于无穷小线段 1 产生的场 $d\mathbf{E}_1$, (b) 由于位于完全相反位置的线段而产生的场 $d\mathbf{E}_1$ 和 $d\mathbf{E}_2$ (例 4-4)。

$d\mathbf{E}_1$ 。因此, 总和中的 $\hat{\mathbf{r}}$ 分量抵消, 而 $\hat{\mathbf{z}}$ 贡献相加。两个贡献的总和是

$$d\mathbf{E} = d\mathbf{E}_1 + d\mathbf{E}_2 = \hat{\mathbf{z}} \frac{\rho_\ell b h}{2\pi\epsilon_0} \frac{d\phi}{(b^2 + h^2)^{3/2}}. \quad (4.22)$$

因为对于在方位角范围 $0 \leq \phi \leq \pi$ (圆环的右半部分) 上定义的半圆中的每个环段, 都有一个位于直径方向的相应段

相反, 在 $(\phi + \pi)$ 处, 我们可以通过积分方程得到环产生的总场。(4.22) 在半圆上为

$$\begin{aligned} \mathbf{E} &= \hat{\mathbf{z}} \frac{\rho_\ell b h}{2\pi\epsilon_0 (b^2 + h^2)^{3/2}} \int_0^\pi d\phi \\ &= \hat{\mathbf{z}} \frac{\rho_\ell b h}{2\epsilon_0 (b^2 + h^2)^{3/2}} = \hat{\mathbf{z}} \frac{h}{4\pi\epsilon_0 (b^2 + h^2)^{3/2}} Q, \end{aligned} \quad (4.23)$$

其中 $Q = 2\pi b \rho_\ell$ 是环上的总电荷。

例 4-5: 带电圆盘的电场

求由于半径为 a 的圆盘和 x - y 平面内均匀电荷密度 ρ_s 导致的笛卡尔坐标 $(0, 0, h)$ 处 P 点的电场 (图 4-7)。另外, 通过设 $a \rightarrow \infty$ 来评估由于无限电荷密度 ρ_s 引起的 $\mathbf{E}$ 。

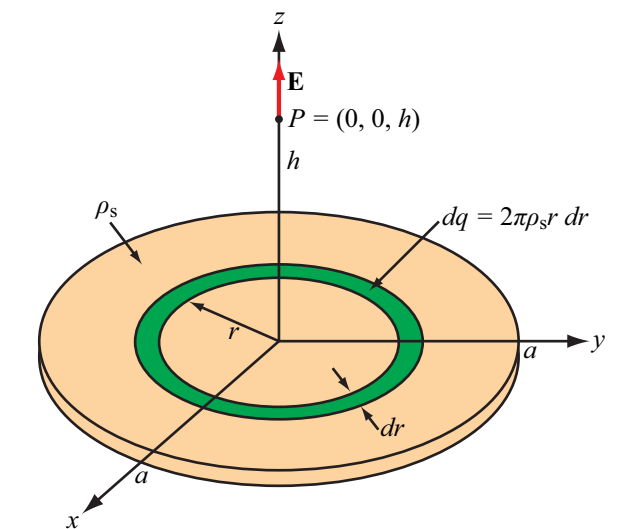


图 4-7 表面电荷密度为 ρ_s 的电荷圆盘。 $P = (0, 0, h)$ 处的电场沿 z 方向指向 (示例 4-5)。

解: 根据例 4-4 中获得的关于圆环电荷产生的轴上电场的表达式, 我们可以通过将圆盘视为一组同心环来确定圆盘产生的电场。半径为 r 、宽度为 dr 的环的面积为 $ds = 2\pi r dr$, 并且包含电荷 $dq = \rho_s ds = 2\pi \rho_s r dr$ 。在使用这个表达式时

Eq. (4.23) and also replacing b with r , we obtain the following expression for the field due to the ring:

$$d\mathbf{E} = \hat{\mathbf{z}} \frac{h}{4\pi\epsilon_0(r^2 + h^2)^{3/2}} (2\pi\rho_s r \, dr).$$

The total field at P is obtained by integrating the expression over the limits $r = 0$ to $r = a$:

$$\mathbf{E} = \hat{\mathbf{z}} \frac{\rho_s h}{2\epsilon_0} \int_0^a \frac{r \, dr}{(r^2 + h^2)^{3/2}} = \pm \hat{\mathbf{z}} \frac{\rho_s}{2\epsilon_0} \left[1 - \frac{|h|}{\sqrt{a^2 + h^2}} \right], \tag{4.24}$$

with the plus sign for $h > 0$ (P above the disk) and the minus sign when $h < 0$ (P below the disk).
For an infinite sheet of charge with $a = \infty$,

$$\mathbf{E} = \pm \hat{\mathbf{z}} \frac{\rho_s}{2\epsilon_0}.$$

(infinite sheet of charge)

(4.25)

We note that for an infinite sheet of charge $\mathbf{E}$ is the same at all points above the x – y plane, and a similar statement applies for points below the x – y plane.

Concept Question 4-4: When characterizing the electrical permittivity of a material, what do the terms *linear* and *isotropic* mean?

Concept Question 4-5: If the electric field is zero at a given point in space, does this imply the absence of electric charges?

Concept Question 4-6: State the principle of linear superposition as it applies to the electric field due to a distribution of electric charge.

Exercise 4-6: An infinite sheet with uniform surface charge density ρ_s is located at $z = 0$ (x – y plane), and another infinite sheet with density $-\rho_s$ is located at $z = 2$ m with both in free space. Determine $\mathbf{E}$ everywhere.

Answer: $\mathbf{E} = 0$ for $z < 0$; $\mathbf{E} = \hat{\mathbf{z}}\rho_s/\epsilon_0$ for $0 < z < 2$ m; and $\mathbf{E} = 0$ for $z > 2$ m. (See [EM](#).)

4-4 Gauss’s Law

In this section, we use Maxwell’s equations to confirm the expressions for the electric field implied by Coulomb’s law,

and propose alternative techniques for evaluating electric fields induced by electric charge. To that end, we restate Eq. (4.1a):

$$\nabla \cdot \mathbf{D} = \rho_v,$$

(differential form of Gauss’s law)

(4.26)

which is referred to as the differential form of ***Gauss’s law***. The adjective “differential” refers to the fact that the divergence operation involves spatial derivatives. As we see shortly, Eq. (4.26) can be converted to an integral form. When solving electromagnetic problems, we often go back and forth between equations in differential and integral form, depending on which of the two happens to be the more applicable or convenient to use. To convert Eq. (4.26) into integral form, we multiply both sides by $d\mathcal{V}$ and evaluate their integrals over an arbitrary volume $\mathcal{V}$:

$$\int_{\mathcal{V}} \nabla \cdot \mathbf{D} \, d\mathcal{V} = \int_{\mathcal{V}} \rho_v \, d\mathcal{V} = Q. \tag{4.27}$$

Here, Q is the total charge enclosed in $\mathcal{V}$. The divergence theorem, given by Eq. (3.98), states that the volume integral of the divergence of any vector over a volume $\mathcal{V}$ equals the total outward flux of that vector through the surface S enclosing $\mathcal{V}$. Thus, for the vector $\mathbf{D}$,

$$\int_{\mathcal{V}} \nabla \cdot \mathbf{D} \, d\mathcal{V} = \oint_S \mathbf{D} \cdot d\mathbf{s}. \tag{4.28}$$

Comparison of Eq. (4.27) with Eq. (4.28) leads to

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q.$$

(integral form of Gauss’s law)

(4.29)

► The integral form of Gauss’s law is illustrated diagrammatically in **Fig. 4-8**; for each differential surface element $d\mathbf{s}$, $\mathbf{D} \cdot d\mathbf{s}$ is the electric field flux flowing outward of $\mathcal{V}$ through $d\mathbf{s}$, and the total flux through surface S equals the enclosed charge Q . The surface S is called a ***Gaussian surface***. ◀

The integral form of Gauss’s law can be applied to determine $\mathbf{D}$ due to a single isolated point charge q by enclosing the latter with a closed, spherical, Gaussian surface S of arbitrary radius R centered at q (**Fig. 4-9**). From symmetry considerations and assuming that q is positive, the direction of $\mathbf{D}$ must be radially outward along the unit vector $\hat{\mathbf{R}}$, and D_R , which is the magnitude of $\mathbf{D}$, must be the same at all points on S . Thus, at any point on S ,

$$\mathbf{D} = \hat{\mathbf{R}}D_R, \tag{4.30}$$

等式。(4.23) 并且将 b 替换为 r , 我们得到以下由环引起的场的表达式:

$$d\mathbf{E} = \hat{\mathbf{z}} \frac{h}{4\pi\epsilon_0(r^2 + h^2)^{3/2}} (2\pi\rho_s r \, dr).$$

P 处的总场通过在 $r = 0$ 到 $r = a$ 的范围内积分表达式来获得:

$$\mathbf{E} = \hat{\mathbf{z}} \frac{\rho_s h}{2\epsilon_0} \int_0^a \frac{r \, dr}{(r^2 + h^2)^{3/2}} = \pm \hat{\mathbf{z}} \frac{\rho_s}{2\epsilon_0} \left[1 - \frac{|h|}{\sqrt{a^2 + h^2}} \right], \tag{4.24}$$

当 $h > 0$ (P 在盘上方) 时为加号, 当 $h < 0$ (P 在盘下方) 时为负号。
对于具有 $= \infty$ 的无限电荷片,

$$\mathbf{E} = \pm \hat{\mathbf{z}} \frac{\rho_s}{2\epsilon_0}.$$

(infinite sheet of charge)

(4.25)

我们注意到, 对于无限电荷片, $\mathbf{E}$ 在 x – y 平面上方的所有点上都是相同的, 并且类似的陈述适用于 x – y 平面下方的点。

概念问题 4-4: 在表征材料的介电常数时, 术语 “线性” 和 “各向同性” 意味着什么?

概念问题 4-5: 如果空间中给定点的电场为零, 这是否意味着不存在电荷?

概念问题 4-6: 陈述线性叠加原理, 因为它适用于电荷分布产生的电场。

练习 4-6: 具有均匀表面电荷密度 ρ_s 的无限片位于 $z = 0$ (x – y 平面), 另一个具有密度 $-\rho_s$ 的无限片位于 $z = 2$ m, 两者都在自由空间中。处处求 $\mathbf{E}$ 。

答案: $\mathbf{E} = 0$ 代表 $z < 0$; $\mathbf{E} = \hat{\mathbf{z}}\rho_s/\epsilon_0$ 为 $0 < z < 2$ m; 和 $\mathbf{E} = 0$ 对于 $z > 2$ m。◯(参见^{EM}。)

4-4 高斯定律

在本节中, 我们使用麦克斯韦方程组来确认库仑定律所隐含的电场表达式,

并提出评估电荷感应电场的替代技术。为此, 我们重申方程。(4.1a):

$$\nabla \cdot \mathbf{D} = \rho_v,$$

(高斯定律的微分形式)

(4.26)

这被称为高斯定律的微分形式。形容词 “微分” 指的是散度运算涉及空间导数。正如我们很快看到的, 等式。(4.26) 可以转化为积分形式。在解决电磁问题时, 我们经常在微分形式和积分形式的方程之间来回切换, 具体取决于两者中哪一个更适用或更方便使用。转换方程。将式 (4.26) 转化为积分形式, 我们将两边乘以 $d\mathcal{V}$ 并计算它们在任意体积 $\mathcal{V}$ 上的积分:

$$\int_{\mathcal{V}} \nabla \cdot \mathbf{D} \, d\mathcal{V} = \int_{\mathcal{V}} \rho_v \, d\mathcal{V} = Q. \tag{4.27}$$

这里, Q 是包含在 $\mathcal{V}$ 中的总电荷。散度定理, 由方程给出。(3.98) 指出, 任何向量在体积 $\mathcal{V}$ 上的散度的体积分等于该向量通过包围 $\mathcal{V}$ 的表面 S 的总向外通量。因此, 对于向量 $\mathbf{D}$,

$$\int_{\mathcal{V}} \nabla \cdot \mathbf{D} \, d\mathcal{V} = \oint_S \mathbf{D} \cdot d\mathbf{s}. \tag{4.28}$$

等式的比较 (4.27) 与方程。(4.28) 导致

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q.$$

(高斯定律的积分形式)

(4.29)

高斯定律的积分形式见图4-8。对于有向面元 $d\mathbf{s}$, $\mathbf{D} \cdot d\mathbf{s}$ 是沿曲面外法向穿过面元的电通量。闭合曲面 S 上的总电通量等于其中包围的电荷 Q ; S 称为高斯面。

高斯定律的积分形式可用于确定由单个孤立点电荷 q 引起的 $\mathbf{D}$, 方法是用一个以 q 为中心、任意半径 R 的闭合球形高斯表面 S 包围后者 (图 4-9)。从对称性考虑, 假设 q 为正, $\mathbf{D}$ 的方向必须沿着单位向量 $\hat{\mathbf{R}}$ 径向向外, 并且 D_R (即 $\mathbf{D}$ 的大小) 在 S 上的所有点上必须相同。因此, 在 S 上的任何点,

$$\mathbf{D} = \hat{\mathbf{R}}D_R, \tag{4.30}$$

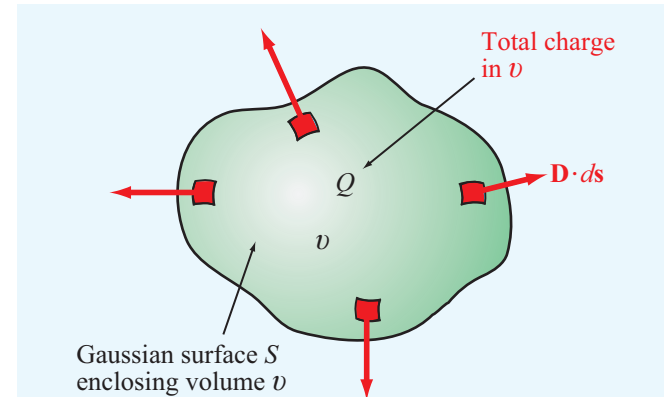


Figure 4-8 The integral form of Gauss's law states that the outward flux of $\mathbf{D}$ through a surface is proportional to the enclosed charge Q .

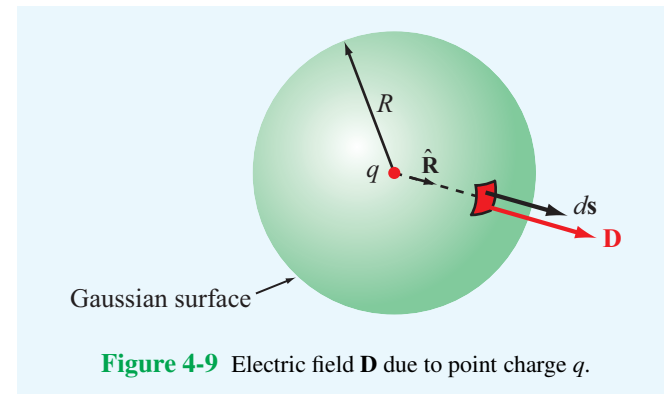


Figure 4-9 Electric field $\mathbf{D}$ due to point charge q .

and $d\mathbf{s} = \hat{\mathbf{R}} ds$. Applying Gauss's law gives

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = \oint_S \hat{\mathbf{R}} D_R \cdot \hat{\mathbf{R}} ds = \oint_S D_R ds = D_R (4\pi R^2) = q. \quad (4.31)$$

Solving for D_R and then inserting the result in Eq. (4.30) gives the expression for the electric field $\mathbf{E}$ induced by an isolated point charge in a medium with permittivity ϵ :

$$\mathbf{E} = \frac{\mathbf{D}}{\epsilon} = \hat{\mathbf{R}} \frac{q}{4\pi\epsilon R^2} \quad (\text{V/m}). \quad (4.32)$$

This is identical with Eq. (4.13) obtained from Coulomb's law; after all, Maxwell's equations incorporate Coulomb's law. For this simple case of an isolated point charge, it does not matter whether Coulomb's law or Gauss's law is used to obtain the expression for $\mathbf{E}$. However, it does matter which approach we follow when we deal with multiple point charges or continuous charge distributions. Even though Coulomb's law can be used

to find $\mathbf{E}$ for any specified distribution of charge, Gauss's law is easier to apply than Coulomb's law, but its utility is limited to symmetrical charge distributions.

► Gauss's law, as given by Eq. (4.29), provides a convenient method for determining the flux density $\mathbf{D}$ when the charge distribution possesses symmetry properties that allow us to infer the variations of the magnitude and direction of $\mathbf{D}$ as a function of spatial location. This facilitates the integration of $\mathbf{D}$ over a cleverly chosen Gaussian surface. ◀

Because at every point on the surface the direction of $d\mathbf{s}$ is along its outward normal, only the normal component of $\mathbf{D}$ at the surface contributes to the integral in Eq. (4.29). To successfully apply Gauss's law, the surface S should be chosen from symmetry considerations so that, across each subsurface of S , $\mathbf{D}$ is constant in magnitude and its direction is either normal or purely tangential to the subsurface. These aspects are illustrated in Example 4-6.

Example 4-6: Electric Field of an Infinite Line Charge

Use Gauss's law to obtain an expression for $\mathbf{E}$ due to an infinitely long line with uniform charge density ρ_ℓ that resides along the z axis in free space.

Solution: Since the charge density along the line is uniform, infinite in extent, and residing along the z axis, symmetry considerations dictate that $\mathbf{D}$ is in the radial $\hat{\mathbf{r}}$ direction and cannot depend on ϕ or z . Thus, $\mathbf{D} = \hat{\mathbf{r}} D_r$. Therefore, we construct a finite cylindrical Gaussian surface of radius r and height h , which is concentric around the line of charge (Fig. 4-10). The total charge contained within the cylinder is $Q = \rho_\ell h$. Since $\mathbf{D}$ is along $\hat{\mathbf{r}}$, the top and bottom surfaces of the cylinder do not contribute to the surface integral on the left-hand side of Eq. (4.29); that is, only the curved surface contributes to the integral. Hence,

$$\int_{z=0}^h \int_{\phi=0}^{2\pi} \hat{\mathbf{r}} D_r \cdot \hat{\mathbf{r}} r d\phi dz = \rho_\ell h$$

or

$$2\pi h D_r r = \rho_\ell h,$$

which yields

$$\mathbf{E} = \frac{\mathbf{D}}{\epsilon_0} = \hat{\mathbf{r}} \frac{D_r}{\epsilon_0} = \hat{\mathbf{r}} \frac{\rho_\ell}{2\pi\epsilon_0 r}. \quad (4.33)$$

(infinite line charge)

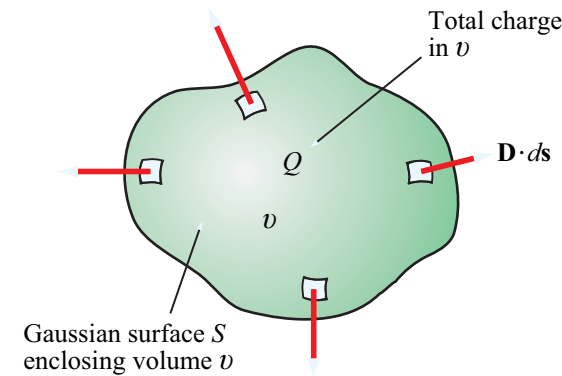


图 4-8 高斯定律的积分形式指出，通过表面的 $\mathbf{D}$ 的向外通量与所包含的电荷 Q 成正比。

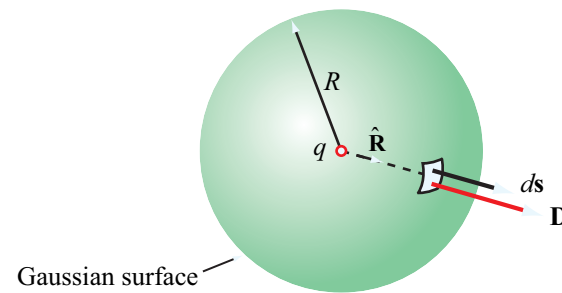


图 4-9 点电荷 q 产生的电场 $\mathbf{D}$ 。

和 $d\mathbf{s} = \hat{\mathbf{R}} ds$ 。应用高斯定律给出

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = \oint_S \hat{\mathbf{R}} D_R \cdot \hat{\mathbf{R}} ds = \oint_S D_R ds = D_R (4\pi R^2) = q. \quad (4.31)$$

求解 D_R ，然后将结果代入方程 (1)。 (4.30) 给出了介电常数为 ϵ 的介质中孤立点电荷感应的电场 $\mathbf{E}$ 的表达式：

$$\mathbf{E} = \frac{\mathbf{D}}{\epsilon} = \hat{\mathbf{R}} \frac{q}{4\pi\epsilon R^2} \quad (\text{V/m}). \quad (4.32)$$

这与等式相同。 (4.13) 由库仑定律获得；毕竟，麦克斯韦方程组包含了库仑定律。对于孤立点电荷的这种简单情况，使用库仑定律还是高斯定律来获得 $\mathbf{E}$ 的表达式并不重要。但是，当我们处理多个点电荷或连续电荷分布时，我们采用哪种方法就很重要了。尽管可以使用库仑定律

对于任何指定的电荷分布，要求 $\mathbf{E}$ ，高斯定律比库仑定律更容易应用，但其实用性仅限于对称电荷分布。

► 高斯定律，如方程所示。 (4.29) 提供了一种在电荷分布具有对称性时确定通量密度 $\mathbf{D}$ 的简便方法，使我们能够推断出 $\mathbf{D}$ 的大小和方向随空间位置的函数的变化。这有利于 $\mathbf{D}$ 在巧妙选择的高斯表面上的积分。 ◀

因为在表面上的每个点， $d\mathbf{s}$ 的方向都是沿着其向外法线的，因此只有表面上 $\mathbf{D}$ 的法线分量对方程 (1) 中的积分有贡献。 (4.29)。为了成功应用高斯定律，应根据对称性考虑选择表面 S ，以便在 S 的每个子表面上， $\mathbf{D}$ 的大小恒定，并且其方向与子表面垂直或纯相切。这些方面在示例 4-6 中进行了说明。

例 4-6：无限线电荷的电场

由于自由空间中沿 z 轴存在一条具有均匀电荷密度 ρ_ℓ 的无限长线，因此可以使用高斯定律获得 $\mathbf{E}$ 的表达式。

解：由于沿线的电荷密度是均匀的、无限大的且位于 z 轴上，对称性考虑表明 $\mathbf{D}$ 位于径向 $\hat{\mathbf{r}}$ 方向，并且不能依赖于 ϕ 或 z 。因此， $\mathbf{D} = \hat{\mathbf{r}} D_r$ 。因此，我们构造一个半径为 r 、高度为 h 的有限圆柱高斯表面，它与电荷线同心（图 4-10）。气缸内的总充气量为 $Q = \rho_\ell h$ 。由于 $\mathbf{D}$ 沿 $\hat{\mathbf{r}}$ ，因此圆柱体的顶面和底面对方程左侧的表面积分没有贡献。 (4.29)；也就是说，只有曲面对积分有贡献。因此，

$$\int_{z=0}^h \int_{\phi=0}^{2\pi} \hat{\mathbf{r}} D_r \cdot \hat{\mathbf{r}} r d\phi dz = \rho_\ell h$$

or

$$2\pi h D_r r = \rho_\ell h,$$

这产生

$$\mathbf{E} = \frac{\mathbf{D}}{\epsilon_0} = \hat{\mathbf{r}} \frac{D_r}{\epsilon_0} = \hat{\mathbf{r}} \frac{\rho_\ell}{2\pi\epsilon_0 r}. \quad (4.33)$$

(infinite line charge)

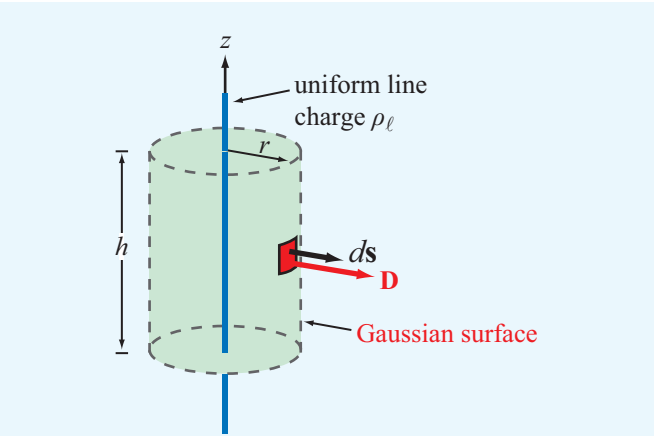


Figure 4-10 Gaussian surface around an infinitely long line of charge (Example 4-6).

Note that Eq. (4.33) is applicable for any infinite line of charge, regardless of its location and direction, as long as $\hat{\mathbf{r}}$ is properly defined as the radial distance vector from the line charge to the observation point (i.e., $\hat{\mathbf{r}}$ is perpendicular to the line of charge).

Example 4-7: Two Infinite Lines of Charge

Figure 4-11 depicts the presence of two infinite lines of charge in free space: one residing in the x - y plane parallel to the $\hat{\mathbf{x}}$ axis

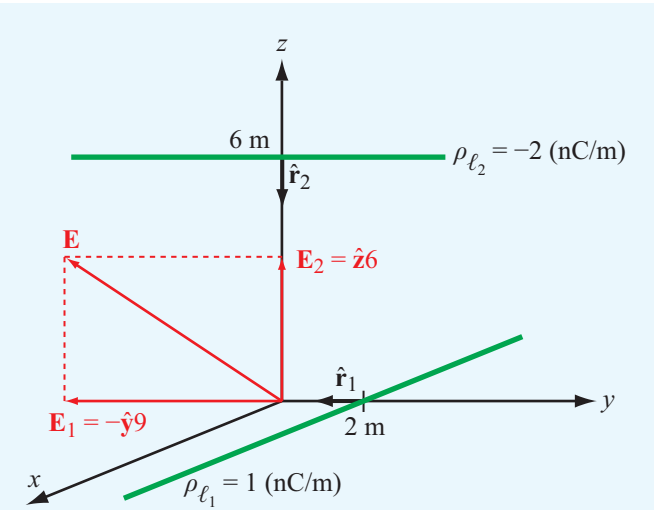


Figure 4-11 Two infinite lines of charge (Example 4-7).

and carrying charge density $\rho_{\ell_1} = 1 \text{ (nC/m)}$, and a second one residing in the y - z plane parallel to the y axis and carrying a charge density $\rho_{\ell_2} = -2 \text{ (nC/m)}$. Determine the electric field at the origin.

Solution: The electric field $\mathbf{E}$ is the sum of two electric field components:

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2,$$

where $\mathbf{E}_1$ and $\mathbf{E}_2$ are the electric fields due to line charges 1 and 2, respectively. According to Eq. (4.33), the direction of the electric field $\hat{\mathbf{r}}$ is perpendicular to the direction of the line charge and points away from the line of charge (if ρ_ℓ is positive). Hence, for the first line of charge, $\rho_{\ell_1} = 1 \text{ (nC/m)}$, $\hat{\mathbf{r}}_1 = -\hat{\mathbf{y}}$, $r_1 = 2$, and

$$\mathbf{E}_1 = \frac{\hat{\mathbf{r}}_1 \rho_{\ell_1}}{2\pi\epsilon_0 r_1} = \frac{-\hat{\mathbf{y}} 10^{-9}}{2\pi \times \frac{1}{36\pi} \times 10^{-9} \times 2} = -\hat{\mathbf{y}} 9 \text{ V/m.}$$

Similarly, for the second line of charge, $\rho_{\ell_2} = -2 \text{ (nC/m)}$, $\hat{\mathbf{r}}_2 = -\hat{\mathbf{z}}$, $r_2 = 6$, and

$$\mathbf{E}_2 = \frac{\hat{\mathbf{r}}_2 \rho_{\ell_2}}{2\pi\epsilon_0 r_2} = \frac{-\hat{\mathbf{z}}(-2) \times 10^{-9}}{2\pi \times \frac{1}{36\pi} \times 10^{-9} \times 6} = \hat{\mathbf{z}} 6 \text{ V/m.}$$

Hence,

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 = (-\hat{\mathbf{y}} 9 + \hat{\mathbf{z}} 6) \text{ V/m.}$$

Concept Question 4-7: Explain Gauss’s law. Under what circumstances is it useful?

Concept Question 4-8: How should one choose a Gaussian surface?

Exercise 4-7: Two infinite lines, each carrying a uniform charge density ρ_ℓ , reside in free space parallel to the z axis at $x = 1$ and $x = -1$. Determine $\mathbf{E}$ at an arbitrary point along the y axis.

Answer: $\mathbf{E} = \hat{\mathbf{y}} \rho_\ell y / [\pi\epsilon_0 (y^2 + 1)]$. (See [EM](#).)

Exercise 4-8: A thin spherical shell of radius a carries a uniform surface charge density ρ_s . Use Gauss’s law to determine $\mathbf{E}$ everywhere in free space.

Answer: $\mathbf{E} = \begin{cases} 0 & \text{for } R < a; \\ \hat{\mathbf{R}} \rho_s a^2 / (\epsilon R^2) & \text{for } R > a. \end{cases}$ (See [EM](#).)

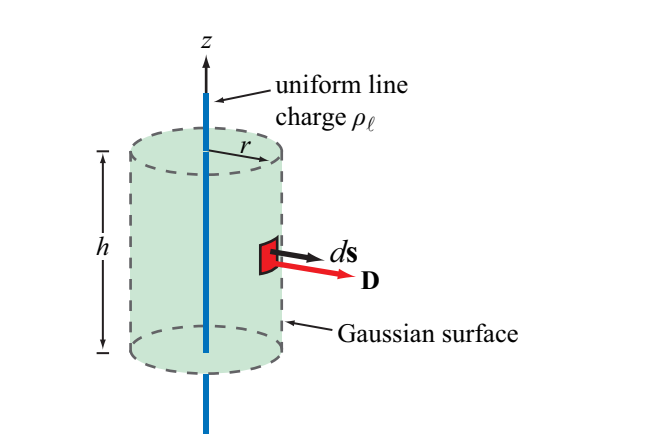


图 4-10 无限长电荷线周围的高斯表面（示例 4-6）。

请注意等式。只要将 $\hat{\mathbf{r}}$ 正确定义为从线电荷到观察点的径向距离矢量（即 $\hat{\mathbf{r}}$ 垂直于电荷线），方程（4.33）适用于任何无限电荷线，无论其位置和方向如何。

例 4-7：两条无限电荷线

图 4-11 描述了自由空间中存在两条无限电荷线：一条位于平行于 $\hat{\mathbf{x}}$ 轴的 x - y 平面中

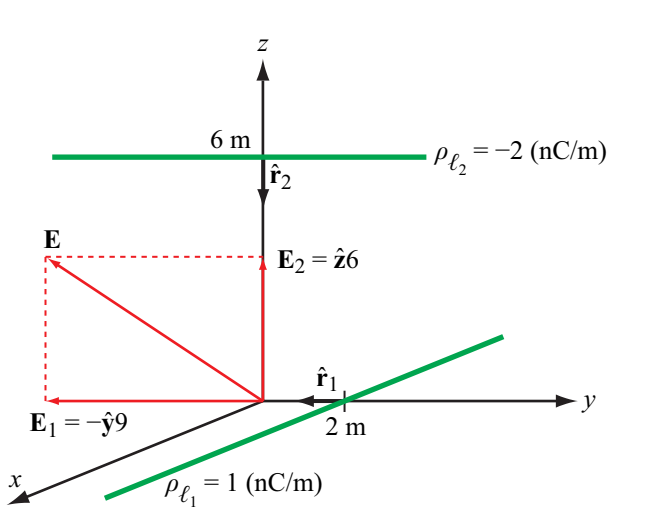


图 4-11 两条无限电荷线（例 4-7）。

并携带电荷密度 $\rho_{\ell_1} = 1 \text{ (nC/m)}$ ，第二个位于平行于 y 轴的 y - z 平面并携带电荷密度 $\rho_{\ell_2} = -2 \text{ (nC/m)}$ 。确定原点处的电场。

解：电场 $\mathbf{E}$ 是两个电场分量之和：

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2,$$

其中 $\mathbf{E}_1$ 和 $\mathbf{E}_2$ 分别是线电荷 1 和 2 产生的电场。根据方程（4.33）中，电场 $\hat{\mathbf{r}}$ 的方向垂直于线电荷的方向，并且指向远离电荷线的方向（如果 ρ_ℓ 为正）。因此，对于第一条电荷线， $\rho_{\ell_1} = 1 \text{ (nC/m)}$ 、 $\hat{\mathbf{r}}_1 = -\hat{\mathbf{y}}$ 、 $r_1 = 2$ 和

$$\mathbf{E}_1 = \frac{\hat{\mathbf{r}}_1 \rho_{\ell_1}}{2\pi\epsilon_0 r_1} = \frac{-\hat{\mathbf{y}} 10^{-9}}{2\pi \times \frac{1}{36\pi} \times 10^{-9} \times 2} = -\hat{\mathbf{y}} 9 \text{ V/m.}$$

同样，对于第二条电荷线， $\rho_{\ell_2} = -2 \text{ (nC/m)}$ 、 $\hat{\mathbf{r}}_2 = -\hat{\mathbf{z}}$ 、 $r_2 = 6$ 和

$$\mathbf{E}_2 = \frac{\hat{\mathbf{r}}_2 \rho_{\ell_2}}{2\pi\epsilon_0 r_2} = \frac{-\hat{\mathbf{z}}(-2) \times 10^{-9}}{2\pi \times \frac{1}{36\pi} \times 10^{-9} \times 6} = \hat{\mathbf{z}} 6 \text{ V/m.}$$

因此，

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 = (-\hat{\mathbf{y}} 9 + \hat{\mathbf{z}} 6) \text{ V/m.}$$

概念问题 4-7：解释高斯定律。什么情况下有用？

概念题4-8：应该如何选择高斯曲面？

练习 4-7：两条无限长的线，每条都带有均匀的电荷密度 ρ_ℓ ，位于平行于 z 轴的自由空间中 $x = 1$ 和 $x = -1$ 处。确定沿 y 轴任意点的 $\mathbf{E}$ 。

Answer: $\mathbf{E} = \hat{\mathbf{y}} \rho_\ell y / [\pi\epsilon_0 (y^2 + 1)]$. (See [EM](#).)

练习 4-8：半径为 a 的薄球壳具有均匀的表面电荷密度 ρ_s 。使用高斯定律确定自由空间中各处的 $\mathbf{E}$ 。

Answer: $\mathbf{E} = \begin{cases} 0 & \text{for } R < a; \\ \hat{\mathbf{R}} \rho_s a^2 / (\epsilon R^2) & \text{for } R > a. \end{cases}$ (See [EM](#).)

Exercise 4-9: A spherical volume of radius a contains a uniform volume charge density ρ_v . Use Gauss's law to determine $\mathbf{D}$ for (a) $R \leq a$ and (b) $R \geq a$.

Answer: (a) $\mathbf{D} = \hat{\mathbf{R}}\rho_v R/3$,
(b) $\mathbf{D} = \hat{\mathbf{R}}\rho_v a^3/(3R^2)$. (See [EM](#).)

4-5 Electric Scalar Potential

The operation of an electric circuit usually is described in terms of the currents flowing through its branches and the voltages at its nodes. The voltage difference V between two points in a circuit represents the amount of work, or *potential* energy, required to move a unit charge from one to the other.

► The term “voltage” is short for “voltage potential” and synonymous with *electric potential*. ◀

Even though when analyzing a circuit we may not consider the electric fields present in the circuit, it is in fact the existence of these fields that gives rise to voltage differences across circuit elements such as resistors or capacitors. The relationship between the electric field $\mathbf{E}$ and the electric potential V is the subject of this section.

4-5.1 Electric Potential as a Function of Electric Field

We begin by considering the simple case of a positive charge q in a uniform electric field $\mathbf{E} = -\hat{\mathbf{y}}E$ in the $-y$ direction ([Fig. 4-12](#)). The presence of the field $\mathbf{E}$ exerts a force $\mathbf{F}_e = q\mathbf{E}$ on the charge in the $-y$ direction. To move the charge along the positive y direction (against the force $\mathbf{F}_e$), we need to provide

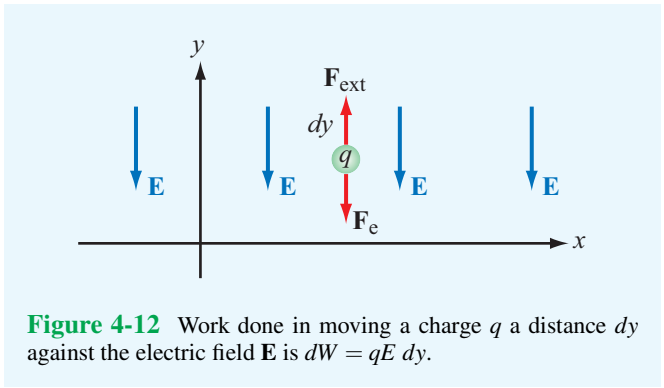


Figure 4-12 Work done in moving a charge q a distance dy against the electric field $\mathbf{E}$ is $dW = qE dy$.

an external force $\mathbf{F}_{\text{ext}}$ to counteract $\mathbf{F}_e$, which requires the expenditure of energy. To move q without acceleration (at constant speed), the net force acting on the charge must be zero, which means that $\mathbf{F}_{\text{ext}} + \mathbf{F}_e = 0$, or

$$\mathbf{F}_{\text{ext}} = -\mathbf{F}_e = -q\mathbf{E}. \quad (4.34)$$

The work done (or energy expended) in moving any object a vector differential distance $d\mathbf{l}$ while exerting a force $\mathbf{F}_{\text{ext}}$ is

$$dW = \mathbf{F}_{\text{ext}} \cdot d\mathbf{l} = -q\mathbf{E} \cdot d\mathbf{l} \quad (\text{J}). \quad (4.35)$$

Work (or energy) is measured in joules (J). If the charge is moved a distance dy along $\hat{\mathbf{y}}$, then

$$dW = -q(-\hat{\mathbf{y}}E) \cdot \hat{\mathbf{y}} dy = qE dy. \quad (4.36)$$

The differential electric potential energy dW per unit charge is called the *differential electric potential* (or differential voltage) dV . That is,

$$dV = \frac{dW}{q} = -\mathbf{E} \cdot d\mathbf{l} \quad (\text{J/C or V}). \quad (4.37)$$

The unit of V is the volt (V) with $1 \text{ V} = 1 \text{ J/C}$, and since V is measured in volts, the electric field is expressed in volts per meter (V/m).

The *potential difference* corresponding to moving a point charge from point P_1 to point P_2 ([Fig. 4-13](#)) is obtained by integrating Eq. (4.37) *along any path* between them. That is,

$$\int_{P_1}^{P_2} dV = - \int_{P_1}^{P_2} \mathbf{E} \cdot d\mathbf{l}, \quad (4.38)$$

or

$$V_{21} = V_2 - V_1 = - \int_{P_1}^{P_2} \mathbf{E} \cdot d\mathbf{l}, \quad (4.39)$$

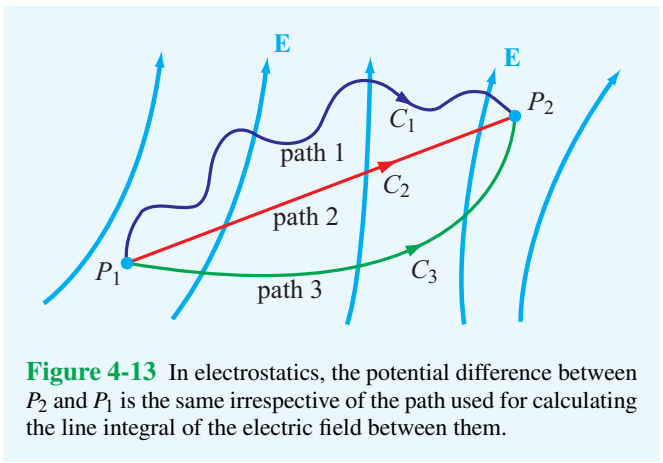


Figure 4-13 In electrostatics, the potential difference between P_2 and P_1 is the same irrespective of the path used for calculating the line integral of the electric field between them.

练习 4-9: 半径为 a 的球形体积包含均匀的体积电荷密度 ρ_v 。使用高斯定律确定 (a) $R \leq a$ 和 (b) $R \geq a$ 的 $\mathbf{D}$ 。

Answer: (a) $\mathbf{D} = \hat{\mathbf{R}}\rho_v R/3$,
(b) $\mathbf{D} = \hat{\mathbf{R}}\rho_v a^3/(3R^2)$. (See [EM](#).)

4-5 电标量势

电路的运行通常用流经其支路的电流和其节点的电压来描述。电路中两点之间的电压差 V 表示将单位电荷从一个点移动到另一个点所需的功量或势能。

► 术语 “电压” 是 “电压电势” 的缩写，与电势同义。 ◀

尽管在分析电路时我们可能不会考虑电路中存在的电场，但实际上这些电场的存在会在电阻器或电容器等电路元件上产生电压差。电场 $\mathbf{E}$ 和电势 V 之间的关系是本节的主题。

4-5.1 电势作为电场的函数

我们首先考虑在 $-y$ 方向均匀电场 $\mathbf{E} = -\hat{\mathbf{y}}E$ 中正电荷 q 的简单情况 ([图 4-12](#))。场 $\mathbf{E}$ 的存在在 $-y$ 方向上对电荷施加力 $\mathbf{F}_e = q\mathbf{E}$ 。为了沿着正 y 方向移动电荷 (对抗力 $\mathbf{F}_e$)，我们需要提供

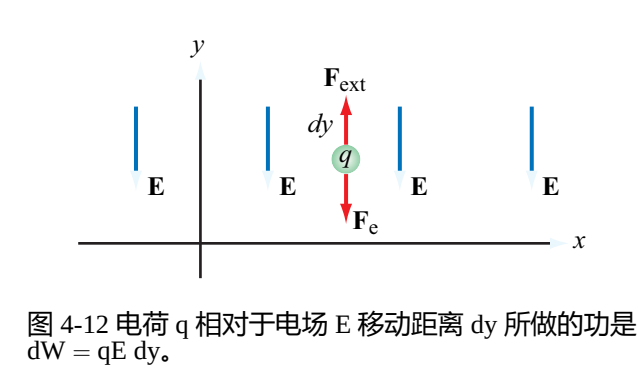


图 4-12 电荷 q 相对于电场 $\mathbf{E}$ 移动距离 dy 所做的功是 $dW = qE dy$ 。

一个外力 $\mathbf{F}_{\text{ext}}$ 来抵消 $\mathbf{F}_e$ ，需要消耗能量。要在没有加速的情况下 (以恒定速度) 移动 q ，作用在电荷上的净力必须为零，这意味着 $\mathbf{F}_{\text{ext}} + \mathbf{F}_e = 0$ ，或

$$\mathbf{F}_{\text{ext}} = -\mathbf{F}_e = -q\mathbf{E}. \quad (4.34)$$

在施加力 $\mathbf{F}_{\text{ext}}$ 的同时将任何物体移动矢量差距离 $d\mathbf{l}$ 所做的功 (或消耗的能量) 为

$$dW = \mathbf{F}_{\text{ext}} \cdot d\mathbf{l} = -q\mathbf{E} \cdot d\mathbf{l} \quad (\text{J}). \quad (4.35)$$

功 (或能量) 以焦耳 (J) 为单位。如果电荷沿 $\hat{\mathbf{y}}$ 移动距离 dy ，则

$$dW = -q(-\hat{\mathbf{y}}E) \cdot \hat{\mathbf{y}} dy = qE dy. \quad (4.36)$$

每单位电荷的微分电势能 dW 称为微分电势 (或微分电压) dV 。那是，

$$dV = \frac{dW}{q} = -\mathbf{E} \cdot d\mathbf{l} \quad (\text{J/C or V}). \quad (4.37)$$

V 的单位是伏特 (V)，其中 $1 \text{ V} = 1 \text{ J/C}$ ，由于 V 以伏特为单位测量，因此电场以伏特每米 (V/m) 表示。

图 1 点电荷从 P_1 点移动到 P_2 (点对应的电势差4-13) 是通过积分方程得到的。(4.37) 沿着它们之间的任何路径。那是，

$$\int_{P_1}^{P_2} dV = - \int_{P_1}^{P_2} \mathbf{E} \cdot d\mathbf{l}, \quad (4.38)$$

或者

$$V_{21} = V_2 - V_1 = - \int_{P_1}^{P_2} \mathbf{E} \cdot d\mathbf{l}, \quad (4.39)$$

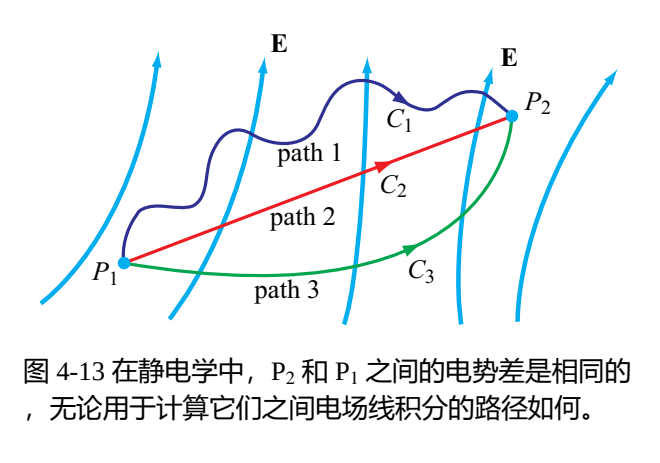


图 4-13 在静电学中， P_2 和 P_1 之间的电势差是相同的，无论用于计算它们之间电场线积分的路径如何。

where V_1 and V_2 are the electric potentials at points P_1 and P_2 , respectively. The result of the line integral on the right-hand side of Eq. (4.39) is independent of the specific integration path that connects points P_1 and P_2 . This follows immediately from the law of conservation of energy. To illustrate with an example, consider a particle in Earth’s gravitational field. If the particle is raised from a height h_1 above Earth’s surface to height h_2 , the particle gains potential energy in an amount proportional to $(h_2 - h_1)$. If instead we were to first raise the particle from height h_1 to a height h_3 greater than h_2 , thereby giving it potential energy proportional to $(h_3 - h_1)$, and then let it drop back to height h_2 by expending an energy amount proportional to $(h_3 - h_2)$, its net gain in potential energy would again be proportional to $(h_2 - h_1)$.

The same principle applies to the electric potential energy W and to the potential difference $(V_2 - V_1)$. The voltage difference between two nodes in an electric circuit has the same value regardless of which path in the circuit we follow between the nodes. Moreover, ***Kirchhoff’s voltage law*** states that the net voltage drop around a closed loop is zero. If we go from P_1 to P_2 by path 1 in **Fig. 4-13** and then return from P_2 to P_1 by path 2, the right-hand side of Eq. (4.39) becomes a closed contour, and the left-hand side vanishes. In fact, ***the line integral of the electrostatic field $\mathbf{E}$ around any closed contour C is zero:***

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = 0. \quad \textbf{(electrostatics)} \quad (4.40)$$

► A vector field whose line integral along any closed path is zero is called a ***conservative*** or an ***irrotational*** field. Hence, the electrostatic field $\mathbf{E}$ is conservative. ◀

As we will see later in Chapter 6, if $\mathbf{E}$ is a time-varying function, it is no longer conservative, and its line integral along a closed path is not necessarily zero.

The conservative property of the electrostatic field can be deduced from Maxwell’s second equation, Eq. (4.1b). If $\partial/\partial t = 0$, then

$$\nabla \times \mathbf{E} = 0. \quad (4.41)$$

If we take the surface integral of $\nabla \times \mathbf{E}$ over an open surface S and then apply Stokes’s theorem expressed by Eq. (3.107) to convert the surface integral into a line integral, we obtain

$$\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{s} = \oint_C \mathbf{E} \cdot d\mathbf{l} = 0, \quad (4.42)$$

where C is a closed contour surrounding S . Thus, Eq. (4.41) is the differential-form equivalent of Eq. (4.40).

We now define what we mean by the electric potential V at a point in space. Before we do so, however, let us revisit our electric-circuit analogue. Just as a node in a circuit cannot be assigned an absolute voltage, a point in space cannot have an absolute electric potential. The voltage of a node in a circuit is measured relative to that of a conveniently chosen reference point to which we have assigned a voltage of zero, which we call ***ground***. The same principle applies to the electric potential V . Usually (but not always), the reference point is chosen to be at infinity. That is, in Eq. (4.39) we assume that $V_1 = 0$ when P_1 is at infinity. Therefore, the electric potential V at any point P is

$$V = - \int_{\infty}^P \mathbf{E} \cdot d\mathbf{l} \quad (\text{V}). \quad (4.43)$$

Example 4-8: Computing V from $\mathbf{E}$ along Two Paths

A vector field is said to be ***conservative*** if its line integral between two points is the same—irrespective of the path taken between them. In a given region of space, the field $\mathbf{E}$ is given by

$$\mathbf{E} = \hat{\mathbf{x}}\,x^2 + \hat{\mathbf{y}}\,y^2 + \hat{\mathbf{z}}\,z^2. \quad (4.44)$$

(a) Confirm that $\mathbf{E}$ is conservative by demonstrating that $\nabla \times \mathbf{E} = 0$. (b) Compute the potential difference V_{21} between points 1 and 2 in **Fig. 4-14** following the direct path between them. (c) Compute V_{21} by following the path $ABCD$ between points 1 and 2.

Solution: (a) The given electric field has components $E_x = x^2$, $E_y = y^2$, and $E_z = z^2$. Applying the curl operator to $\mathbf{E}$ gives

$$\begin{aligned} \nabla \times \mathbf{E} &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & y^2 & z^2 \end{vmatrix} \\ &= \hat{\mathbf{x}} \left(\frac{\partial z^2}{\partial y} - \frac{\partial y^2}{\partial z} \right) - \hat{\mathbf{y}} \left(\frac{\partial z^2}{\partial x} - \frac{\partial x^2}{\partial z} \right) + \hat{\mathbf{z}} \left(\frac{\partial y^2}{\partial x} - \frac{\partial x^2}{\partial y} \right) \\ &= \hat{\mathbf{x}}(0 - 0) - \hat{\mathbf{y}}(0 - 0) + \hat{\mathbf{z}}(0 - 0) = 0. \end{aligned}$$

(b) Voltage V_{21} is given by

$$V_{21} = - \int_{P_1}^{P_2} \mathbf{E} \cdot d\mathbf{l}.$$

其中 V_1 和 V_2 分别是 P_1 和 P_2 点处的电势。等式右侧的线积分结果：(4.39) 独立于连接点 P_1 和 P_2 的特定积分路径。这直接遵循能量守恒定律。为了举例说明，考虑地球引力场中的一个粒子。如果粒子从地球表面上方的高度 h_1 上升到高度 h_2 ，粒子将获得与 $(h_2 - h_1)$ 成比例的势能。相反，如果我们首先将粒子从高度 h_1 提升到大于 h_2 的高度 h_3 ，从而赋予其与 $(h_3 - h_1)$ 成比例的势能，然后通过消耗与 $(h_3 - h_2)$ 成比例的能量让它落回高度 h_2 ，则其势能的净增益将再次与 $(h_2 - h_1)$ 。

同样的原理也适用于电势能 W 和电势差 $(V_2 - V_1)$ 。电路中两个节点之间的电压差具有相同的值，无论我们在节点之间遵循电路中的哪条路径。此外，基尔霍夫电压定律指出，闭环周围的净电压降为零。如果我们通过图 4-13 中的路径 1 从 P_1 到 P_2 ，然后通过路径 2 从 P_2 返回到 P_1 ，则方程的右侧为：(4.39) 变成闭合轮廓，左侧消失。事实上，静电场 $\mathbf{E}$ 围绕任何闭合轮廓 C 的线积分为零：

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = 0. \quad \textbf{(electrostatics)} \quad (4.40)$$

► 沿任何闭合路径的线积分为零的矢量场称为保守场或无旋场。因此，静电场 $\mathbf{E}$ 是保守的。 ◀

正如我们稍后将在第 6 章中看到的，如果 $\mathbf{E}$ 是一个时变函数，它就不再是保守的，并且它沿闭合路径的线积分不一定为零。

静电场的保守性质可以从麦克斯韦第二方程 (Eq. (4.1b))。如果 $\partial/\partial t = 0$ ，则

$$\nabla \times \mathbf{E} = 0. \quad (4.41)$$

如果我们在开放曲面 S 上取 $\nabla \times \mathbf{E}$ 的曲面积分，然后应用方程 1 表示的斯托克斯定理：(3.107) 将面积分转化为线积分，可得

$$\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{s} = \oint_C \mathbf{E} \cdot d\mathbf{l} = 0, \quad (4.42)$$

其中 C 是围绕 S 的闭合轮廓。(4.41) 是等式的微分形式等价物。(4.40)。

现在我们定义空间中一点的电势 V 的含义。然而，在此之前，让我们回顾一下我们的电路模拟。正如电路中的节点无法分配绝对电压一样，空间中的点也不可能具有绝对电势。电路中节点的电压是相对于方便选择的参考点的电压来测量的，我们已将零电压分配给该参考点，我们称之为接地。同样的原理也适用于电势 V 。通常（但并非总是），参考点被选择为无穷远。也就是说，在等式中。(4.39) 我们假设当 P_1 无穷大时 $V_1 = 0$ 。因此，任意点 P 的电势 V 为

$$V = - \int_{\infty}^P \mathbf{E} \cdot d\mathbf{l} \quad (\text{V}). \quad (4.43)$$

示例 4-8：沿着两条路径从 $\mathbf{E}$ 计算 V

如果两个点之间的线积分相同，则称向量场是保守的——无论它们之间的路径如何。在给定的空间区域中，场 $\mathbf{E}$ 由下式给出

$$\mathbf{E} = \hat{\mathbf{x}}\,x^2 + \hat{\mathbf{y}}\,y^2 + \hat{\mathbf{z}}\,z^2. \quad (4.44)$$

(a) 通过证明 $\nabla \times \mathbf{E} = 0$ 来确认 $\mathbf{E}$ 是保守的。(b) 沿着图 4-14 中点 1 和 2 之间的直接路径计算它们之间的电位差 V_{21} 。(c) 沿着点 1 和点 2 之间的路径 $ABCD$ 计算 V_{21} 。

解：(a) 给定电场具有分量 $E_x = x^2$ 、 $E_y = y^2$ 和 $E_z = z^2$ 。将旋度运算符应用于 $\mathbf{E}$ 给出

$$\begin{aligned} \nabla \times \mathbf{E} &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & y^2 & z^2 \end{vmatrix} \\ &= \hat{\mathbf{x}} \left(\frac{\partial z^2}{\partial y} - \frac{\partial y^2}{\partial z} \right) - \hat{\mathbf{y}} \left(\frac{\partial z^2}{\partial x} - \frac{\partial x^2}{\partial z} \right) + \hat{\mathbf{z}} \left(\frac{\partial y^2}{\partial x} - \frac{\partial x^2}{\partial y} \right) \\ &= \hat{\mathbf{x}}(0 - 0) - \hat{\mathbf{y}}(0 - 0) + \hat{\mathbf{z}}(0 - 0) = 0. \end{aligned}$$

(b) 电压 V_{21} 由下式给出

$$V_{21} = - \int_{P_1}^{P_2} \mathbf{E} \cdot d\mathbf{l}.$$

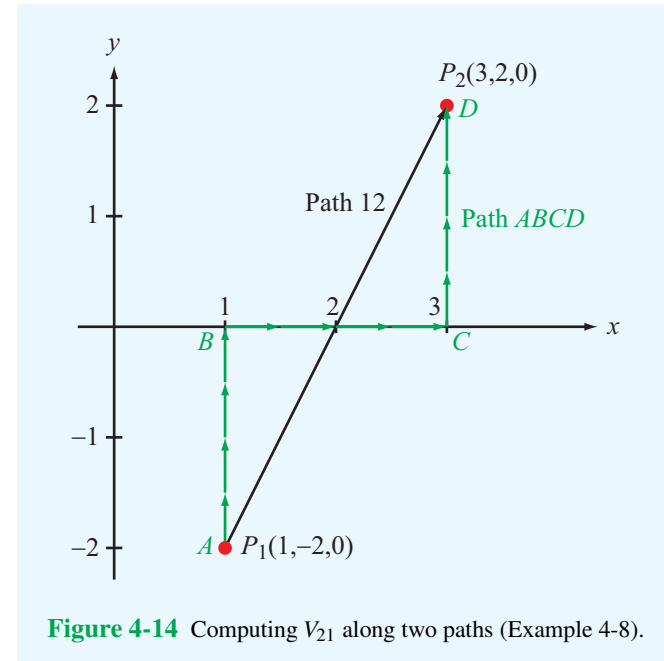


Figure 4-14 Computing V_{21} along two paths (Example 4-8).

The straight-line path resides in the x - y plane, so it is described by the linear form $y = ax + b$. At point 1, $x_1 = 1$ and $y_1 = -2$. Hence,

$$-2 = a + b.$$

Similarly, at point 2, $x_2 = 3$ and $y_2 = 2$, so

$$2 = 3a + b.$$

The two equations lead to $a = 2$, $b = -4$, and

$$y = 2x - 4. \quad (4.45)$$

Since path P_1 - P_2 is entirely in the x - y plane, we can set $z = 0$ in the expression for $\mathbf{E}$. Also, we can use the relation given by Eq. (4.45) to reduce $\mathbf{E}$ to a single variable:

$$\mathbf{E} = \hat{\mathbf{x}} x^2 + \hat{\mathbf{y}} y^2 + \hat{\mathbf{z}} z^2 \Big|_{z=0 \text{ and } y=2x-4} = \hat{\mathbf{x}} x^2 + \hat{\mathbf{y}} (2x-4)^2. \quad (4.46)$$

In general,

$$d\mathbf{l} = \hat{\mathbf{x}} dx + \hat{\mathbf{y}} dy + \hat{\mathbf{z}} dz.$$

In the x - y plane, $dz = 0$, and along the straight-line path given by $y = 2x - 4$,

$$dy = 2 dx. \quad (4.47)$$

Hence,

$$d\mathbf{l} = \hat{\mathbf{x}} dx + \hat{\mathbf{y}} 2 dx. \quad (4.48)$$

The potential difference is then

$$\begin{aligned} V_{21} &= - \int_{P_1}^{P_2} \mathbf{E} \cdot d\mathbf{l} \\ &= - \int_{x=1}^{x=3} [\hat{\mathbf{x}} x^2 + \hat{\mathbf{y}} (2x-4)^2] \cdot [\hat{\mathbf{x}} dx + \hat{\mathbf{y}} 2 dx] \\ &= - \int_{x=1}^3 [x^2 + 2(2x-4)^2] dx \\ &= - \int_{x=1}^3 (9x^2 - 32x + 32) dx = -14 \quad (\text{V}). \end{aligned} \quad (4.49)$$

(c) Path $ABCD$ in Fig. 4-14 consists of three segments.

A to B:

$$\mathbf{E} = \hat{\mathbf{x}} x^2 + \hat{\mathbf{y}} y^2 + \hat{\mathbf{z}} z^2 \Big|_{x=1, z=0} = \hat{\mathbf{x}} 1 + \hat{\mathbf{y}} y^2, \quad (4.50a)$$

and

$$d\mathbf{l} = \hat{\mathbf{y}} dy. \quad (4.50b)$$

B to C:

$$\mathbf{E} = \hat{\mathbf{x}} x^2 + \hat{\mathbf{y}} y^2 + \hat{\mathbf{z}} z^2 \Big|_{y=0, z=0} = \hat{\mathbf{x}} x^2, \quad (4.51a)$$

and

$$d\mathbf{l} = \hat{\mathbf{x}} dx. \quad (4.51b)$$

C to D:

$$\mathbf{E} = \hat{\mathbf{x}} x^2 + \hat{\mathbf{y}} y^2 + \hat{\mathbf{z}} z^2 \Big|_{x=3, z=0} = \hat{\mathbf{x}} 9 + \hat{\mathbf{y}} y^2, \quad (4.52a)$$

and

$$d\mathbf{l} = \hat{\mathbf{y}} dy. \quad (4.52b)$$

Hence,

$$\begin{aligned} V_{21} &= - \int_{P_1}^{P_2} \mathbf{E} \cdot d\mathbf{l} \\ &= - \left[\int_{A@x=1, y=-2}^{B@x=1, y=0} (\hat{\mathbf{x}} 1 + \hat{\mathbf{y}} y^2) \cdot \hat{\mathbf{y}} dy \right. \\ &\quad + \int_{B@x=1, y=0}^{C@x=3, y=0} \hat{\mathbf{x}} x^2 \cdot \hat{\mathbf{x}} dx \\ &\quad \left. + \int_{C@x=3, y=0}^{D@x=3, y=2} (\hat{\mathbf{x}} 9 + \hat{\mathbf{y}} y^2) \cdot \hat{\mathbf{y}} dy \right] \\ &= - \left[\frac{y^3}{3} \Big|_{-2}^0 + \frac{x^3}{3} \Big|_1^3 + \frac{y^3}{3} \Big|_0^2 \right] \\ &= - \left[\frac{8}{3} + \frac{27}{3} - \frac{1}{3} + \frac{8}{3} \right] = -14 \quad (\text{V}), \end{aligned} \quad (4.53)$$

which is identical with the result given by Eq. (4.49) for the line integral along the straight-line path between points 1 and 2.

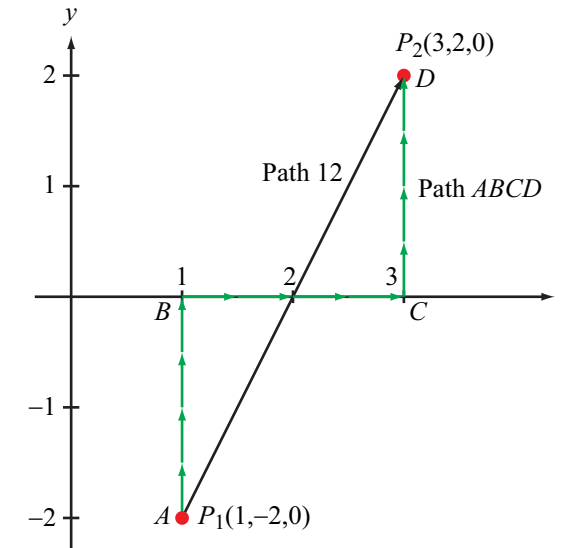


图 4-14 沿两条路径计算 V_{21} (示例 4-8)。

直线路径位于 x - y 平面内，因此由线性形式 $y = ax + b$ 描述。在点 1 处， $x_1 = 1$ 和 $y_1 = -2$ 。因此，

$$-2 = a + b.$$

类似地，在点 2 处， $x_2 = 3$ 且 $y_2 = 2$ ，因此

$$2 = 3a + b.$$

这两个方程可得出 $a = 2$ 、 $b = -4$ ，以及

$$y = 2x - 4. \quad (4.45)$$

由于路径 P_1 - P_2 完全在 x - y 平面上，因此我们可以在 $\mathbf{E}$ 的表达式中将 z 设置为 0。此外，我们可以使用等式 1 给出的关系。(4.45) 将 $\mathbf{E}$ 简化为单个变量：

$$\mathbf{E} = \hat{\mathbf{x}} x^2 + \hat{\mathbf{y}} y^2 + \hat{\mathbf{z}} z^2 \Big|_{z=0 \text{ and } y=2x-4} = \hat{\mathbf{x}} x^2 + \hat{\mathbf{y}} (2x-4)^2. \quad (4.46)$$

一般来说，

$$d\mathbf{l} = \hat{\mathbf{x}} dx + \hat{\mathbf{y}} dy + \hat{\mathbf{z}} dz.$$

在 x - y 平面中， $dz = 0$ ，并沿着 $y = 2x - 4$ 给出的直线路径，

$$dy = 2 dx. \quad (4.47)$$

因此，

$$d\mathbf{l} = \hat{\mathbf{x}} dx + \hat{\mathbf{y}} 2 dx. \quad (4.48)$$

那么电位差就是

$$\begin{aligned} V_{21} &= - \int_{P_1}^{P_2} \mathbf{E} \cdot d\mathbf{l} \\ &= - \int_{x=1}^{x=3} [\hat{\mathbf{x}} x^2 + \hat{\mathbf{y}} (2x-4)^2] \cdot [\hat{\mathbf{x}} dx + \hat{\mathbf{y}} 2 dx] \\ &= - \int_{x=1}^3 [x^2 + 2(2x-4)^2] dx \\ &= - \int_{x=1}^3 (9x^2 - 32x + 32) dx = -14 \quad (\text{V}). \end{aligned} \quad (4.49)$$

(图4-14中的c)路径ABCD由三段组成。

A to B:

$$\mathbf{E} = \hat{\mathbf{x}} x^2 + \hat{\mathbf{y}} y^2 + \hat{\mathbf{z}} z^2 \Big|_{x=1, z=0} = \hat{\mathbf{x}} 1 + \hat{\mathbf{y}} y^2, \quad (4.50a)$$

和

$$d\mathbf{l} = \hat{\mathbf{y}} dy. \quad (4.50b)$$

B to C:

$$\mathbf{E} = \hat{\mathbf{x}} x^2 + \hat{\mathbf{y}} y^2 + \hat{\mathbf{z}} z^2 \Big|_{y=0, z=0} = \hat{\mathbf{x}} x^2, \quad (4.51a)$$

和

$$d\mathbf{l} = \hat{\mathbf{x}} dx. \quad (4.51b)$$

C to D:

$$\mathbf{E} = \hat{\mathbf{x}} x^2 + \hat{\mathbf{y}} y^2 + \hat{\mathbf{z}} z^2 \Big|_{x=3, z=0} = \hat{\mathbf{x}} 9 + \hat{\mathbf{y}} y^2, \quad (4.52a)$$

and

$$d\mathbf{l} = \hat{\mathbf{y}} dy. \quad (4.52b)$$

因此，

$$\begin{aligned} V_{21} &= - \int_{P_1}^{P_2} \mathbf{E} \cdot d\mathbf{l} \\ &= - \left[\int_{A@x=1, y=-2}^{B@x=1, y=0} (\hat{\mathbf{x}} 1 + \hat{\mathbf{y}} y^2) \cdot \hat{\mathbf{y}} dy \right. \\ &\quad + \int_{B@x=1, y=0}^{C@x=3, y=0} \hat{\mathbf{x}} x^2 \cdot \hat{\mathbf{x}} dx \\ &\quad \left. + \int_{C@x=3, y=0}^{D@x=3, y=2} (\hat{\mathbf{x}} 9 + \hat{\mathbf{y}} y^2) \cdot \hat{\mathbf{y}} dy \right] \\ &= - \left[\frac{y^3}{3} \Big|_{-2}^0 + \frac{x^3}{3} \Big|_1^3 + \frac{y^3}{3} \Big|_0^2 \right] \\ &= - \left[\frac{8}{3} + \frac{27}{3} - \frac{1}{3} + \frac{8}{3} \right] = -14 \quad (\text{V}), \end{aligned} \quad (4.53)$$

这与等式给出的结果相同。(4.49) 为沿点 1 和点 2 之间的直线路径的线积分。